# Triangle Inequality Theorem; Converse of the Pythagorean Theorem

*Teacher's Edition*

## The Period

**MA.8.GR.1.3** Use the Triangle Inequality Theorem to determine if a triangle can be formed from a given set of sides. Use the converse of the Pythagorean Theorem to determine if a right triangle can be formed from a given set of sides.

**Must** Decide whether three lengths make a triangle (the two shorter sum to more than the longest) and a right triangle (the converse, on the longest side).

**Must not** Acute or obtuse by comparing squares (Geometry); sides as expressions; finding a missing side (6.03).

**Target** I can decide whether three lengths can make a triangle, and whether they make a right triangle.

**MTR** **MTR.2.1** — Notes I: three lengths as sticks and as one inequality · **MTR.4.1** — boards 3, 5 and 7 split; the sixty-second re-vote · **MTR.6.1** — board 7: a reason checked against the squares, not the sum

| Segment | Slides | Time |
|---|---|---|
| Settle in; post the learning target | 1 | 1 min |
| Warm-Up | 2–4 | 6 min |
| Notes | 5–7 | 11 min |
| Example 1 | 8–9 | 4 min |
| Your Turn | 10–11 | 3 min |
| Example 2 | 12–13 | 4 min |
| Your Turn | 14–15 | 3 min |
| Whiteboards | 16–33 | 10 min |
| Independent Set | 34 | 6 min |
| IXL | 35 | 5 min |
| **Total** | | **53 min** |

### Say these three things out loud today

1. Add the two shorter sides: MORE than the longest is a triangle; equal lies flat.
2. The third side is between the difference and the sum.
3. Right triangle: the shorter sides' squares add up to the longest side's square.

## Slide by Slide

1. **Triangle Inequality Theorem; Converse of the Pythagorean Theorem.** Post the learning target.
2. **Warm-Up — On your own. Four minutes. No notes.** Q3 and Q4 are a pair: $5^2 + 12^2 = 169 = 13^2$, today's converse before it has a name. *1 retrieves 5.10–11 — a unit rate; yesterday was the test, so the band reaches the last lesson (yesterday). 2 retrieves 5.02–03 — area scales by $k^2$ (last week). 3 retrieves 4.05 — a perfect square's root (Unit 4) (two units back). 4 retrieves MA.7.NSO.1.1 — whole-number powers; the converse adds two squares (prior grade — the prerequisite this lesson needs).*
3. **Warm-Up — Answers.** Reveal. 1. 288 mi 2. 45 $cm^2$ 3. 13 4. 169
4. **Vocabulary — The new words, before they are used.** Read each word and its meaning once; the notes use them next.
5. **Notes — Copy the rule.** Equal is not enough. Equal lies flat.
6. **Notes.** Two sides, 4 and 9.
7. **Notes.** Circle the longest side first. It is the only one that can be the hypotenuse.
8. **Example 1 — Can the three lengths make a triangle?** Ask which two sides get added before anyone adds: the two shorter, 5 and 8.
9. **Example 1 — Worked.** 5 + 8 = 13, and 13 is less than 14: the two short sides cannot reach across.
10. **Your Turn — Same steps, your numbers. Boards up when done.** 7 + 9 = 16 > 15, so yes — a board that says no saw the big 15 and did not add.
11. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
12. **Example 2 — Is it a right triangle?** Ask which side is c before anyone squares: 20, the longest, though it is listed first.
13. **Example 2 — Worked.** The longest side is 20, listed first.
14. **Your Turn — Same steps, your numbers. Boards up when done.** 64 + 121 = 185, not 196, so not right — a board that says 'right' compared 185 with something other than $14^2$.
15. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
16. **16–17. Board 1.** Can sides of 2 ft, 7 ft and 10 ft make a triangle? → ***No — 2 + 7 = 9 is less than 10.*** *Yes — added 7 + 10 or 2 + 10 [MA.8.GR.1.3: test the two shorter]; 'yes — the ends just do not meet' [Florida B1G-M, GR.1.3: believing the sides need not meet at the corners]*
18. **18–19. Board 2.** Can sides of 6 m, 6 m and 11 m make a triangle? → ***Yes — 6 + 6 = 12 is more than 11.*** *No — 'two sides cannot be equal' [Notes I: only the sum matters]; 'it is not a right triangle, so the rule does not apply' [Florida B1G-M, GR.1.3: the Triangle Inequality holds for every triangle]*
19. **20–21. Board 3.** Which three lengths can make a triangle? → ***B — 5, 11, 8*** *(A) 7 + 7 = 14 counted — equal lies flat [Notes I]; (C) added the first two listed, 3 + 10 [MA.8.GR.1.3: test the two shorter]; (D) added the longest with another, 12 + 4 [MA.8.GR.1.3: test the two shorter] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*
20. **22–23. Board 4.** Two sides of a triangle are 7 cm and 12 cm. Write an inequality for the length x, in centimeters, of the third side. → ***5 < x < 19 — more than 5 cm***

***and less than 19 cm.*** *5 ≤ x ≤ 19 — the flat cases included [Notes II: never equal]; x < 19 — no bottom end [Notes II: long enough to reach]*

**24–25. Board 5.** Two sides of a triangle are 6 cm and 10 cm. Which inequality shows every possible length x, in centimeters, of the third side? → ***C — 4 < x < 16*** *(A) between the two given sides — the third side can be shorter than both [Notes II: more than the difference]; (B) the flat cases included: at 4 or 16 the sides lie flat [Notes II: never equal]; (D) no bottom end — a third side of 4 or less cannot reach [Notes II: more than the difference] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

**26–27. Board 6.** Do sides of 8 inches, 15 inches and 17 inches make a right triangle? → ***Yes — $8^2 + 15^2 = 289 = 17^2$.*** *No — compared 8 + 15 = 23 with 17 [Notes III: squares, not sides]*

**28–29. Board 7.** A triangle has sides of 25 cm, 7 cm and 24 cm. Is it a right triangle? Which answer has the right reason? → ***A — yes, because $7^2 + 24^2 = 25^2$*** *(B) the right verdict for the wrong reason — the Triangle Inequality shows only a triangle [MA.8.GR.1.3: two tests]; (C) 25 taken as a leg [Florida B1G-M, GR.1.1: misidentifying the hypotenuse]; (D) added the sides, not their squares [Notes III] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

**30–31. Board 8.** A triangular sail has sides of 20 ft, 21 ft and 29 ft, and it weighs 12 pounds. Does the sail have a right angle? → ***Yes — $20^2 + 21^2 = 841 = 29^2$.*** *The 12 pounds is not needed. 'Cannot tell without the angles' [Notes III: the sides are enough]*

**32–33. Board 9.** Ana says sides of 6, 7 and 9 make a right triangle because 6 + 7 > 9. Is she right? Explain. → ***No — 6 + 7 > 9 shows only a triangle; 85 is not 81, so it is not right.*** *'Yes, 13 > 9' — the inequality taken as the converse [MA.8.GR.1.3: two tests]*

**34. Independent Set — Six questions. On your own, in writing.** Hand out the Independent Set; the same six questions are on the board.

**35. IXL — Last five minutes.** IXL: every listed skill is required, SmartScore 67.

**Variation.** Boards 1–2: a triangle? (no, then yes with two equal sides); 3 votes with the equal case and the wrong pair; 4–5 the third side's range, written then voted. The pattern breaks at 6 — a RIGHT triangle?, squares not sums — and 7 lists the longest side first; 8 has the extra number; 9 is written. The independent set walks the same path.

## Misconceptions to Watch

**Board 1.** Yes — added 7 + 10 or 2 + 10 [MA.8.GR.1.3: test the two shorter]

**Board 2.** No — 'two sides cannot be equal' [Notes I: only the sum matters]

**Board 3.** 7 + 7 = 14 counted — equal lies flat [Notes I]

**Board 4.** 5 ≤ x ≤ 19 — the flat cases included [Notes II: never equal]

**Board 5.** between the two given sides — the third side can be shorter than both [Notes II: more than the difference]

**Board 6.** No — compared 8 + 15 = 23 with 17 [Notes III: squares, not sides]

**Board 7.** the right verdict for the wrong reason — the Triangle Inequality shows only a triangle [MA.8.GR.1.3: two tests]

**Board 8.** 'Cannot tell without the angles' [Notes III: the sides are enough]

**Board 9.** 'Yes, 13 > 9' — the inequality taken as the converse [MA.8.GR.1.3: two tests]

**Materials** Whiteboards and markers. Optional: strips cut to 3, 4, 5, 6 and 9 units for Notes I. Calculators allowed, printed nowhere.

# Triangle Inequality Theorem; Converse of the Pythagorean Theorem

*Lesson Plan*

## 1. Lesson Information

| | | | |
|---|---|---|---|
| Course | Grade 7 Accelerated (1205050) | Unit | 6 — Relationships in Triangles |
| Lesson | Lessons 1–2 — Triangle Inequality Theorem; Converse of the Pythagorean Theorem | Date | _____ |
| Length | 53 minutes (one period) | Benchmark(s) | MA.8.GR.1.3 |

## 2. Standards

**MA.8.GR.1.3** Use the Triangle Inequality Theorem to determine if a triangle can be formed from a given set of sides. Use the converse of the Pythagorean Theorem to determine if a right triangle can be formed from a given set of sides.

**Must.** Decide whether three lengths make a triangle (the two shorter sum to more than the longest) and a right triangle (the converse, on the longest side).

**Must not.** Acute or obtuse by comparing squares (Geometry); sides as expressions; finding a missing side (6.03).

## 3. Mathematical Thinking and Reasoning Standards

*the evidence for each, from this period*

| MTR | What it asks | The evidence in this lesson |
|---|---|---|
| MA.K12.MTR.2.1 | Demonstrate understanding by representing problems in multiple ways. | Notes I: three lengths as sticks and as one inequality |
| MA.K12.MTR.4.1 | Engage in discussions that reflect on the mathematical thinking of self and others. | boards 3, 5 and 7 split; the sixty-second re-vote |
| MA.K12.MTR.6.1 | Assess the reasonableness of solutions. | board 7: a reason checked against the squares, not the sum |

## 4. Learning Target and Essential Question

**Learning target.** I can decide whether three lengths can make a triangle, and whether they make a right triangle.

**Essential question.** Why can't any three sticks make a triangle, and how do three numbers tell you a corner is square?

**Building on.** Grade 7 (MA.7.NSO.1.1) — squares of whole numbers; 4.05 — square roots of perfect squares; Grade 5 (MA.5.GR.1.1) — a right triangle has one right angle.

**Working toward.** 6.03 — the Pythagorean Theorem finds a missing side; 6.05–06 — distance on the coordinate plane; Algebra 1 and Geometry — the Triangle Inequality in proofs.

## 5. Sequence

*read from the deck; never typed*

| Segment | Min | What the teacher does | What students do |
|---|---|---|---|
| Settle in; post the learning target | 1 | Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet. | Copy the target. |
| Warm-Up | 6 | Two minutes silent, then reveal. One sentence of reteach per question at most. Then the new words, read once. | Four retrieval questions, alone, no notes. Then copy the new words. |
| Notes | 11 | Teach from the front; say the OFF line out loud. | Copy the notes; answer the checks aloud. |
| Example 1 | 4 | Work the example; name the wrong boards before they appear. | Watch, then work the Your Turn alone. |
| Your Turn | 3 | Circulate; reveal and name what a wrong board most likely did. | Work it alone; boards up on the cue. |
| Example 2 | 4 | Work the example; name the wrong boards before they appear. | Watch, then work the Your Turn alone. |
| Your Turn | 3 | Circulate; reveal and name what a wrong board most likely did. | Work it alone; boards up on the cue. |
| Whiteboards | 10 | One question at a time. Boards down until the cue, all up together; scan the back row first. | Answer on a board; hold it up on three. |
| Independent Set | 6 | Silent. Circulate and mark what you see; do not teach. | Six questions, written, alone. |
| IXL | 5 | Assign the day's IXL skills. | Start the skills; finish tonight. |
| **Total** | **53** | | |

## 6. Gradual Release

**I do.** The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the OFF line is said out loud rather than printed.

**We do.** Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

**You do.** The nine-question whiteboard round (10 minutes) and then the six-question independent set, silent and written.

## 7. Higher-Order Questions

| Question | DOK |
|---|---|
| Two sides of a triangle are 5 and 9. Why can the third side not be 4, and why can it not be 14? | DOK 3 |
| Lengths 6, 8 and 10 make a right triangle. Without computing, do 12, 16 and 20? Why? | DOK 2 |
| A student tests $13^2 + 12^2 = 5^2$ for sides 13, 12 and 5 and decides it is not a right triangle. What should she have tested? | DOK 2 |

## 8. Checks for Understanding

*each board, and what to do about it*

| # | What the board asks | Answer | If the board is wrong |
|---|---|---|---|
| 1 | Can sides of 2 ft, 7 ft and 10 ft make a triangle? | ***No — 2 + 7 = 9 is less than 10.*** | Yes — added 7 + 10 or 2 + 10 [MA.8.GR.1.3: test the two shorter]; 'yes — the ends just do not meet' [Florida B1G-M, GR.1.3: believing the sides need not meet at the corners] |
| 2 | Can sides of 6 m, 6 m and 11 m make a triangle? | ***Yes — 6 + 6 = 12 is more than 11.*** | No — 'two sides cannot be equal' [Notes I: only the sum matters]; 'it is not a right triangle, so the rule does not apply' [Florida B1G-M, GR.1.3: the Triangle Inequality holds for every triangle] |
| 3 | Which three lengths can make a triangle? | ***B — 5, 11, 8*** | (A) 7 + 7 = 14 counted — equal lies flat [Notes I]; (C) added the first two listed, 3 + 10 [MA.8.GR.1.3: test the two shorter]; (D) added the longest with another, 12 + 4 [MA.8.GR.1.3: test the two shorter] |
| 4 | Two sides of a triangle are 7 cm and 12 cm. Write an inequality for the length x, in centimeters, of the third side. | ***5 < x < 19 — more than 5 cm and less than 19 cm.*** | 5 ≤ x ≤ 19 — the flat cases included [Notes II: never equal]; x < 19 — no bottom end [Notes II: long enough to reach] |
| 5 | Two sides of a triangle are 6 cm and 10 cm. Which inequality shows every possible length x, in centimeters, of the third side? | ***C — 4 < x < 16*** | (A) between the two given sides — the third side can be shorter than both [Notes II: more than the difference]; (B) the flat cases included: at 4 or 16 the sides lie flat [Notes II: never equal]; (D) no bottom end — a third side of 4 or less cannot reach [Notes II: more than the difference] |
| 6 | Do sides of 8 inches, 15 inches and 17 inches make a right triangle? | ***Yes — $8^2 + 15^2 = 289 = 17^2$.*** | No — compared 8 + 15 = 23 with 17 [Notes III: squares, not sides] |

| # | What the board asks | Answer | If the board is wrong |
|---|---|---|---|
| 7 | A triangle has sides of 25 cm, 7 cm and 24 cm. Is it a right triangle? Which answer has the right reason? | ***A — yes, because $7^2 + 24^2 = 25^2$*** | (B) the right verdict for the wrong reason — the Triangle Inequality shows only a triangle [MA.8.GR.1.3: two tests]; (C) 25 taken as a leg [Florida B1G-M, GR.1.1: misidentifying the hypotenuse]; (D) added the sides, not their squares [Notes III] |
| 8 | A triangular sail has sides of 20 ft, 21 ft and 29 ft, and it weighs 12 pounds. Does the sail have a right angle? | ***Yes — $20^2 + 21^2 = 841 = 29^2$.*** | 'Cannot tell without the angles' [Notes III: the sides are enough] |
| 9 | Ana says sides of 6, 7 and 9 make a right triangle because 6 + 7 > 9. Is she right? Explain. | ***No — 6 + 7 > 9 shows only a triangle; 85 is not 81, so it is not right.*** | 'Yes, 13 > 9' — the inequality taken as the converse [MA.8.GR.1.3: two tests] |

**MTR.4.1 — the one peer move.** When a board splits — no option holding about two-thirds of the room — sixty seconds of “convince the person next to you”, then re-vote, then reveal. The minute comes out of the whiteboard remainder.

## 9. Differentiation

**ESE / 504.** Two lines stay on the board all period: "the two SHORTER sides, added, beat the longest" and "the two shorter sides' squares, added, EQUAL the longest side's square". Every board starts by circling the longest side. Popsicle-stick sets for Notes I if a picture is not enough.

**ELL.** **Inequality** means "not equal" — here, MORE than. **Hypotenuse**: the long side across from the square corner. Say "longest side" with every board, and point to it.

**Enrichment.** Board 4 finishes early for some: how many whole-number lengths can the third side have when the other two are 7 and 12? (Thirteen: 6 through 18.) Then: which of them make a right triangle? (None is a whole number: √95 and √193.)

## 10. Closure, Vocabulary, Materials, Practice

**Closure.** Board 9 is written work and it is the exit evidence: Ana's test (6 + 7 > 9) proves only that the three sides make a triangle; the converse is a test of squares, and 85 is not 81. A board that says 'she is wrong' without the squares has the verdict and not the reason.

**Vocabulary.** **triangle inequality**; **legs**; **hypotenuse**; **converse**

**Materials.** Whiteboards and markers. Optional: strips cut to 3, 4, 5, 6 and 9 units for Notes I. Calculators allowed, printed nowhere.

**Practice.** The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due by the Unit 6 test: Triangle inequality — 2N5; Converse of the Pythagorean theorem: is it a right triangle? — EQZ.

# Pythagorean Theorem

*Teacher's Edition*

## The Period

**MA.8.GR.1.1** Apply the Pythagorean Theorem to solve mathematical and real-world problems involving unknown side lengths in right triangles.

**Must** Find a missing leg or the hypotenuse of a right triangle from $a^2 + b^2 = c^2$, the hypotenuse found first; an irrational answer exact and, when asked, to the nearest tenth.

**Must not** Stories (6.04); the coordinate plane (6.05); radicands over 225 or with a square factor; 3-D.

**Target** I can find a missing side of a right triangle — the hypotenuse, or a leg.

**MTR** **MTR.6.1** — Notes III and boards 3–4: every leg is checked against c · **MTR.4.1** — boards 4 and 6 split; the sixty-second re-vote · **MTR.5.1** — Notes I and board 5: one theorem however the triangle is turned

| Segment | Slides | Time |
|---|---|---|
| Settle in; post the learning target | 1 | 1 min |
| Warm-Up | 2–4 | 6 min |
| Notes | 5–7 | 11 min |
| Example 1 | 8–9 | 4 min |
| Your Turn | 10–11 | 3 min |
| Example 2 | 12–13 | 4 min |
| Your Turn | 14–15 | 3 min |
| Whiteboards | 16–33 | 10 min |
| Independent Set | 34 | 6 min |
| IXL | 35 | 5 min |
| **Total** | | **53 min** |

### Say these three things out loud today

1. Find the square corner first: the side across from it is c, the longest side.
2. The hypotenuse: square, add, then the root.
3. A leg: subtract the squares — and check it is shorter than the hypotenuse.

## Slide by Slide

1. **Pythagorean Theorem.** Post the learning target.
2. **Warm-Up — On your own. Four minutes. No notes.** Q4 is Notes III's move before it has a name: $10^2 - 6^2 = 64 = 8^2$, the missing leg of the 6-8-10 triangle Notes II builds. *1 retrieves 6.01–02 — the Triangle Inequality, the last class (yesterday). 2 retrieves 5.10–11 — a unit rate used once (last week). 3 retrieves 4.05 — a perfect square's root (Unit 4) (two units back). 4 retrieves MA.7.NSO.1.1 — whole-number powers; a missing leg subtracts two squares like this (prior grade — the prerequisite this lesson needs).*
3. **Warm-Up — Answers.** Reveal. 1. No — 4 + 7 = 11 < 12 2. 81 pages 3. 14 4. 64
4. **Vocabulary — The new words, before they are used.** Read each word and its meaning once; the notes use them next.
5. **Notes — Copy the theorem and the picture.** Find the square corner. The side across from it is c.
6. **Notes.** Square, add, then the root. The root is the last step.
7. **Notes.** A leg is shorter than the hypotenuse. Always check.
8. **Example 1 — Find the hypotenuse.** Ask where the square corner is and which side is c before anyone squares.
9. **Example 1 — Worked.** 49 + 81 = 130, not a perfect square: c = √130, between 11 and 12 (121 < 130 < 144), and 11.40 to two places, so 11.4.
10. **Your Turn — Same steps, your numbers. Boards up when done.** 25 + 121 = 146 and √146 ≈ 12.08, so 12.1 — a board with 16 added the legs; a board with 146 stopped before the root.
11. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
12. **Example 2 — Find a leg.** Ask which number is c before anyone squares: 15, the longest — so this time a square is subtracted.
13. **Example 2 — Worked.** 15 is c, so $15^2$ stands alone: $b^2 + 64 = 225$, $b^2 = 161$, b = √161 ≈ 12.69, so 12.7 — and 12.7 is shorter than 15, as a leg must be.
14. **Your Turn — Same steps, your numbers. Boards up when done.** 121 − 36 = 85 and √85 ≈ 9.22, so 9.2 — a board with 12.5 added the squares (√157), longer than the hypotenuse.
15. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).

**16–17. Board 1.** A right triangle has legs of 9 cm and 12 cm. How long is the hypotenuse? → ***15 cm*** *21 — added the legs, not their squares [MA.8.GR.1.1: the theorem is about squares]*

**18–19. Board 2.** A right triangle has legs of 4 inches and 9 inches. How long is the hypotenuse, to the nearest tenth? → ***√97 in ≈ 9.8 in*** *13 — added the legs [MA.8.GR.1.1]; 9.9 — rounded twice [Notes II: round once, at the end]*

**20–21. Board 3.** A right triangle has a hypotenuse of 25 ft and one leg of 20 ft. How long is the other leg? → ***15 ft*** *32.0 — added the squares, longer than the hypotenuse [Notes III: a leg is shorter than c]*

**22–23. Board 4.** A right triangle has a hypotenuse of 41 m and one leg of 40 m. How long is the other leg? → ***A — 9 m*** *(B) stopped at $b^2 = 81$ and forgot the square root [Math Nation names skipping the final root]; (C) subtracted the sides, 41 − 40 [MA.8.GR.1.1: the theorem is about squares]; (D) added the squares instead of subtracting: about 57.3, longer than the hypotenuse [Notes III: for a leg, subtract] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

**24–25. Board 5.** The right triangle is drawn turned. Its legs are 5 cm and 7 cm. How long is the hypotenuse, to the nearest tenth? → ***√74 cm ≈ 8.6 cm*** *4.9 — subtracted, taking the 7 as the hypotenuse [Florida B1G-M, GR.1.1: misidentifying the hypotenuse]*

**26–27. Board 6.** A right triangle's hypotenuse is 20 inches; its legs are 16 inches and x. Which equation finds x? → ***D — and x = 12 in*** *(A) the sides added, not their squares [MA.8.GR.1.1]; (B) the hypotenuse put with the legs [Florida B1G-M, GR.1.1: misidentifying the hypotenuse]; (C) doubled the 20 instead of squaring it — $20^2$ means 20 × 20, not 2 × 20 [Notes I: the theorem is about squares] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

**28–29. Board 7.** A right triangle has a hypotenuse of 14 m and one leg of 9 m. How long is the other leg, to the nearest tenth? → ***√115 m ≈ 10.7 m*** *16.6 — added the squares, longer than the hypotenuse [Notes III]*

**30–31. Board 8.** A pennant is cut as a right triangle with legs of 8 inches and 11 inches. The fabric costs 6 dollars. How long is its longest side, to the nearest tenth? → ***√185 in ≈ 13.6 in*** *The 6 dollars is not needed. 19 — added the legs [MA.8.GR.1.1]; 'cannot tell' — the longest side is the hypotenuse [Notes I]*

**32–33. Board 9.** Jo finds a leg of a right triangle. The hypotenuse is 13 cm and the other leg is 5 cm. She writes $\sqrt{(5^2 + 13^2)} \approx 13.9$. What went wrong? How long is the leg? → ***She added the squares; a leg must be shorter than 13 cm. Subtract: the leg is 12 cm.*** *'13.9 is fine' — no check against the hypotenuse [Notes III]*

**34. Independent Set — Six questions. On your own, in writing.** Hand out the Independent Set; the same six questions are on the board.

**35. IXL — Last five minutes.** IXL: every listed skill is required, SmartScore 67.

**Variation.** Boards 1–2: the hypotenuse, whole then to the tenth; 3–4 a leg, whole then the vote. The pattern breaks at 5 — the triangle turned — and 6 asks only for the equation; 7 is a leg to the tenth; 8 has the extra number; 9 is the check in writing. The independent set: a whole hypotenuse, one to the tenth, a whole leg from a big hypotenuse, a leg to the tenth, a turned triangle, the equation and its value.

## Misconceptions to Watch

**Board 1.** 21 — added the legs, not their squares [MA.8.GR.1.1: the theorem is about squares]

**Board 2.** 13 — added the legs [MA.8.GR.1.1]

**Board 3.** 32.0 — added the squares, longer than the hypotenuse [Notes III: a leg is shorter than c]

**Board 4.** stopped at $b^2 = 81$ and forgot the square root [Math Nation names skipping the final root]

**Board 5.** 4.9 — subtracted, taking the 7 as the hypotenuse [Florida B1G-M, GR.1.1: misidentifying the hypotenuse]

**Board 6.** the sides added, not their squares [MA.8.GR.1.1]

**Board 7.** 16.6 — added the squares, longer than the hypotenuse [Notes III]

**Board 8.** 19 — added the legs [MA.8.GR.1.1]

**Board 9.** '13.9 is fine' — no check against the hypotenuse [Notes III]

**Materials** Whiteboards and markers. $a^2 + b^2 = c^2$ on the board for the period (memorized by the test). Calculators allowed, printed nowhere.

# Pythagorean Theorem

*Lesson Plan*

## 1. Lesson Information

| | | | |
|---|---|---|---|
| Course | Grade 7 Accelerated (1205050) | Unit | 6 — Relationships in Triangles |
| Lesson | Lesson 3 — Pythagorean Theorem | Date | _____ |
| Length | 53 minutes (one period) | Benchmark(s) | MA.8.GR.1.1 |

## 2. Standards

**MA.8.GR.1.1** Apply the Pythagorean Theorem to solve mathematical and real-world problems involving unknown side lengths in right triangles.

**Must.** Find a missing leg or the hypotenuse of a right triangle from $a^2 + b^2 = c^2$, the hypotenuse found first; an irrational answer exact and, when asked, to the nearest tenth.

**Must not.** Stories (6.04); the coordinate plane (6.05); radicands over 225 or with a square factor; 3-D.

## 3. Mathematical Thinking and Reasoning Standards

*the evidence for each, from this period*

| MTR | What it asks | The evidence in this lesson |
|---|---|---|
| MA.K12.MTR.6.1 | Assess the reasonableness of solutions. | Notes III and boards 3–4: every leg is checked against c — a leg is always shorter than the hypotenuse |
| MA.K12.MTR.4.1 | Engage in discussions that reflect on the mathematical thinking of self and others. | boards 4 and 6 split; the sixty-second re-vote |
| MA.K12.MTR.5.1 | Use patterns and structure to help understand and connect mathematical concepts. | Notes I and board 5: one theorem however the triangle is turned — the square corner first |

## 4. Learning Target and Essential Question

**Learning target.** I can find a missing side of a right triangle — the hypotenuse, or a leg.

**Essential question.** If you know two sides of a right triangle, how do you find the third — and how do you know which side is c?

**Building on.** 6.01–02 — the converse and the 3-4-5 squares; 4.05 — square roots of perfect squares; Unit 2 (MA.8.NSO.1.2) — a square root between two whole numbers.

**Working toward.** 6.04 — the theorem inside real situations; 6.05–06 — distance on the coordinate plane; Geometry — right-triangle trigonometry.

## 5. Sequence

*read from the deck; never typed*

| Segment | Min | What the teacher does | What students do |
|---|---|---|---|
| Settle in; post the learning target | 1 | Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet. | Copy the target. |
| Warm-Up | 6 | Two minutes silent, then reveal. One sentence of reteach per question at most. Then the new words, read once. | Four retrieval questions, alone, no notes. Then copy the new words. |
| Notes | 11 | Teach from the front; say the OFF line out loud. | Copy the notes; answer the checks aloud. |
| Example 1 | 4 | Work the example; name the wrong boards before they appear. | Watch, then work the Your Turn alone. |
| Your Turn | 3 | Circulate; reveal and name what a wrong board most likely did. | Work it alone; boards up on the cue. |
| Example 2 | 4 | Work the example; name the wrong boards before they appear. | Watch, then work the Your Turn alone. |
| Your Turn | 3 | Circulate; reveal and name what a wrong board most likely did. | Work it alone; boards up on the cue. |
| Whiteboards | 10 | One question at a time. Boards down until the cue, all up together; scan the back row first. | Answer on a board; hold it up on three. |
| Independent Set | 6 | Silent. Circulate and mark what you see; do not teach. | Six questions, written, alone. |
| IXL | 5 | Assign the day's IXL skills. | Start the skills; finish tonight. |
| **Total** | **53** | | |

## 6. Gradual Release

**I do.** The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the OFF line is said out loud rather than printed.

**We do.** Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

**You do.** The nine-question whiteboard round (10 minutes) and then the six-question independent set, silent and written.

## 7. Higher-Order Questions

| Question | DOK |
|---|---|
| A right triangle has legs 5 and 12. Why is the hypotenuse 13 and not 17? | DOK 2 |
| Without computing, which is longer: the hypotenuse when the legs are 6 and 8, or a leg when the hypotenuse is 10 and the other leg is 6? How do you know? | DOK 3 |
| Why can a missing leg never be found by adding two squares? | DOK 3 |

## 8. Checks for Understanding

*each board, and what to do about it*

| # | What the board asks | Answer | If the board is wrong |
|---|---|---|---|
| 1 | A right triangle has legs of 9 cm and 12 cm. How long is the hypotenuse? | ***15 cm*** | 21 — added the legs, not their squares [MA.8.GR.1.1: the theorem is about squares] |
| 2 | A right triangle has legs of 4 inches and 9 inches. How long is the hypotenuse, to the nearest tenth? | ***$\sqrt{97}$ in ≈ 9.8 in*** | 13 — added the legs [MA.8.GR.1.1]; 9.9 — rounded twice [Notes II: round once, at the end] |
| 3 | A right triangle has a hypotenuse of 25 ft and one leg of 20 ft. How long is the other leg? | ***15 ft*** | 32.0 — added the squares, longer than the hypotenuse [Notes III: a leg is shorter than c] |
| 4 | A right triangle has a hypotenuse of 41 m and one leg of 40 m. How long is the other leg? | ***A — 9 m*** | (B) stopped at $b^2 = 81$ and forgot the square root [Math Nation names skipping the final root]; (C) subtracted the sides, 41 − 40 [MA.8.GR.1.1: the theorem is about squares]; (D) added the squares instead of subtracting: about 57.3, longer than the hypotenuse [Notes III: for a leg, subtract] |
| 5 | The right triangle is drawn turned. Its legs are 5 cm and 7 cm. How long is the hypotenuse, to the nearest tenth? | ***$\sqrt{74}$ cm ≈ 8.6 cm*** | 4.9 — subtracted, taking the 7 as the hypotenuse [Florida B1G-M, GR.1.1: misidentifying the hypotenuse] |
| 6 | A right triangle's hypotenuse is 20 inches; its legs are 16 inches and x. Which equation finds x? | ***D — and x = 12 in*** | (A) the sides added, not their squares [MA.8.GR.1.1]; (B) the hypotenuse put with the legs [Florida B1G-M, GR.1.1: misidentifying the hypotenuse]; (C) doubled the 20 instead of squaring it — $20^2$ means 20 × 20, not 2 × 20 [Notes I: the theorem is about squares] |
| 7 | A right triangle has a hypotenuse of 14 m and one leg of 9 m. How long is the other leg, to the nearest tenth? | ***$\sqrt{115}$ m ≈ 10.7 m*** | 16.6 — added the squares, longer than the hypotenuse [Notes III] |
| 8 | A pennant is cut as a right triangle with legs of 8 inches and 11 inches. The fabric costs 6 dollars. How long is its longest side, to the nearest tenth? | ***$\sqrt{185}$ in ≈ 13.6 in*** | 19 — added the legs [MA.8.GR.1.1]; 'cannot tell' — the longest side is the hypotenuse [Notes I] |
| 9 | Jo finds a leg of a right triangle. The hypotenuse is 13 cm and the other leg is 5 cm. She writes $\sqrt{(5^2 + 13^2)} \approx 13.9$. What went wrong? How long is the leg? | ***She added the squares; a leg must be shorter than 13 cm. Subtract: the leg is 12 cm.*** | '13.9 is fine' — no check against the hypotenuse [Notes III] |

**MTR.4.1 — the one peer move.** When a board splits — no option holding about two-thirds of the room — sixty seconds of “convince the person next to you”, then re-vote, then reveal. The minute comes out of the whiteboard remainder.

## 9. Differentiation

**ESE / 504.** $\text{Leg}^2 + \text{leg}^2 = \text{hypotenuse}^2$ stays on the board all period. Every board starts by marking the square corner and the side across from it. A set-up with the numbers already squared for boards 1–2 if the squaring is the obstacle.

**ELL.** **Hypotenuse**: the long side across from the square corner. **Leg**: one of the two sides that make the square corner. Say "square it" and "square root" every time, with the gesture of a square.

**Enrichment.** Legs 1 and 1 give $\sqrt{2}$; legs $\sqrt{2}$ and 1 give $\sqrt{3}$ — keep going (the Wheel of Theodorus): which leg lengths give $\sqrt{7}$? And why is every hypotenuse in the wheel $\sqrt{n}$?

## 10. Closure, Vocabulary, Materials, Practice

**Closure.** Board 9 is written work and it is the exit evidence: Jo added the squares to find a leg, and her answer is longer than the 13-cm hypotenuse — impossible. A board that finds 12 without saying why 13.9 cannot be a leg has the number and not the check.

**Vocabulary.** **Pythagorean Theorem**; **hypotenuse**; **square root**

**Materials.** Whiteboards and markers. $a^2 + b^2 = c^2$ on the board for the period (memorized by the test). Calculators allowed, printed nowhere.

**Practice.** The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due by the Unit 6 test: Pythagorean theorem: find the missing leg or hypotenuse length — MTM.

# Pythagorean Theorem in Real-World Problems

*Teacher's Edition*

## The Period

**MA.8.GR.1.1** Apply the Pythagorean Theorem to solve mathematical and real-world problems involving unknown side lengths in right triangles.

**Must** Find a missing side in a real situation: the right angle located, the hypotenuse named, then the theorem; the answer in the story's unit, rounded once where it measures.

**Must not** 3-D (a box's space diagonal — the book's 6.4.5); radicands over 225; coordinates (6.05–06).

**Target** I can find a length in a real situation by finding the right triangle inside it.

**MTR** **MTR.7.1** — every board is a real object: a ladder, a ramp, a screen, a lawn, a kite · **MTR.4.1** — boards 4 and 6 split; the sixty-second re-vote · **MTR.6.1** — boards 7 and 9: the answer is checked against the story

| Segment | Slides | Time |
|---|---|---|
| Settle in; post the learning target | 1 | 1 min |
| Warm-Up | 2–4 | 6 min |
| Notes | 5–7 | 11 min |
| Example 1 | 8–9 | 4 min |
| Your Turn | 10–11 | 3 min |
| Example 2 | 12–13 | 4 min |
| Your Turn | 14–15 | 3 min |
| Whiteboards | 16–33 | 10 min |
| Independent Set | 34 | 6 min |
| IXL | 35 | 5 min |
| **Total** | | **53 min** |

### Say these three things out loud today

1. Sketch the story and mark the square corner before any number.
2. A ladder, a diagonal, a ramp, a kite string: each is the hypotenuse.
3. How high, or how far along the ground, is a leg: subtract.

## Slide by Slide

1. **Pythagorean Theorem in Real-World Problems.** Post the learning target.
2. **Warm-Up — On your own. Four minutes. No notes.** Q4 is today's first move: every rectangle — a screen, a lawn, a box — has four right angles, so its diagonal is a hypotenuse. *1 retrieves 6.03 — a missing leg: subtract the squares (yesterday). 2 retrieves 6.01–02 — the converse, on the longest side (last week). 3 retrieves 5.04 — a real length from a plan's scale (last unit). 4 retrieves Grade 3–4 (MA.4.GR.1) — a rectangle's corners are right angles; today every diagonal cuts off a right triangle (prior grade — the prerequisite this lesson needs).*
3. **Warm-Up — Answers.** Reveal. 1. 7 ft 2. Yes — 81 + 1,600 = 1,681 = $41^2$ 3. 14 ft 4. 4
4. **Vocabulary — The new words, before they are used.** Read each word and its meaning once; the notes use them next.
5. **Notes — Copy the four triangles' names.** Sketch it. Mark the square corner. Name c.
6. **Notes — A garden 5 m by 12 m, and a path from corner to corner.** A rectangle's diagonal is always a hypotenuse.
7. **Notes.** How high is a leg. Subtract.
8. **Example 1 — A ramp's length.** Ask where the square corner is before any number: under the high end, where the rise meets the ground.
9. **Example 1 — Worked.** The rise and the run are the legs; the ramp is c.
10. **Your Turn — Same steps, your numbers. Boards up when done.** A screen's corner is square, so the diagonal is c: 49 + 100 = 149 and √149 ≈ 12.21, so 12.2 in — a board with 17 added the sides.
11. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
12. **Example 2 — How high a ladder reaches.** Ask which length is c before anyone squares: the ladder, 12 ft.
13. **Example 2 — Worked.** The ladder is c: $h^2$ = 144 − 25 = 119, √119 ≈ 10.91, so 10.9 ft — less than 12, as a leg must be.
14. **Your Turn — Same steps, your numbers. Boards up when done.** 169 − 36 = 133 and √133 ≈ 11.53, so 11.5 ft — a board with 14.3 added the squares (√205), higher than the ladder is long.
15. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).

**16–17. Board 1.** A tablet screen is 6 inches wide and 8 inches tall. How long is its diagonal? → ***10 in*** *14 — added the sides [MA.8.GR.1.1: the theorem is about squares]*

**18–19. Board 2.** A 15-ft ladder leans on a wall. Its foot is 9 ft from the wall. How high up the wall does it reach? → ***12 ft*** *17.5 — added the squares: a 15-ft ladder cannot reach 17.5 ft [Notes III]*

**20–21. Board 3.** A ramp rises 4 ft over a run of 11 ft. How long is the ramp, to the nearest tenth of a foot? → ***About 11.7 ft*** *15 — added the rise and the run [MA.8.GR.1.1]; '11, the same as the run' — the ramp is the hypotenuse, longer [Notes I]*

**22–23. Board 4.** A 20-ft ladder reaches 16 ft up a wall. How far from the wall is its foot? → ***A — 12 ft*** *(B) added the squares: about 25.6 ft, farther than the ladder is long*

*[Notes III: for a leg, subtract]; (C) subtracted the lengths, 20 − 16 [MA.8.GR.1.1: the theorem is about squares]; (D) stopped at $a^2 = 144$ and forgot the square root [Math Nation names skipping the final root]  Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

**24–25. Board 5.** A lawn is 9 m by 12 m. Ben walks along two sides; Ava walks straight across, corner to corner. How much shorter is Ava's walk? → ***6 m shorter*** *15 — Ava's walk, not the difference [the question asks how much shorter]; 21 — Ben's walk*

**26–27. Board 6.** A kite flies on 30 m of string, straight above a spot 18 m from where the string is held. Which length is the hypotenuse? → ***C — the 30-m string*** *(A) took the ground distance as c — it is a leg [Florida B1G-M, GR.1.1: misidentifying the hypotenuse]; (B) took the height as c — the height is the unknown leg [Florida B1G-M, GR.1.1: misidentifying the hypotenuse]; (D) did not see the right angle under the kite [Florida B1G-M, GR.1.1: failing to visualize the triangle in a real-world problem]  Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

**28–29. Board 7.** A box is 7 inches wide and 12 inches long. Will a 14-in umbrella lie flat in it, corner to corner? → ***No — the diagonal is about 13.9 in.*** *'Yes, 19 > 14' — added the sides [MA.8.GR.1.1]; 'Yes, 12 < 14' — compared with one side [Notes II: the diagonal is longest]*

**30–31. Board 8.** A 10-ft ladder that weighs 25 pounds has its foot 7 ft from a wall. How high does it reach, to the nearest tenth of a foot? → ***About 7.1 ft*** *The 25 pounds is not needed.  12.2 — added the squares (√149): higher than the ladder is long [Notes III]*

**32–33. Board 9.** A window is 12 ft above the ground. Why can an 11-ft ladder never reach it, wherever its foot stands? → ***The height is a leg — always shorter than the 11-ft ladder.*** *'Move the foot closer' — closer, the height grows toward 11 ft but never reaches it, and the window is at 12 [Notes I: the ladder is c]*

**34. Independent Set — Six questions. On your own, in writing.** Hand out the Independent Set; the same six questions are on the board.

**35. IXL — Last five minutes.** IXL: every listed skill is required, SmartScore 67.

**Variation.** Boards 1–2: a diagonal and a ladder's height, both whole; 3 a ramp to the tenth; 4 the ladder's foot, voted. The pattern breaks at 5 — two walks and their difference — and 6, which asks only which length is c; 7 is will-it-fit by a tenth; 8 has the extra number; 9 is the reason in writing. The independent set: a diagonal, a ladder's foot, a ramp, along two sides against across, will it fit, a kite's height.

## Misconceptions to Watch

**Board 1.** 14 — added the sides [MA.8.GR.1.1: the theorem is about squares]

**Board 2.** 17.5 — added the squares: a 15-ft ladder cannot reach 17.5 ft [Notes III]

**Board 3.** 15 — added the rise and the run [MA.8.GR.1.1]

**Board 4.** added the squares: about 25.6 ft, farther than the ladder is long [Notes III: for a leg, subtract]

**Board 5.** 15 — Ava's walk, not the difference [the question asks how much shorter]

**Board 6.** took the ground distance as c — it is a leg [Florida B1G-M, GR.1.1: misidentifying the hypotenuse]

**Board 7.** 'Yes, 19 > 14' — added the sides [MA.8.GR.1.1]

**Board 8.** 12.2 — added the squares (√149): higher than the ladder is long [Notes III]

**Board 9.** 'Move the foot closer' — closer, the height grows toward 11 ft but never reaches it, and the window is at 12 [Notes I: the ladder is c]

**Materials** Whiteboards and markers. A meter stick or a ladder picture for Notes I if the room needs it. Calculators allowed, printed nowhere.

# Pythagorean Theorem in Real-World Problems

*Lesson Plan*

## 1. Lesson Information

| | | | |
|---|---|---|---|
| Course | Grade 7 Accelerated (1205050) | Unit | 6 — Relationships in Triangles |
| Lesson | Lesson 4 — Pythagorean Theorem in Real-World Problems | Date | _____ |
| Length | 53 minutes (one period) | Benchmark(s) | MA.8.GR.1.1 |

## 2. Standards

**MA.8.GR.1.1** Apply the Pythagorean Theorem to solve mathematical and real-world problems involving unknown side lengths in right triangles.

**Must.** Find a missing side in a real situation: the right angle located, the hypotenuse named, then the theorem; the answer in the story's unit, rounded once where it measures.

**Must not.** 3-D (a box's space diagonal — the book's 6.4.5); radicands over 225; coordinates (6.05–06).

## 3. Mathematical Thinking and Reasoning Standards

*the evidence for each, from this period*

| MTR | What it asks | The evidence in this lesson |
|---|---|---|
| MA.K12.MTR.7.1 | Apply mathematics to real-world contexts. | every board is a real object: a ladder, a ramp, a screen, a lawn, a kite — the triangle is found in it before any number |
| MA.K12.MTR.4.1 | Engage in discussions that reflect on the mathematical thinking of self and others. | boards 4 and 6 split; the sixty-second re-vote |
| MA.K12.MTR.6.1 | Assess the reasonableness of solutions. | boards 7 and 9: the answer is checked against the story — does it fit, can it reach |

## 4. Learning Target and Essential Question

**Learning target.** I can find a length in a real situation by finding the right triangle inside it.

**Essential question.** Where is the right angle in a ladder leaning on a wall — and why does that decide which length is c?

**Building on.** 6.03 — a missing hypotenuse or leg; 5.04 — real lengths from a plan; Grade 3–4 — a rectangle's corners are right angles.

**Working toward.** 6.05–06 — the same right triangle drawn on a grid between two points; Geometry — right-triangle trigonometry.

## 5. Sequence

*read from the deck; never typed*

| Segment | Min | What the teacher does | What students do |
|---|---|---|---|
| Settle in; post the learning target | 1 | Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet. | Copy the target. |
| Warm-Up | 6 | Two minutes silent, then reveal. One sentence of reteach per question at most. Then the new words, read once. | Four retrieval questions, alone, no notes. Then copy the new words. |
| Notes | 11 | Teach from the front; say the OFF line out loud. | Copy the notes; answer the checks aloud. |
| Example 1 | 4 | Work the example; name the wrong boards before they appear. | Watch, then work the Your Turn alone. |
| Your Turn | 3 | Circulate; reveal and name what a wrong board most likely did. | Work it alone; boards up on the cue. |
| Example 2 | 4 | Work the example; name the wrong boards before they appear. | Watch, then work the Your Turn alone. |
| Your Turn | 3 | Circulate; reveal and name what a wrong board most likely did. | Work it alone; boards up on the cue. |
| Whiteboards | 10 | One question at a time. Boards down until the cue, all up together; scan the back row first. | Answer on a board; hold it up on three. |
| Independent Set | 6 | Silent. Circulate and mark what you see; do not teach. | Six questions, written, alone. |
| IXL | 5 | Assign the day's IXL skills. | Start the skills; finish tonight. |
| **Total** | **53** | | |

## 6. Gradual Release

**I do.** The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the OFF line is said out loud rather than printed.

**We do.** Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

**You do.** The nine-question whiteboard round (10 minutes) and then the six-question independent set, silent and written.

## 7. Higher-Order Questions

| Question | DOK |
|---|---|
| A ladder's foot slides farther from the wall. Does the top go up or down? Why? | DOK 3 |
| Why is the diagonal of a rectangle always longer than either of its sides? | DOK 2 |
| A 10-ft ladder reaches 9.5 ft up a wall. Is its foot more or less than 3 ft from the wall? How can you tell without the theorem? | DOK 3 |

## 8. Checks for Understanding

*each board, and what to do about it*

| # | What the board asks | Answer | If the board is wrong |
|---|---|---|---|
| 1 | A tablet screen is 6 inches wide and 8 inches tall. How long is its diagonal? | ***10 in*** | 14 — added the sides [MA.8.GR.1.1: the theorem is about squares] |
| 2 | A 15-ft ladder leans on a wall. Its foot is 9 ft from the wall. How high up the wall does it reach? | ***12 ft*** | 17.5 — added the squares: a 15-ft ladder cannot reach 17.5 ft [Notes III] |
| 3 | A ramp rises 4 ft over a run of 11 ft. How long is the ramp, to the nearest tenth of a foot? | ***About 11.7 ft*** | 15 — added the rise and the run [MA.8.GR.1.1]; '11, the same as the run' — the ramp is the hypotenuse, longer [Notes I] |
| 4 | A 20-ft ladder reaches 16 ft up a wall. How far from the wall is its foot? | ***A — 12 ft*** | (B) added the squares: about 25.6 ft, farther than the ladder is long [Notes III: for a leg, subtract]; (C) subtracted the lengths, 20 − 16 [MA.8.GR.1.1: the theorem is about squares]; (D) stopped at $a^2 = 144$ and forgot the square root [Math Nation names skipping the final root] |
| 5 | A lawn is 9 m by 12 m. Ben walks along two sides; Ava walks straight across, corner to corner. How much shorter is Ava's walk? | ***6 m shorter*** | 15 — Ava's walk, not the difference [the question asks how much shorter]; 21 — Ben's walk |
| 6 | A kite flies on 30 m of string, straight above a spot 18 m from where the string is held. Which length is the hypotenuse? | ***C — the 30-m string*** | (A) took the ground distance as c — it is a leg [Florida B1G-M, GR.1.1: misidentifying the hypotenuse]; (B) took the height as c — the height is the unknown leg [Florida B1G-M, GR.1.1: misidentifying the hypotenuse]; (D) did not see the right angle under the kite [Florida B1G-M, GR.1.1: failing to visualize the triangle in a real-world problem] |
| 7 | A box is 7 inches wide and 12 inches long. Will a 14-in umbrella lie flat in it, corner to corner? | ***No — the diagonal is about 13.9 in.*** | 'Yes, 19 > 14' — added the sides [MA.8.GR.1.1]; 'Yes, 12 < 14' — compared with one side [Notes II: the diagonal is longest] |
| 8 | A 10-ft ladder that weighs 25 pounds has its foot 7 ft from a wall. How high does it reach, to the nearest tenth of a foot? | ***About 7.1 ft*** | 12.2 — added the squares (√149): higher than the ladder is long [Notes III] |
| 9 | A window is 12 ft above the ground. Why can an 11-ft ladder never reach it, wherever its foot stands? | ***The height is a leg — always shorter than the 11-ft ladder.*** | 'Move the foot closer' — closer, the height grows toward 11 ft but never reaches it, and the window is at 12 [Notes I: the ladder is c] |

**MTR.4.1 — the one peer move.** When a board splits — no option holding about two-thirds of the room — sixty seconds of “convince the person next to you”, then re-vote, then reveal. The minute comes out of the whiteboard remainder.

## 9. Differentiation

**ESE / 504.** Every story starts with a sketch: the right angle marked first, then c. Boards 1–3 come with the sketch printed; the student labels it. $Leg^2 + leg^2 = hypotenuse^2$ stays on the board.

**ELL.** **Ladder**, **wall**, **ramp**, **diagonal**, **kite string**: point at the picture for each before the board. "How high" is a leg; "how long is the ladder" is the hypotenuse.

**Enrichment.** A 25-ft ladder stands 7 ft from a wall. Its foot slides 8 ft farther out. How far does the top slide down? (24 − 20 = 4 ft — both heights whole.)

## 10. Closure, Vocabulary, Materials, Practice

**Closure.** Board 9 is written work and it is the exit evidence: the ladder is the hypotenuse, the longest side, and the height it reaches is a leg — so an 11-ft ladder can never touch a 12-ft window, wherever its foot stands. A board that tries numbers has not named the hypotenuse.

**Vocabulary.** **diagonal**; **rise and run**

**Materials.** Whiteboards and markers. A meter stick or a ladder picture for Notes I if the room needs it. Calculators allowed, printed nowhere.

**Practice.** The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due by the Unit 6 test: Pythagorean theorem: word problems — 87U.

# Distance on the Coordinate Plane

*Teacher's Edition*

## The Period

**MA.8.GR.1.2** Apply the Pythagorean Theorem to solve mathematical and real-world problems involving the distance between two points in a coordinate plane.

**Must** Find the distance between two points: on one line by subtracting (across an axis too), slanted by the right triangle whose legs are read from the coordinates; an irrational answer exact and, when asked, to the tenth.

**Must not** The distance formula (Clarification 2); radicands over 225; perimeters and maps with several legs (6.06).

**Target** I can find the distance between two points: subtract on a line, and draw the right triangle when the segment is slanted.

**MTR** **MTR.2.1** — Notes II–III: one distance as a slanted segment, as a right triangle's legs and as $a^2 + b^2 = c^2$ · **MTR.4.1** — boards 3 and 6 split; the sixty-second re-vote · **MTR.6.1** — Notes I and board 9: a distance across zero checked by counting the squares

| Segment | Slides | Time |
|---|---|---|
| Settle in; post the learning target | 1 | 1 min |
| Warm-Up | 2–4 | 6 min |
| Notes | 5–7 | 11 min |
| Example 1 | 8–9 | 3 min |
| Your Turn | 10–11 | 3 min |
| Example 2 | 12–13 | 4 min |
| Your Turn | 14–15 | 3 min |
| Whiteboards | 16–33 | 11 min |
| Independent Set | 34 | 6 min |
| IXL | 35 | 5 min |
| **Total** | | **53 min** |

### Say these three things out loud today

1. Same y: subtract the x's; same x: subtract the y's — across zero the two pieces add.
2. Slanted: draw the right triangle; its legs come from the coordinates.
3. A distance is never negative, and there is no new formula: the triangle is the method.

## Slide by Slide

1. **Distance on the Coordinate Plane.** Post the learning target.
2. **Warm-Up — On your own. Four minutes. No notes.** Q4 is Notes I's rule before it is written down: same x, so subtract the y's — and 5 − (−3) is 8, across zero. *1 retrieves 6.04 — a leg in a story (yesterday). 2 retrieves 6.01–02 — the converse (last week). 3 retrieves 5.01 and 5.04 — a real distance from a map's scale (last unit). 4 retrieves MA.6.GR.1.2 — the distance between points with the same x-coordinate (prior grade — the prerequisite this lesson needs).*
3. **Warm-Up — Answers.** Reveal. 1. 15 ft 2. Yes — 144 + 1,225 = 1,369 = $37^2$ 3. 17 km 4. 8
4. **Vocabulary — The new words, before they are used.** Read each word and its meaning once; the notes use them next.
5. **Notes — Copy the rule and the example.** Across zero, the two pieces add.
6. **Notes.** Slanted? Draw the triangle. Across, then up.
7. **Notes.** Across: the x's. Up or down: the y's. Then the theorem.
8. **Example 1 — On one line.** Ask what the two points share before anyone subtracts: the x, so the distance is in the y's.
9. **Example 1 — Worked.** Same x, so the y's: 7 − (−2) = 9 — 2 below the axis and 7 above it.
10. **Your Turn — Same steps, your numbers. Boards up when done.** Same y: 8 − (−3) = 11 — a board with 5 subtracted 8 − 3.
11. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
12. **Example 2 — Slanted.** Ask whether the points share an x or a y: neither — so draw the right triangle first.
13. **Example 2 — Worked.** Across 4 − (−2) = 6, up 9 − 1 = 8; 36 + 64 = 100, so 10 units — whole, because 6, 8, 10 is a triple.
14. **Your Turn — Same steps, your numbers. Boards up when done.** Across 2 − (−5) = 7, up 1 − (−3) = 4; 49 + 16 = 65 and √65 ≈ 8.06, so 8.1 — a board with 11 added the legs.
15. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).

**16–17. Board 1.** How far apart are $(-7, -4)$ and $(-7, 6)$ ? → ***10 units*** *2 — subtracted 6 − 4 across zero [Florida B1G-M, GR.1.2: integer errors across an axis]*

**18–19. Board 2.** How far apart are $(-9, 2)$ and $(3, 2)$ ? → ***12 units*** *6 — subtracted 9 − 3 across zero [Florida B1G-M, GR.1.2: integer errors across an axis]*

**20–21. Board 3.** What is the distance between (−5, 1) and (3, 1)? → ***B — 8 units***
*(A) counted the grid lines, not the spaces [Math Nation names counting gridlines]; (C) subtracted 5 − 3 across zero [Florida B1G-M, GR.1.2: (−2, 0) to (3, 0) is not 1]; (D) −5 − 3: a distance is never negative [Notes I] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

**22–23. Board 4.** How far apart are $A(-4, -3)$ and $B(5, 9)$ ? → ***15 units*** *21 — added the legs [MA.8.GR.1.2: the theorem, not a sum]; 'cannot, no triangle' — draw one [Math Nation names 'no triangle drawn']*

**24–25. Board 5.** How far apart are $C(-2, -5)$ and $D(3, 3)$, to the nearest tenth? → ***√89 ≈ 9.4 units*** *13 — added the legs [MA.8.GR.1.2]; √(1² + 8²) — subtracted 3 − 2 across zero [Florida B1G-M, GR.1.2]*

**26–27. Board 6.** Which gives the distance between (4, 5) and (−3, −1)? → ***C — √(7² + 6²) = √85*** *(A) 4 − 3 = 1 for the across leg — the sign of −3 dropped [Florida B1G-M, GR.1.2: integer errors across an axis]; (B) one point's coordinates taken as the legs [Math Nation names wrong horizontal and vertical lengths]; (D) added the legs [MA.8.GR.1.2: the theorem, not a sum] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

**28–29. Board 7.** How far is $(5, -12)$ from the origin? → ***13 units*** *17 — added the legs [MA.8.GR.1.2]; 'the square root of −119' — subtracted 25 − 144 [the theorem adds the legs' squares]*

**30–31. Board 8.** On a town map in miles, the school is at $(-3, 2)$, the library at $(4, 2)$ and the town hall at $(0, 5)$. How far is the school from the library? → ***7 mi*** *The town hall at (0, 5) is not needed. 1 — subtracted 4 − 3 across zero [Florida B1G-M, GR.1.2]; a distance to the town hall — the wrong pair of points*

**32–33. Board 9.** Ty says (−3, 0) and (7, 0) are 4 units apart. What did he do? What is the distance? → ***He subtracted 7 − 3. Across zero the pieces add: 10 units.*** *'4 is right' — the axis-crossing error [Florida B1G-M, GR.1.2]*

**34. Independent Set — Six questions. On your own, in writing.** Hand out the Independent Set; the same six questions are on the board.

**35. IXL — Last five minutes.** IXL: every listed skill is required, SmartScore 67.

**Variation.** Boards 1–2: one line, vertical then horizontal, both across zero; 3 votes on the axis error. The pattern breaks at 4 — slanted, with only the points on the grid — whole, then to the tenth (5), then only the expression (6), then from the origin (7); 8 carries an extra point; 9 is written. The independent set: horizontal, vertical, slanted whole, slanted to the tenth, from the origin, a map.

## Misconceptions to Watch

**Board 1.** 2 — subtracted 6 − 4 across zero [Florida B1G-M, GR.1.2: integer errors across an axis]

**Board 2.** 6 — subtracted 9 − 3 across zero [Florida B1G-M, GR.1.2: integer errors across an axis]

**Board 3.** counted the grid lines, not the spaces [Math Nation names counting gridlines]

**Board 4.** 21 — added the legs [MA.8.GR.1.2: the theorem, not a sum]

**Board 5.** 13 — added the legs [MA.8.GR.1.2]

**Board 6.** 4 − 3 = 1 for the across leg — the sign of −3 dropped [Florida B1G-M, GR.1.2: integer errors across an axis]

**Board 7.** 17 — added the legs [MA.8.GR.1.2]

**Board 8.** 1 — subtracted 4 − 3 across zero [Florida B1G-M, GR.1.2]

**Board 9.** '4 is right' — the axis-crossing error [Florida B1G-M, GR.1.2]

**Materials** Whiteboards and markers; a printed grid for any student whose board has none. Calculators allowed, printed nowhere.

# Distance on the Coordinate Plane

*Lesson Plan*

## 1. Lesson Information

| | | | |
|---|---|---|---|
| Course | Grade 7 Accelerated (1205050) | Unit | 6 — Relationships in Triangles |
| Lesson | Lesson 5 — Distance on the Coordinate Plane | Date | _____ |
| Length | 53 minutes (one period) | Benchmark(s) | MA.8.GR.1.2 |

## 2. Standards

**MA.8.GR.1.2** Apply the Pythagorean Theorem to solve mathematical and real-world problems involving the distance between two points in a coordinate plane.

**Must.** Find the distance between two points: on one line by subtracting (across an axis too), slanted by the right triangle whose legs are read from the coordinates; an irrational answer exact and, when asked, to the tenth.

**Must not.** The distance formula (Clarification 2); radicands over 225; perimeters and maps with several legs (6.06).

## 3. Mathematical Thinking and Reasoning Standards

*the evidence for each, from this period*

| MTR | What it asks | The evidence in this lesson |
|---|---|---|
| MA.K12.MTR.2.1 | Demonstrate understanding by representing problems in multiple ways. | Notes II–III: one distance as a slanted segment, as a right triangle's legs and as $a^2 + b^2 = c^2$ |
| MA.K12.MTR.4.1 | Engage in discussions that reflect on the mathematical thinking of self and others. | boards 3 and 6 split; the sixty-second re-vote |
| MA.K12.MTR.6.1 | Assess the reasonableness of solutions. | Notes I and board 9: a distance across zero checked by counting the squares |

## 4. Learning Target and Essential Question

**Learning target.** I can find the distance between two points: subtract on a line, and draw the right triangle when the segment is slanted.

**Essential question.** How can two pairs of numbers tell you how far apart two points are?

**Building on.** 6.03–04 — the Pythagorean Theorem for a missing side; Grade 6 (MA.6.GR.1.2) — the distance between points with the same x or the same y; Grade 7 — subtracting integers.

**Working toward.** 6.06 — perimeters and maps on the grid; Algebra 1 and Geometry — the distance formula, which is this triangle written as one line.

## 5. Sequence

*read from the deck; never typed*

| Segment | Min | What the teacher does | What students do |
|---|---|---|---|
| Settle in; post the learning target | 1 | Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet. | Copy the target. |
| Warm-Up | 6 | Two minutes silent, then reveal. One sentence of reteach per question at most. Then the new words, read once. | Four retrieval questions, alone, no notes. Then copy the new words. |
| Notes | 11 | Teach from the front; say the OFF line out loud. | Copy the notes; answer the checks aloud. |
| Example 1 | 3 | Work the example; name the wrong boards before they appear. | Watch, then work the Your Turn alone. |
| Your Turn | 3 | Circulate; reveal and name what a wrong board most likely did. | Work it alone; boards up on the cue. |
| Example 2 | 4 | Work the example; name the wrong boards before they appear. | Watch, then work the Your Turn alone. |
| Your Turn | 3 | Circulate; reveal and name what a wrong board most likely did. | Work it alone; boards up on the cue. |
| Whiteboards | 11 | One question at a time. Boards down until the cue, all up together; scan the back row first. | Answer on a board; hold it up on three. |
| Independent Set | 6 | Silent. Circulate and mark what you see; do not teach. | Six questions, written, alone. |
| IXL | 5 | Assign the day's IXL skills. | Start the skills; finish tonight. |
| **Total** | **53** | | |

## 6. Gradual Release

**I do.** The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the OFF line is said out loud rather than printed.

**We do.** Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

**You do.** The nine-question whiteboard round (11 minutes) and then the six-question independent set, silent and written.

## 7. Higher-Order Questions

| Question | DOK |
|---|---|
| Why is the distance from (−4, 3) to (5, 3) not 5 − 4? | DOK 2 |
| Two points have different x's and different y's. Why can you not just subtract? | DOK 3 |
| Does it matter which point you start from? Why does the answer come out the same? | DOK 2 |

## 8. Checks for Understanding

*each board, and what to do about it*

| # | What the board asks | Answer | If the board is wrong |
|---|---|---|---|
| 1 | How far apart are $(-7, -4)$ and $(-7, 6)$ ? | ***10 units*** | 2 — subtracted 6 − 4 across zero [Florida B1G-M, GR.1.2: integer errors across an axis] |
| 2 | How far apart are $(-9, 2)$ and $(3, 2)$ ? | ***12 units*** | 6 — subtracted 9 − 3 across zero [Florida B1G-M, GR.1.2: integer errors across an axis] |
| 3 | What is the distance between (−5, 1) and (3, 1)? | ***B — 8 units*** | (A) counted the grid lines, not the spaces [Math Nation names counting gridlines]; (C) subtracted 5 − 3 across zero [Florida B1G-M, GR.1.2: (−2, 0) to (3, 0) is not 1]; (D) −5 − 3: a distance is never negative [Notes I] |
| 4 | How far apart are $A(-4, -3)$ and $B(5, 9)$ ? | ***15 units*** | 21 — added the legs [MA.8.GR.1.2: the theorem, not a sum]; 'cannot, no triangle' — draw one [Math Nation names 'no triangle drawn'] |
| 5 | How far apart are $C(-2, -5)$ and $D(3, 3)$ , to the nearest tenth? | ***$\sqrt{89} \approx 9.4$ units*** | 13 — added the legs [MA.8.GR.1.2]; $\sqrt{(1^2 + 8^2)}$ — subtracted 3 − 2 across zero [Florida B1G-M, GR.1.2] |
| 6 | Which gives the distance between (4, 5) and (−3, −1)? | ***C — $\sqrt{(7^2 + 6^2)} = \sqrt{85}$*** | (A) 4 − 3 = 1 for the across leg — the sign of −3 dropped [Florida B1G-M, GR.1.2: integer errors across an axis]; (B) one point's coordinates taken as the legs [Math Nation names wrong horizontal and vertical lengths]; (D) added the legs [MA.8.GR.1.2: the theorem, not a sum] |
| 7 | How far is $(5, -12)$ from the origin? | ***13 units*** | 17 — added the legs [MA.8.GR.1.2]; 'the square root of −119' — subtracted 25 − 144 [the theorem adds the legs' squares] |
| 8 | On a town map in miles, the school is at $(-3, 2)$ , the library at $(4, 2)$ and the town hall at $(0, 5)$ . How far is the school from the library? | ***7 mi*** | 1 — subtracted 4 − 3 across zero [Florida B1G-M, GR.1.2]; a distance to the town hall — the wrong pair of points |
| 9 | Ty says (−3, 0) and (7, 0) are 4 units apart. What did he do? What is the distance? | ***He subtracted 7 − 3. Across zero the pieces add: 10 units.*** | '4 is right' — the axis-crossing error [Florida B1G-M, GR.1.2] |

**MTR.4.1 — the one peer move.** When a board splits — no option holding about two-thirds of the room — sixty seconds of "convince the person next to you", then re-vote, then reveal. The minute comes out of the whiteboard remainder.

## 9. Differentiation

**ESE / 504.** A printed grid for every board. The dashed legs are drawn on Notes II–III and the examples; from board 4 the student draws them. "Across, then up or down" said aloud with every slanted board.

**ELL.** **Horizontal**: side to side, like the horizon. **Vertical**: up and down. **Distance**: how far apart — never negative. Point at the grid with each word.

**Enrichment.** Find every point on the grid exactly 5 units from the origin with whole-number coordinates. (There are 12: (±5, 0), (0, ±5), (±3, ±4), (±4, ±3).)

## 10. Closure, Vocabulary, Materials, Practice

**Closure.** Board 9 is written work and it is the exit evidence: Ty subtracted 7 − 3, but −3 and 7 are on opposite sides of zero, so the two pieces add: 3 + 7 = 10. A board that says only '10' without naming Ty's step has the answer and not the error.

**Vocabulary.** **horizontal distance**; **vertical distance**; **legs on the grid**

**Materials.** Whiteboards and markers; a printed grid for any student whose board has none. Calculators allowed, printed nowhere.

**Practice.** The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due by the Unit 6 test: Find the distance between two points on a horizontal or vertical line — L9V; Find the distance between two points — ZBP.

# Real-World Coordinate-Plane Problems; Perimeter

*Teacher's Edition*

## The Period

**MA.8.GR.1.2** Apply the Pythagorean Theorem to solve mathematical and real-world problems involving the distance between two points in a coordinate plane.

**Must** Find the perimeter of a figure on the coordinate plane, each slanted side by its right triangle (a triangle with no right angle split by its height); the straight-line and along-the-grid distances on a map, and how much shorter the straight line is; an irrational total kept exact and rounded once.

**Must not** The distance formula (Clarification 2); radicands over 225 or with a square factor; combining radicals (√34 + √34 is two lengths added, not 2√34 — Algebra 1).

**Target** I can find the perimeter of a figure on the coordinate plane, and a straight-line distance on a map.

**MTR** **MTR.2.1** — Notes I and III: one perimeter as sides on the grid, as right triangles and as a sum · **MTR.4.1** — boards 4 and 6 split; the sixty-second re-vote · **MTR.7.1** — Notes II and boards 5, 7 and 8: maps, the streets against the straight line

| Segment | Slides | Time |
|---|---|---|
| Settle in; post the learning target | 1 | 1 min |
| Warm-Up | 2–4 | 6 min |
| Notes | 5–7 | 11 min |
| Example 1 | 8–9 | 3 min |
| Your Turn | 10–11 | 3 min |
| Example 2 | 12–13 | 4 min |
| Your Turn | 14–15 | 3 min |
| Whiteboards | 16–33 | 11 min |
| Independent Set | 34 | 6 min |
| IXL | 35 | 5 min |
| **Total** | | **53 min** |

### Say these three things out loud today

1. A perimeter adds every side; a slanted side is found by its own right triangle.
2. On a map, the streets add the legs and the straight line is the hypotenuse.
3. No right angle? The height makes two. Round once, at the end.

## Slide by Slide

1. **Real-World Coordinate-Plane Problems; Perimeter.** Post the learning target.
2. **Warm-Up — On your own. Four minutes. No notes.** Q4 is today's idea with only straight sides: a perimeter adds every side — 7, 3, 7 and 3, across zero for both. *1 retrieves 6.05 — a slanted distance by its triangle (yesterday). 2 retrieves 6.03 — a hypotenuse from the legs (√53: a whole radicand up to 225, GR.1.1 Clarification 3) (earlier this week). 3 retrieves 5.04 — a real distance from a map's scale (last unit). 4 retrieves MA.6.GR.1.3 — the perimeter of a rectangle on the coordinate plane (prior grade — the prerequisite this lesson needs).*
3. **Warm-Up — Answers.** Reveal. 1. 10 units 2. √53 ≈ 7.3 m 3. 75 mi 4. 20 units
4. **Vocabulary — The new words, before they are used.** Read each word and its meaning once; the notes use them next.
5. **Notes — Copy the rule and the example.** Every side counts. A slanted side gets its triangle.
6. **Notes.** The streets add the legs. The straight line is the hypotenuse.
7. **Notes.** No square corner? Make two. Round once, at the end.
8. **Example 1 — A perimeter on the grid.** Ask which sides lie on grid lines before anyone measures: DE and EF.
9. **Example 1 — Worked.** DE: 3 − (−3) = 6; EF: 6 − (−2) = 8; DF: 36 + 64 = 100, 10.
10. **Your Turn — Same steps, your numbers. Boards up when done.** Across 4 − (−5) = 9 and up 5 − (−7) = 12, so the slanted side is √225 = 15 and P = 9 + 12 + 15 = 36 — a board with 21 left the slanted side out.
11. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
12. **Example 2 — A map: the streets and the straight line.** Ask what 'along the streets' means on this map: on the grid lines — across, then up.
13. **Example 2 — Worked.** Across 4 − (−3) = 7, up 3 − (−2) = 5.
14. **Your Turn — Same steps, your numbers. Boards up when done.** Across −2 − (−5) = 3 and down 4 − (−4) = 8: by the streets 11 blocks, straight √73 ≈ 8.54, so 8.5 — a board with 11 for the straight line added the legs.
15. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
16. **Board 1.** Triangle ABC has corners $A(-6, -2)$, $B(6, -2)$ and $C(6, 3)$. Find its perimeter. → ***30 units*** *17 — left the slanted side out [Notes I: every side]; 34 — took AC as 12 + 5 [MA.8.GR.1.2: the theorem, not a sum]*
17. **Board 2.** A triangle has corners $(-3, -4)$, $(1, -4)$ and $(1, 5)$. Find its perimeter, to the nearest tenth. → ***22.8 units*** *13 — left the slanted side out [Notes I]; 22.9 — rounded 9.849 to 9.85 and then to 9.9 [one rounding, at the end — Notes III]; 26 — the slanted side as 4 + 9 [MA.8.GR.1.2]*
18. **Board 3.** Triangle $A(-4, -6)$, $B(10, -6)$, $C(1, 6)$ has no right angle. Find its perimeter. → ***42 units*** *40 — took both slanted sides as 13, as if the height split the base in half [the foot is at (1, −6): 5 and 9]; a perimeter with 12 in it — the height is inside the triangle, not a side [B1G-M, GR.1.2: the theorem on a triangle with no right angle]*

**22–23. Board 4.** What is the perimeter of the triangle with corners (−2, 3), (2, 3) and (2, −4), to the nearest tenth? → ***D — 19.1 units*** *(A) 4 + 7: left the slanted side out [Notes I: every side]; (B) 4 + 7 + 11: the slanted side taken as the sum of the legs [MA.8.GR.1.2: the theorem, not a sum]; (C) 4 + 7 + 65: forgot the square root [B1G-M, GR.1.1: root errors] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

**24–25. Board 5.** On a map in blocks, the library is at $(-7, -1)$ and the pool at $(5, 4)$. How much shorter is the straight line than the streets? → ***4 blocks shorter*** *13 or 17 — one path, not how much shorter [the question asks a difference]; 30 — added the two paths [the question asks a difference]; 2 across, from 7 − 5 [Florida B1G-M, GR.1.2: integer errors across an axis]*

**26–27. Board 6.** How long is a diagonal of the rectangle with corners (−3, −4), (6, −4), (6, 8) and (−3, 8)? → ***A — 15 units*** *(B) 9 + 12: added two sides [MA.8.GR.1.2: the theorem, not a sum]; (C) 9 + 12 + 9 + 12: the perimeter, not one diagonal [Notes II: the straight line is one segment]; (D) 6 and 8, one corner's coordinates, taken as the legs [Math Nation names wrong horizontal and vertical lengths] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

**28–29. Board 7.** On a park map in meters, the fountain is at $(-5, 2)$ and the gate at $(4, -3)$. How far is it straight across, to the nearest tenth? → ***√106 ≈ 10.3 m*** *14 — added the legs: the paths, not the straight line [MA.8.GR.1.2]; √26 — 4 − 5 = 1 for the across leg [Florida B1G-M, GR.1.2: integer errors across an axis]*

**30–31. Board 8.** On a trail map in miles, the camp is at $(-3, -6)$, the lake at $(2, 6)$ and the trailhead at $(-1, 0)$. How far is the camp from the lake, in a straight line? → ***13 mi*** *The trailhead at (−1, 0) is not needed. 17 — added the legs [MA.8.GR.1.2]; a distance to the trailhead — the wrong pair of points*

**32–33. Board 9.** Mia says the segment from (0, 0) to (3, 1) is 3 units long, because it crosses 3 grid squares. Is she right? Explain, and find its length to the nearest tenth. → ***No: it is the hypotenuse of a triangle with legs 3 and 1, √10 ≈ 3.2 units.*** *'Yes, 3' — counted the squares [Notes I: a slanted side is measured by its triangle]; 4 — added the legs [MA.8.GR.1.2]*

**34. Independent Set — Six questions. On your own, in writing.** Hand out the Independent Set; the same six questions are on the board.

**35. IXL — Last five minutes.** IXL: every listed skill is required, SmartScore 67.

**Variation.** Boards 1–2: a right triangle's perimeter, whole then to the tenth; 3 has no right angle and is split by its height, not in half; 4 votes on the perimeter errors. The pattern breaks at 5 — a map, and how much shorter the straight line is — then a rectangle's diagonal voted (6), a park map to the tenth (7), a map with an extra point (8); 9 is written. The independent set: a perimeter whole, a perimeter to the tenth, a parallelogram, straight against the streets, a map to the tenth, a rectangle's missing corner and its diagonal.

## Misconceptions to Watch

**Board 1.** 17 — left the slanted side out [Notes I: every side]

**Board 2.** 13 — left the slanted side out [Notes I]

**Board 3.** 40 — took both slanted sides as 13, as if the height split the base in half [the foot is at (1, −6): 5 and 9]

**Board 4.** 4 + 7: left the slanted side out [Notes I: every side]

**Board 5.** 13 or 17 — one path, not how much shorter [the question asks a difference]

**Board 6.** 9 + 12: added two sides [MA.8.GR.1.2: the theorem, not a sum]

**Board 7.** 14 — added the legs: the paths, not the straight line [MA.8.GR.1.2]

**Board 8.** 17 — added the legs [MA.8.GR.1.2]

**Board 9.** 'Yes, 3' — counted the squares [Notes I: a slanted side is measured by its triangle]

**Materials** Whiteboards and markers; a printed grid for any student whose board has none. Calculators allowed, printed nowhere.

# Real-World Coordinate-Plane Problems; Perimeter

*Lesson Plan*

## 1. Lesson Information

| | | | |
|---|---|---|---|
| Course | Grade 7 Accelerated (1205050) | Unit | 6 — Relationships in Triangles |
| Lesson | Lesson 6 — Real-World Coordinate-Plane Problems; Perimeter | Date | _____ |
| Length | 53 minutes (one period) | Benchmark(s) | MA.8.GR.1.2 |

## 2. Standards

**MA.8.GR.1.2** Apply the Pythagorean Theorem to solve mathematical and real-world problems involving the distance between two points in a coordinate plane.

**Must.** Find the perimeter of a figure on the coordinate plane, each slanted side by its right triangle (a triangle with no right angle split by its height); the straight-line and along-the-grid distances on a map, and how much shorter the straight line is; an irrational total kept exact and rounded once.

**Must not.** The distance formula (Clarification 2); radicands over 225 or with a square factor; combining radicals (√34 + √34 is two lengths added, not 2√34 — Algebra 1).

## 3. Mathematical Thinking and Reasoning Standards

*the evidence for each, from this period*

| MTR | What it asks | The evidence in this lesson |
|---|---|---|
| MA.K12.MTR.2.1 | Demonstrate understanding by representing problems in multiple ways. | Notes I and III: one perimeter as sides on the grid, as right triangles and as a sum |
| MA.K12.MTR.4.1 | Engage in discussions that reflect on the mathematical thinking of self and others. | boards 4 and 6 split; the sixty-second re-vote |
| MA.K12.MTR.7.1 | Apply mathematics to real-world contexts. | Notes II and boards 5, 7 and 8: maps, the streets against the straight line |

## 4. Learning Target and Essential Question

**Learning target.** I can find the perimeter of a figure on the coordinate plane, and a straight-line distance on a map.

**Essential question.** How do you measure around a shape whose sides do not follow the grid lines?

**Building on.** 6.05 — the distance between two points on the grid; 6.03 — the Pythagorean Theorem; Grade 6 (MA.6.GR.1.3) — the perimeter of a rectangle on the coordinate plane.

**Working toward.** 6.07–08 — similar triangles; Geometry — perimeters on the plane, and the distance formula, which is this triangle written as one line.

## 5. Sequence

*read from the deck; never typed*

| Segment | Min | What the teacher does | What students do |
|---|---|---|---|
| Settle in; post the learning target | 1 | Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet. | Copy the target. |
| Warm-Up | 6 | Two minutes silent, then reveal. One sentence of reteach per question at most. Then the new words, read once. | Four retrieval questions, alone, no notes. Then copy the new words. |
| Notes | 11 | Teach from the front; say the OFF line out loud. | Copy the notes; answer the checks aloud. |
| Example 1 | 3 | Work the example; name the wrong boards before they appear. | Watch, then work the Your Turn alone. |
| Your Turn | 3 | Circulate; reveal and name what a wrong board most likely did. | Work it alone; boards up on the cue. |
| Example 2 | 4 | Work the example; name the wrong boards before they appear. | Watch, then work the Your Turn alone. |
| Your Turn | 3 | Circulate; reveal and name what a wrong board most likely did. | Work it alone; boards up on the cue. |
| Whiteboards | 11 | One question at a time. Boards down until the cue, all up together; scan the back row first. | Answer on a board; hold it up on three. |
| Independent Set | 6 | Silent. Circulate and mark what you see; do not teach. | Six questions, written, alone. |
| IXL | 5 | Assign the day's IXL skills. | Start the skills; finish tonight. |
| **Total** | **53** | | |

## 6. Gradual Release

**I do.** The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the OFF line is said out loud rather than printed.

**We do.** Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

**You do.** The nine-question whiteboard round (11 minutes) and then the six-question independent set, silent and written.

## 7. Higher-Order Questions

| Question | DOK |
|---|---|
| Why is the straight path shorter than the streets unless the two points share an x or a y? | DOK 2 |
| Notes III rounded once, at the end. When does rounding each side first change the answer? | DOK 3 |
| A triangle has no right angle. Where would you draw a line to give it two? | DOK 2 |

## 8. Checks for Understanding

*each board, and what to do about it*

| # | What the board asks | Answer | If the board is wrong |
|---|---|---|---|
| 1 | Triangle ABC has corners $A(-6, -2)$, $B(6, -2)$ and $C(6, 3)$. Find its perimeter. | ***30 units*** | 17 — left the slanted side out [Notes I: every side]; 34 — took AC as 12 + 5 [MA.8.GR.1.2: the theorem, not a sum] |
| 2 | A triangle has corners $(-3, -4)$, $(1, -4)$ and $(1, 5)$. Find its perimeter, to the nearest tenth. | ***22.8 units*** | 13 — left the slanted side out [Notes I]; 22.9 — rounded 9.849 to 9.85 and then to 9.9 [one rounding, at the end — Notes III]; 26 — the slanted side as 4 + 9 [MA.8.GR.1.2] |
| 3 | Triangle $A(-4, -6)$, $B(10, -6)$, $C(1, 6)$ has no right angle. Find its perimeter. | ***42 units*** | 40 — took both slanted sides as 13, as if the height split the base in half [the foot is at (1, −6): 5 and 9]; a perimeter with 12 in it — the height is inside the triangle, not a side [B1G-M, GR.1.2: the theorem on a triangle with no right angle] |
| 4 | What is the perimeter of the triangle with corners (−2, 3), (2, 3) and (2, −4), to the nearest tenth? | ***D — 19.1 units*** | (A) 4 + 7: left the slanted side out [Notes I: every side]; (B) 4 + 7 + 11: the slanted side taken as the sum of the legs [MA.8.GR.1.2: the theorem, not a sum]; (C) 4 + 7 + 65: forgot the square root [B1G-M, GR.1.1: root errors] |
| 5 | On a map in blocks, the library is at $(-7, -1)$ and the pool at $(5, 4)$. How much shorter is the straight line than the streets? | ***4 blocks shorter*** | 13 or 17 — one path, not how much shorter [the question asks a difference]; 30 — added the two paths [the question asks a difference]; 2 across, from 7 − 5 [Florida B1G-M, GR.1.2: integer errors across an axis] |
| 6 | How long is a diagonal of the rectangle with corners (−3, −4), (6, −4), (6, 8) and (−3, 8)? | ***A — 15 units*** | (B) 9 + 12: added two sides [MA.8.GR.1.2: the theorem, not a sum]; (C) 9 + 12 + 9 + 12: the perimeter, not one diagonal [Notes II: the straight line is one segment]; (D) 6 and 8, one corner's coordinates, taken as the legs [Math Nation names wrong horizontal and vertical lengths] |
| 7 | On a park map in meters, the fountain is at $(-5, 2)$ and the gate at $(4, -3)$. How far is it straight across, to the nearest tenth? | ***$\sqrt{106} \approx 10.3$ m*** | 14 — added the legs: the paths, not the straight line [MA.8.GR.1.2]; √26 — 4 − 5 = 1 for the across leg [Florida B1G-M, GR.1.2: integer errors across an axis] |
| 8 | On a trail map in miles, the camp is at $(-3, -6)$, the lake at $(2, 6)$ and the trailhead at $(-1, 0)$. How far is the camp from the lake, in a straight line? | ***13 mi*** | 17 — added the legs [MA.8.GR.1.2]; a distance to the trailhead — the wrong pair of points |

| # | What the board asks | Answer | If the board is wrong |
|---|---|---|---|
| 9 | Mia says the segment from (0, 0) to (3, 1) is 3 units long, because it crosses 3 grid squares. Is she right? Explain, and find its length to the nearest tenth. | ***No: it is the hypotenuse of a triangle with legs 3 and 1, √10 ≈ 3.2 units.*** | 'Yes, 3' — counted the squares [Notes I: a slanted side is measured by its triangle]; 4 — added the legs [MA.8.GR.1.2] |

**MTR.4.1 — the one peer move.** When a board splits — no option holding about two-thirds of the room — sixty seconds of “convince the person next to you”, then re-vote, then reveal. The minute comes out of the whiteboard remainder.

## 9. Differentiation

**ESE / 504.** A printed grid for every board. Find the sides on grid lines first and write them on the figure; then the slanted side, by its triangle. "Every side counts" said aloud with every perimeter.

**ELL.** **Perimeter**: around the outside — every side. **Along the streets**: on the grid lines. **Straight line**: one segment, as a bird flies. Trace each with a finger on the grid.

**Enrichment.** Plot (0, 0), (6, 0) and (3, 4). Is it a right triangle? (No: 25 + 25 is not 36.) Find its perimeter. (5 + 5 + 6 = 16.)

## 10. Closure, Vocabulary, Materials, Practice

**Closure.** Board 9 is written work and it is the exit evidence: Mia counted the squares her segment crosses, but a slanted segment is the hypotenuse of a right triangle — legs 3 and 1, √10 ≈ 3.2, longer than 3. A board that says only '3.2' has the length and not the reason.

**Vocabulary.** **perimeter**; **height of a triangle**; **straight-line distance**

**Materials.** Whiteboards and markers; a printed grid for any student whose board has none. Calculators allowed, printed nowhere.

**Practice.** The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due by the Unit 6 test: Pythagorean theorem: find the perimeter — VGE.

# Similar Triangles; Indirect Measurement

*Teacher's Edition*

## The Period

**MA.8.GR.2.4** Solve mathematical and real-world problems involving proportional relationships between similar triangles.

**Must** Decide similarity from all three side ratios; match the sides of a turned pair (by size, or by a name's letters); find missing sides with k = copy ÷ original; measure with a shadow, a mirror or a ramp when the words say the triangles are similar.

**Must not** AA named or proved (Geometry); angle sums (Thread B); a mirror or ramp item that does not say its triangles are similar.

**Target** I can find a missing side of similar triangles with the scale factor, and use similar triangles to measure what I cannot reach.

**MTR** **MTR.2.1** — Notes I: one pair as a picture, three ratios and one k · **MTR.4.1** — boards 4 and 5 split; the re-vote · **MTR.7.1** — Notes III, Example 2 and boards 6–8: heights measured without climbing

| Segment | Slides | Time |
|---|---|---|
| Settle in; post the learning target | 1 | 1 min |
| Warm-Up | 2–4 | 6 min |
| Notes | 5–7 | 12 min |
| Example 1 | 8–9 | 3 min |
| Your Turn | 10–11 | 3 min |
| Example 2 | 12–13 | 4 min |
| Your Turn | 14–15 | 3 min |
| Whiteboards | 16–33 | 10 min |
| Independent Set | 34 | 6 min |
| IXL | 35 | 5 min |
| **Total** | | **53 min** |

### Say these three things out loud today

1. Similar triangles have equal angles, and every side is multiplied by the same k.
2. Match by size or by angle, never by place on the page; k = copy ÷ original.
3. Same sun, same angle: shadows, mirrors and ramps make similar triangles.

## Slide by Slide

1. **Similar Triangles; Indirect Measurement.** Post the learning target.
2. **Warm-Up — On your own. Four minutes. No notes.** Q3 is today's k: copy ÷ original, 4 ÷ 10 = 0.4 — a copy smaller than the original. *1 retrieves 6.06 — a perimeter, the slanted side by its triangle (√13: a whole radicand up to 225, GR.1.2 Clarification 3) (yesterday). 2 retrieves 6.04 — a ladder: the ladder is c, so subtract (last week). 3 retrieves 5.01 — k = copy ÷ original; a k under 1 shrinks (last unit). 4 retrieves MA.7.GR.1.5 — a scale drawing: every length times one number (prior grade — the prerequisite this lesson needs).*
3. **Warm-Up — Answers.** Reveal. 1. 8.6 units 2. 12 ft 3. k = 0.4 4. 21 ft
4. **Vocabulary — The new words, before they are used.** Read each word and its meaning once; the notes use them next.
5. **Notes — Copy the rule and the example.** Similar: one k on every side. Match by size, not by place.
6. **Notes.** The names say the matching. One matching pair gives k.
7. **Notes.** Same sun, same angle: similar triangles.
8. **Example 1 — A turned pair.** Ask which side of the second triangle matches the 5 before anyone divides: the shortest, 15 — wherever it sits.
9. **Example 1 — Worked.** Sorted: 5, 7, 9 and 15, x, y — the 5 and the 15 are both shortest.
10. **Your Turn — Same steps, your numbers. Boards up when done.** Sorted, the copy is 12, x, 28, so x matches the 5: k = 12 ÷ 3 = 4 and x = 20 — a board with about 46.7 matched by the order in the list (3 with 28).
11. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
12. **Example 2 — A mirror on the ground.** Ask which lengths match: Omar's eye height with the tree's height, and his 4 ft to the mirror with the tree's 18 ft — the words say the two right triangles are similar.
13. **Example 2 — Worked.** Ground to ground: k = 18 ÷ 4 = 4.5, so the tree is 5 × 4.5 = 22.5 ft — a board with about 1.1 used k upside down (4 ÷ 18), one with 19 added 14.
14. **Your Turn — Same steps, your numbers. Boards up when done.** k = 12 ÷ 1.5 = 8, so the flagpole is 4 × 8 = 32 ft — a board with 0.5 used k upside down (1.5 ÷ 12).
15. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).

**16–17. Board 1.** Are triangles with sides 8, 12, 14 and 12, 18, 21 similar? → ***Yes — every ratio is 1.5*** *'No — the sides are different' [Florida B1G-M, GR.2.4: similar triangles assumed to have equal sides]; 'yes' from two ratios only [Notes I: all three]*

**18–19. Board 2.** Triangle ABC is similar to triangle DEF. Find x. → ***x = 24*** *13.5 — matched the 8 with the 18 by place on the page, k = 18 ÷ 8 [Florida B1G-M, GR.2.4: mismatching corresponding sides]; 20 — added 12 to the 8 [Notes I: similar means multiplied by one k]*

**20–21. Board 3.** A triangle has sides of 12, 16 and 20 inches. A similar triangle's longest side is 5 inches. Find its other two sides. → ***3 inches and 4 inches*** *48 and 64 — multiplied by 20 ÷ 5 = 4, k upside down [Notes II: k = copy ÷ original]*

**22–23. Board 4.** The two triangles are similar (the second is turned). Which side of the second triangle matches the 9? → ***C — the 12*** *(A) the side in the same place on the page — the second triangle is turned [Florida B1G-M, GR.2.4: mismatching corresponding sides]; (B) 16, the longest, for the middle side [Notes I: match by size]; (D) looked for a side equal to the 9 — similar taken as equal [Florida B1G-M, GR.2.4: similar triangles assumed to have equal sides] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

**24–25. Board 5.** Which triangle is similar to one with sides 5, 6 and 8? → ***B — 10, 12, 16*** *(A) added 2 to every side [Notes I: similar means multiplied by one k, not added]; (C) checked two ratios, not three: 15 ÷ 8 is not 2 [Notes I: all three ratios]; (D) kept two sides the same [Florida B1G-M, GR.2.4: similar triangles assumed to have equal sides] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

**26–27. Board 6.** A 4-ft post casts a 2.5-ft shadow as a tree casts a 15-ft shadow. How tall is the tree? → ***24 ft*** *about 9.4 — height matched to shadow crosswise, 15 × 2.5 ÷ 4 [Florida B1G-M, GR.2.4: mismatching corresponding sides]; 16.5 — added 12.5 [similar means multiplied]*

**28–29. Board 7.** A ramp rises 0.6 m over its first 2 m from A. The right triangles are similar. How high is the ramp 8 m from A? → ***2.4 m*** *1.8 — used the 6 m between the uprights, not the 8 m from A [Math Nation names the overlapping-triangles error]; 6.6 — added 6 [similar means multiplied]*

**30–31. Board 8.** Lena's eyes are 5 ft up. In a mirror on the ground 4 ft away she sees the top of a 45-year-old flagpole, 16 ft past the mirror. The two right triangles are similar. How tall is the flagpole? → ***20 ft*** *The flagpole's age (45 years) is not needed. 180 or 225 — used the 45 years [the extra number]; 1.25 — divided the other way, 4 × 5 ÷ 16 [Florida B1G-M, GR.2.4: mismatching corresponding sides]*

**32–33. Board 9.** Kim says triangles with sides 3, 4, 5 and 5, 6, 7 are similar, because each side grew by 2. Is she right? Explain. → ***No — similar means × one k, not + 2.*** *'Yes — each side grew by 2' [Notes I: similar means multiplied by one k]*

**34. Independent Set — Six questions. On your own, in writing.** Hand out the Independent Set; the same six questions are on the board.

**35. IXL — Last five minutes.** IXL: every listed skill is required, SmartScore 67.

**Variation.** Boards 1–3: similar or not by three ratios, a named turned pair's missing side (k = 3), then a k under 1; 4 votes on matching in a turned pair, 5 on which triangle is similar. The pattern breaks at 6 — a shadow — then a ramp's two triangles from their shared corner (7) and a mirror with an extra number (8); 9 is written. The independent set: similar or not (it is not), named triangles with no picture (the letters say the matching), a k under 1, a shadow, a nested pair from the shared corner (with its picture), which is similar.

## Misconceptions to Watch

**Board 1.** 'No — the sides are different' [Florida B1G-M, GR.2.4: similar triangles assumed to have equal sides]

**Board 2.** 13.5 — matched the 8 with the 18 by place on the page, k = 18 ÷ 8 [Florida B1G-M, GR.2.4: mismatching corresponding sides]

**Board 3.** 48 and 64 — multiplied by 20 ÷ 5 = 4, k upside down [Notes II: k = copy ÷ original]

**Board 4.** the side in the same place on the page — the second triangle is turned [Florida B1G-M, GR.2.4: mismatching corresponding sides]

**Board 5.** added 2 to every side [Notes I: similar means multiplied by one k, not added]

**Board 6.** about 9.4 — height matched to shadow crosswise, 15 × 2.5 ÷ 4 [Florida B1G-M, GR.2.4: mismatching corresponding sides]

**Board 7.** 1.8 — used the 6 m between the uprights, not the 8 m from A [Math Nation names the overlapping-triangles error]

**Board 8.** 180 or 225 — used the 45 years [the extra number]

**Board 9.** 'Yes — each side grew by 2' [Notes I: similar means multiplied by one k]

**Materials** Whiteboards and markers. Calculators allowed, printed nowhere. The Unit Review, one per student, goes home today.

# Similar Triangles; Indirect Measurement

*Lesson Plan*

## 1. Lesson Information

| | | | |
|---|---|---|---|
| Course | Grade 7 Accelerated (1205050) | Unit | 6 — Relationships in Triangles |
| Lesson | Lessons 7–8 — Similar Triangles; Indirect Measurement | Date | _____ |
| Length | 53 minutes (one period) | Benchmark(s) | MA.8.GR.2.4 |

## 2. Standards

**MA.8.GR.2.4** Solve mathematical and real-world problems involving proportional relationships between similar triangles.

**Must.** Decide similarity from all three side ratios; match the sides of a turned pair (by size, or by a name's letters); find missing sides with k = copy ÷ original; measure with a shadow, a mirror or a ramp when the words say the triangles are similar.

**Must not.** AA named or proved (Geometry); angle sums (Thread B); a mirror or ramp item that does not say its triangles are similar.

## 3. Mathematical Thinking and Reasoning Standards

*the evidence for each, from this period*

| MTR | What it asks | The evidence in this lesson |
|---|---|---|
| MA.K12.MTR.2.1 | Demonstrate understanding by representing problems in multiple ways. | Notes I: one pair as a picture, three ratios and one k |
| MA.K12.MTR.4.1 | Engage in discussions that reflect on the mathematical thinking of self and others. | boards 4 and 5 split; the re-vote |
| MA.K12.MTR.7.1 | Apply mathematics to real-world contexts. | Notes III, Example 2 and boards 6–8: heights measured without climbing |

## 4. Learning Target and Essential Question

**Learning target.** I can find a missing side of similar triangles with the scale factor, and use similar triangles to measure what I cannot reach.

**Essential question.** How can a shadow tell you the height of something you cannot climb?

**Building on.** 5.01 — the scale factor k = copy ÷ original; Grade 7 (MA.7.GR.1.5) — scale drawings; 6.03 — right triangles.

**Working toward.** The Unit 6 test tomorrow; Unit 10 — slope, where similar triangles on one line give one rise over run; Geometry — similarity proved (AA, SAS, SSS).

## 5. Sequence

*read from the deck; never typed*

| Segment | Min | What the teacher does | What students do |
|---|---|---|---|
| Settle in; post the learning target | 1 | Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet. | Copy the target. |
| Warm-Up | 6 | Two minutes silent, then reveal. One sentence of reteach per question at most. Then the new words, read once. | Four retrieval questions, alone, no notes. Then copy the new words. |
| Notes | 12 | Teach from the front; say the OFF line out loud. | Copy the notes; answer the checks aloud. |
| Example 1 | 3 | Work the example; name the wrong boards before they appear. | Watch, then work the Your Turn alone. |
| Your Turn | 3 | Circulate; reveal and name what a wrong board most likely did. | Work it alone; boards up on the cue. |
| Example 2 | 4 | Work the example; name the wrong boards before they appear. | Watch, then work the Your Turn alone. |
| Your Turn | 3 | Circulate; reveal and name what a wrong board most likely did. | Work it alone; boards up on the cue. |
| Whiteboards | 10 | One question at a time. Boards down until the cue, all up together; scan the back row first. | Answer on a board; hold it up on three. |
| Independent Set | 6 | Silent. Circulate and mark what you see; do not teach. | Six questions, written, alone. |
| IXL | 5 | Assign the day's IXL skills. | Start the skills; finish tonight. |
| **Total** | **53** | | |

## 6. Gradual Release

**I do.** The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the OFF line is said out loud rather than printed.

**We do.** Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

**You do.** The nine-question whiteboard round (10 minutes) and then the six-question independent set, silent and written.

## 7. Higher-Order Questions

| Question | DOK |
|---|---|
| Every side of a triangle is doubled. What happens to its angles, and to its shape? | DOK 2 |
| Adding 2 to every side of a 3, 4, 5 triangle makes a triangle that is not similar. Is there a triangle where adding the same amount to every side does make a similar one? (Yes: an equilateral triangle — its sides stay equal to each other.) | DOK 3 |
| The shadow method needs both shadows measured at the same time. What would go wrong an hour apart? | DOK 3 |

## 8. Checks for Understanding

*each board, and what to do about it*

| # | What the board asks | Answer | If the board is wrong |
|---|---|---|---|
| 1 | Are triangles with sides 8, 12, 14 and 12, 18, 21 similar? | ***Yes — every ratio is 1.5*** | 'No — the sides are different' [Florida B1G-M, GR.2.4: similar triangles assumed to have equal sides]; 'yes' from two ratios only [Notes I: all three] |
| 2 | Triangle ABC is similar to triangle DEF. Find x. | ***x = 24*** | 13.5 — matched the 8 with the 18 by place on the page, k = 18 ÷ 8 [Florida B1G-M, GR.2.4: mismatching corresponding sides]; 20 — added 12 to the 8 [Notes I: similar means multiplied by one k] |
| 3 | A triangle has sides of 12, 16 and 20 inches. A similar triangle's longest side is 5 inches. Find its other two sides. | ***3 inches and 4 inches*** | 48 and 64 — multiplied by 20 ÷ 5 = 4, k upside down [Notes II: k = copy ÷ original] |
| 4 | The two triangles are similar (the second is turned). Which side of the second triangle matches the 9? | ***C — the 12*** | (A) the side in the same place on the page — the second triangle is turned [Florida B1G-M, GR.2.4: mismatching corresponding sides]; (B) 16, the longest, for the middle side [Notes I: match by size]; (D) looked for a side equal to the 9 — similar taken as equal [Florida B1G-M, GR.2.4: similar triangles assumed to have equal sides] |
| 5 | Which triangle is similar to one with sides 5, 6 and 8? | ***B — 10, 12, 16*** | (A) added 2 to every side [Notes I: similar means multiplied by one k, not added]; (C) checked two ratios, not three: 15 ÷ 8 is not 2 [Notes I: all three ratios]; (D) kept two sides the same [Florida B1G-M, GR.2.4: similar triangles assumed to have equal sides] |
| 6 | A 4-ft post casts a 2.5-ft shadow as a tree casts a 15-ft shadow. How tall is the tree? | ***24 ft*** | about 9.4 — height matched to shadow crosswise, 15 × 2.5 ÷ 4 [Florida B1G-M, GR.2.4: mismatching corresponding sides]; 16.5 — added 12.5 [similar means multiplied] |
| 7 | A ramp rises 0.6 m over its first 2 m from A. The right triangles are similar. How high is the ramp 8 m from A? | ***2.4 m*** | 1.8 — used the 6 m between the uprights, not the 8 m from A [Math Nation names the overlapping-triangles error]; 6.6 — added 6 [similar means multiplied] |
| 8 | Lena's eyes are 5 ft up. In a mirror on the ground 4 ft away she sees the top of a 45-year-old flagpole, 16 ft past the mirror. The two right triangles are similar. How tall is the flagpole? | ***20 ft*** | 180 or 225 — used the 45 years [the extra number]; 1.25 — divided the other way, 4 × 5 ÷ 16 [Florida B1G-M, GR.2.4: mismatching corresponding sides] |
| 9 | Kim says triangles with sides 3, 4, 5 and 5, 6, 7 are similar, because each side grew by 2. Is she right? Explain. | ***No — similar means × one k, not + 2.*** | 'Yes — each side grew by 2' [Notes I: similar means multiplied by one k] |

**MTR.4.1 — the one peer move.** When a board splits — no option holding about two-thirds of the room — sixty seconds of “convince the person next to you”, then re-vote, then reveal. The minute comes out of the whiteboard remainder.

## 9. Differentiation

**ESE / 504.** A three-row table on every board — shortest, middle, longest — with the original's sides, the copy's sides and copy ÷ original. Sort both triangles' sides before matching anything.

**ELL.** **Similar**: the same shape, any size. **Corresponding**: matching. **Scale factor**: the number every side is multiplied by. Point to each matching pair as it is said.

**Enrichment.** Notes III's tree is 30 ft. A cell tower's shadow at the same moment is 80 ft: how tall is the tower? (120 ft.) Is that realistic? (Yes: most cell towers are 50–200 ft — the state guide's own question.)

## 10. Closure, Vocabulary, Materials, Practice

**Closure.** Board 9 is written work and it is the exit evidence: Kim added 2 to every side, but similar sides are MULTIPLIED by one k — 5/3, 6/4 and 7/5 are three different numbers. A board that says only 'not similar' has the verdict and not the reason.

**Vocabulary.** **similar triangles**; **corresponding sides**; **scale factor k**; **indirect measurement**

**Materials.** Whiteboards and markers. Calculators allowed, printed nowhere. The Unit Review, one per student, goes home today.

**Practice.** The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due by the Unit 6 test: Side lengths of similar triangles — QSK; Similar triangles and indirect measurement — 88W; Identify similar triangles — 6PD.

# Angle Pairs: Supplementary, Complementary, Vertical, Adjacent

*Teacher's Edition*

## The Period

**MA.8.GR.1.4** Solve mathematical problems involving the relationships between supplementary, complementary, vertical or adjacent angles.

**Must** Name adjacent, vertical, complementary and supplementary pairs (a linear pair is supplementary); find a missing angle from each; write and solve the equation when the angles are expressions, and report the angle the question asks for.

**Must not** Parallel lines and transversals (Geometry); angle sums of polygons (MA.8.GR.1.6); reflex angles.

**Target** I can name an angle pair and use it to find a missing angle, with an equation when the angles are expressions.

**MTR** **MTR.2.1** — Notes II–III: one pair as a picture, a name and an equation · **MTR.4.1** — boards 4 and 7 split; the re-vote · **MTR.5.1** — Notes III and boards 5–6: the same equation shape behind every pair

| Segment | Slides | Time |
|---|---|---|
| Settle in; post the learning target | 1 | 1 min |
| Warm-Up | 2–4 | 6 min |
| Notes | 5–7 | 10 min |
| Example 1 | 8–9 | 3 min |
| Your Turn | 10–11 | 3 min |
| Example 2 | 12–13 | 4 min |
| Your Turn | 14–15 | 3 min |
| Whiteboards | 16–33 | 12 min |
| Independent Set | 34 | 6 min |
| IXL | 35 | 5 min |
| **Total** | | **53 min** |

### Say these three things out loud today

1. Adjacent angles share a side; vertical angles sit across a crossing and are equal.
2. Complementary makes 90 — a corner; supplementary makes 180 — a straight line.
3. Name the pair, write the equation, solve — then put x back: the question asks for the angle.

## Slide by Slide

1. **Angle Pairs: Supplementary, Complementary, Vertical, Adjacent.** Post the learning target.
2. **Warm-Up — On your own. Four minutes. No notes.** Q4 is today's 'supplementary' before it has a name: a straight angle is 180°, so the other is 50°. *1 retrieves 6.07 — a shadow; yesterday was the test, so the band reaches the last lesson (yesterday). 2 retrieves 6.05 — same y, across zero (last week). 3 retrieves 5.10–11 — a unit rate (last unit). 4 retrieves MA.4.GR.1.3 — angle measures add; a straight angle is 180° (prior grade — the prerequisite this lesson needs).*
3. **Warm-Up — Answers.** Reveal. 1. 27 ft 2. 9 units 3. 42 4. 50°
4. **Vocabulary — The new words, before they are used.** Read each word and its meaning once; the notes use them next.
5. **Notes — Copy the names and the picture.** Next to, sharing a side: adjacent. Across: vertical, and equal.
6. **Notes.** Complementary: corner, 90. Supplementary: straight, 180.
7. **Notes.** Name it, write it, solve it, put x back.
8. **Example 1 — A right angle, split.** Ask what the square mark says before anyone writes: the two angles make 90 — complementary.
9. **Example 1 — Worked.** Complementary: 3x − 1 + 2x + 6 = 90, so 5x + 5 = 90, x = 17.
10. **Your Turn — Same steps, your numbers. Boards up when done.** 4x + x + 10 = 90, 5x = 80 and x = 16, so 64° and 26° — a board with 16 stopped at x; one with 34 used 180.
11. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
12. **Example 2 — Vertical angles.** Ask what vertical angles do: they are equal — so the two expressions are set equal.
13. **Example 2 — Worked.** Vertical, so equal: 5x − 12 = 2x + 18; take 2x off both sides, 3x − 12 = 18, so 3x = 30 and x = 10.
14. **Your Turn — Same steps, your numbers. Boards up when done.** 4x + 3 = x + 39, and taking x off both sides gives 3x + 3 = 39, so x = 12 and each angle is 51° — a board with 12 reported x.
15. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
16. **16–17. Board 1.** What is the complement of 34°? → ***56°*** *146° — used 180 [Florida B1G-M, GR.1.4: complementary and supplementary swapped]*
17. **18–19. Board 2.** What is the supplement of 112°? → ***68°*** *−22 or 'none' — used 90 [Florida B1G-M, GR.1.4: complementary and supplementary swapped]*
18. **20–21. Board 3.** Two lines cross. One angle is 47°. Find a and b. → ***a = 47°, b = 133°*** *a = 133°, b = 47° — vertical and adjacent swapped [Math Nation names thinking every adjacent pair has a special sum]*
19. **22–23. Board 4.** $\angle P$ and $\angle Q$ are complementary, and $m\angle P = 28^\circ$. What is $m\angle Q$ ? → ***B — 62°*** *(A) 180 − 28: supplementary for complementary [Florida B1G-M, GR.1.4: complementary and supplementary swapped]; (C) took the two as equal — that is vertical angles [Notes I]; (D) 90 + 28: added instead of subtracting [Notes II: the two add*

*to 90] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

**24–25. Board 5.** A linear pair measures $(3x+15)^\circ$ and $105^\circ$. Find x. → ***x = 20*** *30 — set 3x + 15 = 105, as if vertical [Math Nation names setting a linear pair equal]; −10 — used 90 [Florida B1G-M, GR.1.4]*

**26–27. Board 6.** Vertical angles measure $(6x-20)^\circ$ and $(2x+28)^\circ$. Find the measure of each angle. → ***52°*** *12 — reported x as the angle [Math Nation names reporting x]; 21.5 — added them to 180 [vertical angles are equal, not supplementary]*

**28–29. Board 7.** In the figure, lines AD and BE cross at O, and ray OC is drawn. Which two angles are adjacent? → ***C — angles AOB and BOC*** *(A) a vertical pair: across the crossing, no shared side [Notes I]; (B) angle AOB lies inside angle AOC — the two overlap [Notes I: no overlap]; (D) these share only the vertex O, not a side [Notes I: a shared side] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

**30–31. Board 8.** Above a straight line a ray makes angles of 38° and x°; below it another ray makes an angle of 50°. Find x. → ***x = 142*** *The 50° angle below the line is not needed. 92 — used the 50° (180 − 38 − 50) [the extra number]; 52 — used 90 [Florida B1G-M, GR.1.4]*

**32–33. Board 9.** Can two vertical angles be complementary? Always, sometimes or never? Explain. → ***Sometimes — only when each is 45°.*** *'Never — they are not next to each other' [complementary is about the sum, not about being adjacent]; 'always' [Florida B1G-M, GR.1.4]*

**34. Independent Set — Six questions. On your own, in writing.** Hand out the Independent Set; the same six questions are on the board.

**35. IXL — Last five minutes.** IXL: every listed skill is required, SmartScore 67.

**Variation.** Boards 1–2: a complement, then a supplement; 3 reads a crossing (vertical and a linear pair at once); 4 votes on 90 against 180. The pattern breaks at 5 — a linear pair with x — then vertical angles with x where the angle is asked (6), which pair is adjacent voted (7), and a picture with one angle too many (8); 9 is written. The independent set: a complement, a supplement, a crossing, a linear pair with x, vertical angles with x, a pair named and used.

## Misconceptions to Watch

**Board 1.** 146° — used 180 [Florida B1G-M, GR.1.4: complementary and supplementary swapped]

**Board 2.** −22 or 'none' — used 90 [Florida B1G-M, GR.1.4: complementary and supplementary swapped]

**Board 3.** a = 133°, b = 47° — vertical and adjacent swapped [Math Nation names thinking every adjacent pair has a special sum]

**Board 4.** 180 − 28: supplementary for complementary [Florida B1G-M, GR.1.4: complementary and supplementary swapped]

**Board 5.** 30 — set 3x + 15 = 105, as if vertical [Math Nation names setting a linear pair equal]

**Board 6.** 12 — reported x as the angle [Math Nation names reporting x]

**Board 7.** a vertical pair: across the crossing, no shared side [Notes I]

**Board 8.** 92 — used the 50° (180 − 38 − 50) [the extra number]

**Board 9.** 'Never — they are not next to each other' [complementary is about the sum, not about being adjacent]

**Materials** Whiteboards and markers. Calculators allowed, printed nowhere.

# Angle Pairs: Supplementary, Complementary, Vertical, Adjacent

*Lesson Plan*

## 1. Lesson Information

| | | | |
|---|---|---|---|
| Course | Grade 7 Accelerated (1205050) | Unit | 6 — Relationships in Triangles |
| Lesson | Thread B · Day 1 — Angle Pairs: Supplementary, Complementary, Vertical, Adjacent | Date | _____ |
| Length | 53 minutes (one period) | Benchmark(s) | MA.8.GR.1.4 |

## 2. Standards

**MA.8.GR.1.4** Solve mathematical problems involving the relationships between supplementary, complementary, vertical or adjacent angles.

**Must.** Name adjacent, vertical, complementary and supplementary pairs (a linear pair is supplementary); find a missing angle from each; write and solve the equation when the angles are expressions, and report the angle the question asks for.

**Must not.** Parallel lines and transversals (Geometry); angle sums of polygons (MA.8.GR.1.6); reflex angles.

## 3. Mathematical Thinking and Reasoning Standards

*the evidence for each, from this period*

| MTR | What it asks | The evidence in this lesson |
|---|---|---|
| MA.K12.MTR.2.1 | Demonstrate understanding by representing problems in multiple ways. | Notes II–III: one pair as a picture, a name and an equation |
| MA.K12.MTR.4.1 | Engage in discussions that reflect on the mathematical thinking of self and others. | boards 4 and 7 split; the re-vote |
| MA.K12.MTR.5.1 | Use patterns and structure to help understand and connect mathematical concepts. | Notes III and boards 5–6: the same equation shape behind every pair |

## 4. Learning Target and Essential Question

**Learning target.** I can name an angle pair and use it to find a missing angle, with an equation when the angles are expressions.

**Essential question.** When two angles are neighbours or opposites, what does one tell you about the other?

**Building on.** Grade 4 (MA.4.GR.1.3) — angle measures add, a straight angle is 180°; Unit 1 — two-step equations (MA.7.AR.2.2). An x on both sides is new: Example 2 takes it off one side as its own line.

**Working toward.** T-B2 tomorrow — the angles of a triangle; Geometry — angle pairs made by parallel lines and a transversal.

## 5. Sequence

*read from the deck; never typed*

| Segment | Min | What the teacher does | What students do |
|---|---|---|---|
| Settle in; post the learning target | 1 | Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet. | Copy the target. |
| Warm-Up | 6 | Two minutes silent, then reveal. One sentence of reteach per question at most. Then the new words, read once. | Four retrieval questions, alone, no notes. Then copy the new words. |
| Notes | 10 | Teach from the front; say the OFF line out loud. | Copy the notes; answer the checks aloud. |
| Example 1 | 3 | Work the example; name the wrong boards before they appear. | Watch, then work the Your Turn alone. |
| Your Turn | 3 | Circulate; reveal and name what a wrong board most likely did. | Work it alone; boards up on the cue. |
| Example 2 | 4 | Work the example; name the wrong boards before they appear. | Watch, then work the Your Turn alone. |
| Your Turn | 3 | Circulate; reveal and name what a wrong board most likely did. | Work it alone; boards up on the cue. |
| Whiteboards | 12 | One question at a time. Boards down until the cue, all up together; scan the back row first. | Answer on a board; hold it up on three. |
| Independent Set | 6 | Silent. Circulate and mark what you see; do not teach. | Six questions, written, alone. |
| IXL | 5 | Assign the day's IXL skills. | Start the skills; finish tonight. |
| **Total** | **53** | | |

## 6. Gradual Release

**I do.** The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the OFF line is said out loud rather than printed.

**We do.** Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

**You do.** The nine-question whiteboard round (12 minutes) and then the six-question independent set, silent and written.

## 7. Higher-Order Questions

| Question | DOK |
|---|---|
| Can two adjacent angles be vertical angles? Why or why not? | DOK 2 |
| Two angles are supplementary and equal. What are they, and what do they make? | DOK 2 |
| Board 6's x is 12 and its angles are 52°. Which does the question ask for, and how do you know? | DOK 2 |

## 8. Checks for Understanding

*each board, and what to do about it*

| # | What the board asks | Answer | If the board is wrong |
|---|---|---|---|
| 1 | What is the complement of 34°? | ***56°*** | 146° — used 180 [Florida B1G-M, GR.1.4: complementary and supplementary swapped] |
| 2 | What is the supplement of 112°? | ***68°*** | −22 or 'none' — used 90 [Florida B1G-M, GR.1.4: complementary and supplementary swapped] |
| 3 | Two lines cross. One angle is 47°. Find a and b. | ***a = 47°, b = 133°*** | a = 133°, b = 47° — vertical and adjacent swapped [Math Nation names thinking every adjacent pair has a special sum] |
| 4 | $\angle P$ and $\angle Q$ are complementary, and $m\angle P = 28^\circ$. What is $m\angle Q$ ? | ***B — 62°*** | (A) 180 − 28: supplementary for complementary [Florida B1G-M, GR.1.4: complementary and supplementary swapped]; (C) took the two as equal — that is vertical angles [Notes I]; (D) 90 + 28: added instead of subtracting [Notes II: the two add to 90] |
| 5 | A linear pair measures $(3x + 15)^\circ$ and $105^\circ$. Find x. | ***x = 20*** | 30 — set 3x + 15 = 105, as if vertical [Math Nation names setting a linear pair equal]; −10 — used 90 [Florida B1G-M, GR.1.4] |
| 6 | Vertical angles measure $(6x - 20)^\circ$ and $(2x + 28)^\circ$. Find the measure of each angle. | ***52°*** | 12 — reported x as the angle [Math Nation names reporting x]; 21.5 — added them to 180 [vertical angles are equal, not supplementary] |
| 7 | In the figure, lines AD and BE cross at O, and ray OC is drawn. Which two angles are adjacent? | ***C — angles AOB and BOC*** | (A) a vertical pair: across the crossing, no shared side [Notes I]; (B) angle AOB lies inside angle AOC — the two overlap [Notes I: no overlap]; (D) these share only the vertex O, not a side [Notes I: a shared side] |
| 8 | Above a straight line a ray makes angles of 38° and x°; below it another ray makes an angle of 50°. Find x. | ***x = 142*** | 92 — used the 50° (180 − 38 − 50) [the extra number]; 52 — used 90 [Florida B1G-M, GR.1.4] |
| 9 | Can two vertical angles be complementary? Always, sometimes or never? Explain. | ***Sometimes — only when each is 45°.*** | 'Never — they are not next to each other' [complementary is about the sum, not about being adjacent]; 'always' [Florida B1G-M, GR.1.4] |

**MTR.4.1 — the one peer move.** When a board splits — no option holding about two-thirds of the room — sixty seconds of “convince the person next to you”, then re-vote, then reveal. The minute comes out of the whiteboard remainder.

## 9. Differentiation

**ESE / 504.** One card on the desk: Corner (90) — complementary; Straight (180) — supplementary; across — vertical, equal; next to, sharing a side — adjacent. Name the pair before writing a number.

**ELL.** **Adjacent**: next to. **Vertical** here means across from (not up and down). **Complement / supplement**: the piece that completes 90 / 180. Point and say each one.

**Enrichment.** Three lines cross at one point and make six angles. Two of them are 40° and 75°. Find the other four. (40°, 75°, 65°, 65°: vertical pairs are equal, and the three around one side make 180.)

## 10. Closure, Vocabulary, Materials, Practice

**Closure.** Board 9 is written work and it is the exit evidence: vertical angles are equal, so they are complementary only when each is 45° — sometimes, never always. A board that says 'yes' or 'no' without the 45 has the verdict and not the reason.

**Vocabulary.** **adjacent angles**; **vertical angles**; **complementary angles**; **supplementary angles**

**Materials.** Whiteboards and markers. Calculators allowed, printed nowhere.

**Practice.** The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due by the Unit 11 test: Identify complementary, supplementary, vertical, adjacent, and congruent angles — HGV; Find measures of complementary, supplementary, vertical, and adjacent angles — R2B; Write and solve equations using angle relationships — DBV.

# Interior and Exterior Angles of a Triangle

*Teacher's Edition*

## The Period

**MA.8.GR.1.5** Solve problems involving the relationships of interior and exterior angles of a triangle.

**Must** Find a missing interior angle with the Triangle Sum Theorem; find an exterior angle, or a remote interior angle, with the Exterior Angle Theorem; write and solve the equation when the angles are expressions.

**Must not** Angle sums of polygons (MA.8.GR.1.6); parallel lines and transversals; reflex angles (the angle around the outside is a named error, not an answer).

**Target** I can find a missing angle of a triangle, inside or outside, with the Triangle Sum and the Exterior Angle Theorem.

**MTR** **MTR.2.1** — Notes I–II: 180° as torn-off corners, as a sum and as a linear pair · **MTR.4.1** — boards 4 and 7 split; the re-vote · **MTR.5.1** — Notes II: the theorem from the triangle sum and the linear pair together

| Segment | Slides | Time |
|---|---|---|
| Settle in; post the learning target | 1 | 1 min |
| Warm-Up | 2–4 | 6 min |
| Notes | 5–7 | 10 min |
| Example 1 | 8–9 | 3 min |
| Your Turn | 10–11 | 3 min |
| Example 2 | 12–13 | 4 min |
| Your Turn | 14–15 | 3 min |
| Whiteboards | 16–33 | 12 min |
| Independent Set | 34 | 6 min |
| IXL | 35 | 5 min |
| **Total** | | **53 min** |

### Say these three things out loud today

1. The three interior angles of a triangle add to 180°.
2. An exterior angle equals the two remote interior angles added — not the angle beside it.
3. With expressions: write the sum or the theorem, solve, put x back, check 180.

## Slide by Slide

1. **Interior and Exterior Angles of a Triangle.** Post the learning target.
2. **Warm-Up — On your own. Four minutes. No notes.** Q4 is Notes I before it is said: three angles along a straight line make 180°. *1 retrieves T-B1 — supplementary: the two make 180 (yesterday). 2 retrieves 6.05 — a slanted distance by its triangle (last week). 3 retrieves 5.10–11 — a unit rate (last unit). 4 retrieves MA.4.GR.1.3 — angle measures add; three angles along a straight line make 180° (prior grade — the prerequisite this lesson needs).*
3. **Warm-Up — Answers.** Reveal. 1. 97° 2. 10 units 3. $0.75 4. 70
4. **Vocabulary — The new words, before they are used.** Read each word and its meaning once; the notes use them next.
5. **Notes — Copy the rule and the example.** Inside a triangle, the three make 180.
6. **Notes.** Outside equals the two far ones added.
7. **Notes.** Write the sum, solve, put x back, check 180.
8. **Example 1 — The third angle.** Ask what the three angles of any triangle add to before anyone subtracts: 180.
9. **Example 1 — Worked.** 58 + 71 = 129, and 180 − 129 = 51° — a board with 231 used 360; one with 129 stopped at the sum.
10. **Your Turn — Same steps, your numbers. Boards up when done.** 39 + 84 = 123, so 180 − 123 = 57° — a board with 123 stopped at the sum.
11. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
12. **Example 2 — An exterior angle.** Ask which two angles are remote — not touching the exterior angle: the 64° and the 49°.
13. **Example 2 — Worked.** The Exterior Angle Theorem: x = 64 + 49 = 113°.
14. **Your Turn — Same steps, your numbers. Boards up when done.** 72 + 35 = 107° — a board with 73 found the interior angle beside it.
15. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).

**16–17. Board 1.** Two angles of a triangle are 47° and 68°. Find the third angle. → ***65°*** *115 — stopped at the sum [Math Nation names misremembering the sum]; 245 — used 360 [the Triangle Sum is 180]*

**18–19. Board 2.** An isosceles triangle's two base angles are equal. Each is 52°. Find the third angle. → ***76°*** *128 — used one 52 [the two base angles are equal: both count]; 52 — took all three as equal [isosceles: two equal, not three]*

**20–21. Board 3.** A triangle's remote interior angles are 38° and 77°. Find the exterior angle x. → ***115°*** *65 — the interior angle beside it, 180 − 115 [Notes II: the exterior angle is not the interior angle beside it]; 295 — the angle around the outside, 360 − 65 [Florida B1G-M, GR.1.5: the reflex angle]*

**22–23. Board 4.** A triangle's angles are 45°, 60° and 75°. What is the exterior angle beside the 75°? → ***B — 105°*** *(A) equals the adjacent interior angle [Notes II: the exterior angle is not the interior angle beside it]; (C) the angle around the outside, 360 − 75 [Florida B1G-M, GR.1.5: the exterior angle taken as the reflex angle]; (D) only one remote angle [the Exterior Angle Theorem adds both] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

**24–25. Board 5.** An exterior angle of a triangle is 130°. One remote interior angle is 55°. Find the other remote interior angle. → ***75°*** *185 — added the exterior and the remote angle [the theorem: the remote angles add TO the exterior one]; 50 — the interior angle beside the exterior one [Notes II: the exterior angle is not the interior angle beside it]*

**26–27. Board 6.** A triangle's angles are $x^\circ$, $3x^\circ$ and $(2x+12)^\circ$. Find each angle. → ***28°, 84° and 68°*** *28 alone — reported x, one angle of three [Math Nation names reporting x]*

**28–29. Board 7.** An exterior angle measures (5x + 4)°; its remote interior angles measure 52° and (3x)°. Which equation finds x? → ***C — 5x + 4 = 52 + 3x*** *(A) the exterior angle and one remote angle taken as a linear pair [Notes II: the exterior angle's linear pair is the interior angle beside it]; (B) one remote angle only [the theorem adds both]; (D) the exterior angle used as an interior angle in the Triangle Sum [Math Nation names writing the equation for the wrong relationship] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

**30–31. Board 8.** A triangle has angles of 41° and 66°, and its longest side is 9 cm. Find its third angle. → ***73°*** *The 9-cm side is not needed. 64 — subtracted the 9 as well [the extra number]; 107 — stopped at the sum*

**32–33. Board 9.** Why is an exterior angle of a triangle always bigger than either of its remote interior angles? Explain. → ***It is the two remote angles added, each more than 0°.*** *'Because it is on the outside' [no reason]; 'it is not always bigger' [the remote angles are each more than 0°]*

**34. Independent Set — Six questions. On your own, in writing.** Hand out the Independent Set; the same six questions are on the board.

**35. IXL — Last five minutes.** IXL: every listed skill is required, SmartScore 67.

**Variation.** Boards 1–2: the third angle, then an isosceles triangle; 3 is an exterior angle; 4 votes on the angle beside the 75°. The pattern breaks at 5 — a remote angle found backwards from the exterior one — then three expressions (6), which equation fits the words voted (7), a side length that is not an angle (8); 9 is written. The independent set: a third angle, an isosceles triangle's base angles, an exterior angle, a remote angle, three expressions, an equation from the theorem.

## Misconceptions to Watch

**Board 1.** 115 — stopped at the sum [Math Nation names misremembering the sum]

**Board 2.** 128 — used one 52 [the two base angles are equal: both count]

**Board 3.** 65 — the interior angle beside it, 180 − 115 [Notes II: the exterior angle is not the interior angle beside it]

**Board 4.** equals the adjacent interior angle [Notes II: the exterior angle is not the interior angle beside it]

**Board 5.** 185 — added the exterior and the remote angle [the theorem: the remote angles add TO the exterior one]

**Board 6.** 28 alone — reported x, one angle of three [Math Nation names reporting x]

**Board 7.** the exterior angle and one remote angle taken as a linear pair [Notes II: the exterior angle's linear pair is the interior angle beside it]

**Board 8.** 64 — subtracted the 9 as well [the extra number]

**Board 9.** 'Because it is on the outside' [no reason]

**Materials** Whiteboards and markers; one paper triangle to tear for Notes I, if there is time. Calculators allowed, printed nowhere.

# Interior and Exterior Angles of a Triangle

*Lesson Plan*

## 1. Lesson Information

| | | | |
|---|---|---|---|
| Course | Grade 7 Accelerated (1205050) | Unit | 6 — Relationships in Triangles |
| Lesson | Thread B · Day 2 — Interior and Exterior Angles of a Triangle | Date | _____ |
| Length | 53 minutes (one period) | Benchmark(s) | MA.8.GR.1.5 |

## 2. Standards

**MA.8.GR.1.5** Solve problems involving the relationships of interior and exterior angles of a triangle.

**Must.** Find a missing interior angle with the Triangle Sum Theorem; find an exterior angle, or a remote interior angle, with the Exterior Angle Theorem; write and solve the equation when the angles are expressions.

**Must not.** Angle sums of polygons (MA.8.GR.1.6); parallel lines and transversals; reflex angles (the angle around the outside is a named error, not an answer).

## 3. Mathematical Thinking and Reasoning Standards

*the evidence for each, from this period*

| MTR | What it asks | The evidence in this lesson |
|---|---|---|
| MA.K12.MTR.2.1 | Demonstrate understanding by representing problems in multiple ways. | Notes I–II: 180° as torn-off corners, as a sum and as a linear pair |
| MA.K12.MTR.4.1 | Engage in discussions that reflect on the mathematical thinking of self and others. | boards 4 and 7 split; the re-vote |
| MA.K12.MTR.5.1 | Use patterns and structure to help understand and connect mathematical concepts. | Notes II: the theorem from the triangle sum and the linear pair together |

## 4. Learning Target and Essential Question

**Learning target.** I can find a missing angle of a triangle, inside or outside, with the Triangle Sum and the Exterior Angle Theorem.

**Essential question.** Why do the three angles of every triangle add to the same number?

**Building on.** T-B1 — supplementary angles and the linear pair; Grade 4 (MA.4.GR.1.3) — angle measures add; Unit 1 — equations in one unknown.

**Working toward.** Unit 11 — equations and systems; Geometry — the angle sums of polygons and triangle proofs.

## 5. Sequence

*read from the deck; never typed*

| Segment | Min | What the teacher does | What students do |
|---|---|---|---|
| Settle in; post the learning target | 1 | Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet. | Copy the target. |
| Warm-Up | 6 | Two minutes silent, then reveal. One sentence of reteach per question at most. Then the new words, read once. | Four retrieval questions, alone, no notes. Then copy the new words. |
| Notes | 10 | Teach from the front; say the OFF line out loud. | Copy the notes; answer the checks aloud. |
| Example 1 | 3 | Work the example; name the wrong boards before they appear. | Watch, then work the Your Turn alone. |
| Your Turn | 3 | Circulate; reveal and name what a wrong board most likely did. | Work it alone; boards up on the cue. |
| Example 2 | 4 | Work the example; name the wrong boards before they appear. | Watch, then work the Your Turn alone. |
| Your Turn | 3 | Circulate; reveal and name what a wrong board most likely did. | Work it alone; boards up on the cue. |
| Whiteboards | 12 | One question at a time. Boards down until the cue, all up together; scan the back row first. | Answer on a board; hold it up on three. |
| Independent Set | 6 | Silent. Circulate and mark what you see; do not teach. | Six questions, written, alone. |
| IXL | 5 | Assign the day's IXL skills. | Start the skills; finish tonight. |
| **Total** | **53** | | |

## 6. Gradual Release

**I do.** The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the OFF line is said out loud rather than printed.

**We do.** Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

**You do.** The nine-question whiteboard round (12 minutes) and then the six-question independent set, silent and written.

## 7. Higher-Order Questions

| Question | DOK |
|---|---|
| Can a triangle have two right angles? Two obtuse angles? Why not? | DOK 2 |
| Why does an exterior angle equal the two remote interior angles? Use the linear pair. | DOK 3 |
| Each corner of a triangle has two exterior angles. Why are they equal? | DOK 2 |

## 8. Checks for Understanding

*each board, and what to do about it*

| # | What the board asks | Answer | If the board is wrong |
|---|---|---|---|
| 1 | Two angles of a triangle are 47° and 68°. Find the third angle. | ***65°*** | 115 — stopped at the sum [Math Nation names misremembering the sum]; 245 — used 360 [the Triangle Sum is 180] |
| 2 | An isosceles triangle's two base angles are equal. Each is 52°. Find the third angle. | ***76°*** | 128 — used one 52 [the two base angles are equal: both count]; 52 — took all three as equal [isosceles: two equal, not three] |
| 3 | A triangle's remote interior angles are 38° and 77°. Find the exterior angle x. | ***115°*** | 65 — the interior angle beside it, 180 − 115 [Notes II: the exterior angle is not the interior angle beside it]; 295 — the angle around the outside, 360 − 65 [Florida B1G-M, GR.1.5: the reflex angle] |
| 4 | A triangle's angles are 45°, 60° and 75°. What is the exterior angle beside the 75°? | ***B — 105°*** | (A) equals the adjacent interior angle [Notes II: the exterior angle is not the interior angle beside it]; (C) the angle around the outside, 360 − 75 [Florida B1G-M, GR.1.5: the exterior angle taken as the reflex angle]; (D) only one remote angle [the Exterior Angle Theorem adds both] |
| 5 | An exterior angle of a triangle is 130°. One remote interior angle is 55°. Find the other remote interior angle. | ***75°*** | 185 — added the exterior and the remote angle [the theorem: the remote angles add TO the exterior one]; 50 — the interior angle beside the exterior one [Notes II: the exterior angle is not the interior angle beside it] |
| 6 | A triangle's angles are $x^\circ$, $3x^\circ$ and $(2x+12)^\circ$. Find each angle. | ***28°, 84° and 68°*** | 28 alone — reported x, one angle of three [Math Nation names reporting x] |
| 7 | An exterior angle measures (5x + 4)°; its remote interior angles measure 52° and (3x)°. Which equation finds x? | ***C — 5x + 4 = 52 + 3x*** | (A) the exterior angle and one remote angle taken as a linear pair [Notes II: the exterior angle's linear pair is the interior angle beside it]; (B) one remote angle only [the theorem adds both]; (D) the exterior angle used as an interior angle in the Triangle Sum [Math Nation names writing the equation for the wrong relationship] |
| 8 | A triangle has angles of 41° and 66°, and its longest side is 9 cm. Find its third angle. | ***73°*** | 64 — subtracted the 9 as well [the extra number]; 107 — stopped at the sum |
| 9 | Why is an exterior angle of a triangle always bigger than either of its remote interior angles? Explain. | ***It is the two remote angles added, each more than 0°.*** | 'Because it is on the outside' [no reason]; 'it is not always bigger' [the remote angles are each more than 0°] |

**MTR.4.1 — the one peer move.** When a board splits — no option holding about two-thirds of the room — sixty seconds of "convince the person next to you", then re-vote, then reveal. The minute comes out of the whiteboard remainder.

## 9. Differentiation

**ESE / 504.** Two lines on the desk: inside, the three make 180; outside, the two far ones added. Shade the two remote angles before writing the equation.

**ELL.** **Interior**: inside. **Exterior**: outside. **Remote**: far away — not touching. Point to each with the word.

**Enrichment.** A triangle's angles are in the ratio 2 : 3 : 4. Find each angle, then the exterior angle at the largest one. (40°, 60°, 80°; 100°.)

## 10. Closure, Vocabulary, Materials, Practice

**Closure.** Board 9 is written work and it is the exit evidence: an exterior angle is the two remote interior angles added, and each of them is more than 0°, so the sum is bigger than either one. A board that says only 'because it is outside' has the claim and not the reason.

**Vocabulary.** **interior angle**; **exterior angle**; **remote interior angles**; **Triangle Sum Theorem**

**Materials.** Whiteboards and markers; one paper triangle to tear for Notes I, if there is time. Calculators allowed, printed nowhere.

**Practice.** The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due by the Unit 11 test: Find missing angles in triangles — JFJ; Exterior Angle Theorem — FMP; Triangle Angle-Sum Theorem — 6Q6.

# Unit 6 Review

*Relationships in Triangles*

Name_____
Date_____
Period_____

*Show your work. Circle your final answer.*

*HOW TO USE THIS. This review covers every learning target of Lessons 1–8 — more than the test does. Work it in parts. If you miss two or more in a part, that part is where to spend your study time. The reference sheet has everything you need for each part. Thread B's two angle days come after the test and are not on it.*

## PART A. The Triangle Inequality and the Converse

*Lessons 1–2 · 4 questions*

**1.** Can the three lengths be the sides of a triangle? Show the test.

**a.** 5 cm, 9 cm and 15 cm

**b.** 6 inches, 10 inches and 16 inches

**c.** 11 m, 4 m and 8 m

**2.** Two sides of a triangle are 6 ft and 14 ft. Write an inequality for every possible length x of the third side.

**3.** Is a triangle with these sides a right triangle? Show the test.

**a.** 65, 33 and 56

**b.** 7, 8 and 11

**4.** Select ALL of the sets of side lengths that make a right triangle.

- ☐ A. 6, 7, 8
- ☐ B. 10, 24, 26
- ☐ C. 3, 4, 7
- ☐ D. 8, 9, 12
- ☐ E. 16, 30, 34

## PART B. The Pythagorean Theorem

*Lesson 3 · 6 questions*

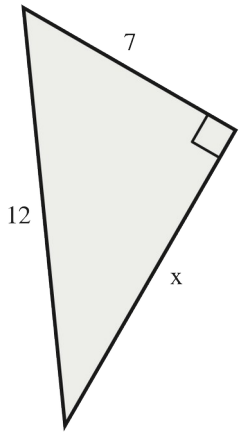
**5.** A right triangle has legs of 12 ft and 5 ft. How long is the hypotenuse?

**6.** A right triangle has legs of 3 inches and 13 inches. How long is the hypotenuse, exactly and to the nearest tenth?

**7.** A right triangle has a hypotenuse of 85 cm and one leg of 84 cm. How long is the other leg?

**8.** A right triangle has a hypotenuse of 11 ft and one leg of 8 ft. How long is the other leg, exactly and to the nearest tenth?

**9.** Which equation can be used to find x?




- ○ A. $7^2 + 12^2 = x^2$
- ○ B. $7 + x = 12$
- ○ C. $7^2 + x^2 = 12^2$
- ○ D. $7^2 + x^2 = 2 \cdot 12$

**10.** Ana finds the missing leg of a right triangle whose hypotenuse is 5 cm and whose other leg is 3 cm. She writes $3^2 + 5^2 = 34$ , so the leg is about 5.8 cm. What did she do wrong? Find the leg.

## PART C. The Pythagorean Theorem in Real-World Problems

*Lesson 4 · 5 questions*

**11.** A 10-ft ladder leans against a wall with its foot 6 ft from the wall. How high up the wall does it reach?

**12.** A rectangular garden is 10 m long and 11 m wide. How long is a path straight across it from corner to corner, to the nearest tenth of a meter?

**13.** A ramp rises 1 ft over a run of 9 ft along the ground. How long is the ramp, to the nearest tenth of a foot?

**14.** A park is a rectangle 9 m by 11 m. How much shorter is it to walk straight across the park from corner to corner than to walk along two of its sides? Give the answer to the nearest tenth.

**15.** A 9-ft ladder stands with its foot 2 ft from a wall. Will it reach a window ledge 8.5 ft up the wall?

## PART D. Distance on the Coordinate Plane

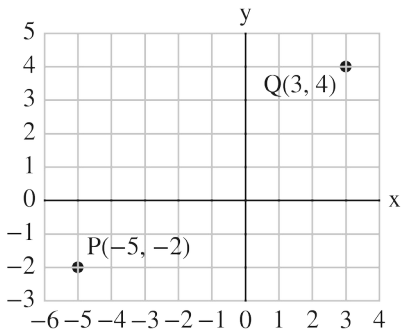
*Lesson 5 · 5 questions*

**16.** Find the distance between the two points.

**a.** $(-6, 2)$ and $(5, 2)$

**b.** $(3, -7)$ and $(3, 2)$

**17.** Find the distance between points P and Q.




**18.** Find the distance between (−6, −1) and (1, 5), to the nearest tenth.

**19.** How far is the point (−1, 6) from the origin? Round your answer to the nearest tenth.

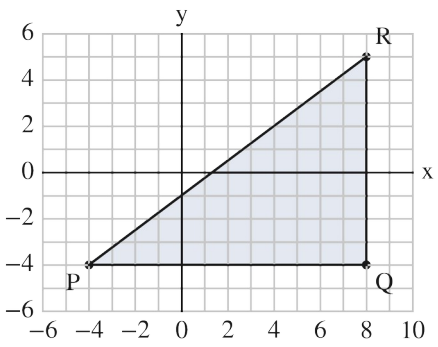
**20.** What is the distance between (−2, −6) and (−2, 3)?

- ○ A. 3 units
- ○ B. 9 units
- ○ C. 10 units
- ○ D. −9 units

## PART E. Coordinate-Plane Problems and Perimeter

*Lesson 6 · 4 questions*

**21.** Every part below is about triangle PQR on the grid, with corners P(−4, −4), Q(8, −4) and R(8, 5).




**a.** Find the length of PQ.

**b.** Is triangle PQR a right triangle? How do you know?

**c.** Find the length of PR.

**d.** A triangle similar to PQR has a shortest side of 3. How long is its longest side?

**22.** Find the perimeter of the triangle with corners (−4, 2), (6, 2) and (6, −1), to the nearest tenth.

**23.** On a town map in blocks, the school is at (−2, −4) and the park is at (2, 1).

**a.** How far is it from the school to the park along the streets?

**b.** How far is it in a straight line? Round your answer to the nearest tenth.

**24.** A rectangle has corners (−3, −2), (5, −2), (5, 4) and (−3, 4).

**a.** Find its perimeter.

**b.** Find the length of one of its diagonals.

## PART F. Similar Triangles and Indirect Measurement

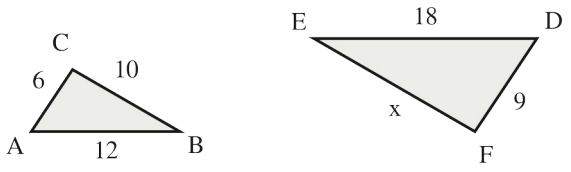
*Lessons 7–8 · 6 questions*

**25.** Are the two triangles similar? Show the three ratios.

**a.** Sides 5, 7, 8 and sides 15, 21, 24

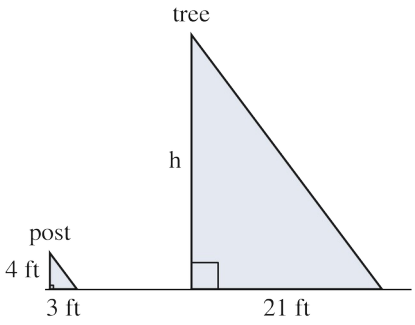
**b.** Sides 5, 8, 10 and sides 10, 16, 18

**26.** Triangle ABC is similar to triangle DEF. Find x.



**27.** A triangle has sides 15, 20 and 30. A similar triangle's middle side is 8. Find its other two sides.

**28.** A 4-ft fence post casts a 3-ft shadow. At the same moment, a tree casts a 21-ft shadow. How tall is the tree?




**29.** Kai's eyes are 5 ft above the ground. He stands 3 ft from a mirror on the ground and sees in it the top of a pole that stands 12 ft beyond the mirror. The two right triangles are similar. How tall is the pole?

**30.** Which triangle is similar to one with sides 6, 8 and 9?

- ○ A. 8, 10, 11
- ○ B. 12, 16, 20
- ○ C. 9, 12, 13.5
- ○ D. 6, 8, 12

GRADE 7 ACCELERATED · UNIT 6

# Unit 6 Review

*Relationships in Triangles* **ANSWER KEY**

Name
Date
Period

*Show your work. Circle your final answer.*

*HOW TO USE THIS. This review covers every learning target of Lessons 1–8 — more than the test does. Work it in parts. If you miss two or more in a part, that part is where to spend your study time. The reference sheet has everything you need for each part. Thread B's two angle days come after the test and are not on it.*

## PART A. The Triangle Inequality and the Converse

*Lessons 1–2 · MA.8.GR.1.3 · 4 questions*

**1.** Can the three lengths be the sides of a triangle? Show the test.

**a.** 5 cm, 9 cm and 15 cm

***No — 5 + 9 = 14 is less than 15.***

*the two shorter sides against the longest*

**b.** 6 inches, 10 inches and 16 inches

***No — 6 + 10 = 16 is equal to the longest side, so the sides lie flat.***

*equal is not more*

**c.** 11 m, 4 m and 8 m

***Yes — 4 + 8 = 12 is more than 11.***

*the longest is listed first; 11 + 4 tests the wrong pair*

**2.** Two sides of a triangle are 6 ft and 14 ft. Write an inequality for every possible length x of the third side.

***$8 < x < 20$***

*between the difference, 14 − 6, and the sum, 14 + 6 — never equal to either*

**3.** Is a triangle with these sides a right triangle? Show the test.

**a.** 65, 33 and 56

***Yes — $33^2 + 56^2 = 1{,}089 + 3{,}136 = 4{,}225 = 65^2$.***

*the longest, 65, is c, though it is listed first*

**b.** 7, 8 and 11

***No — $49 + 64 = 113$, and $11^2 = 121$.***

*a triangle (7 + 8 > 11), but not a right one*

**4.** Select ALL of the sets of side lengths that make a right triangle.

- ☐ A. 6, 7, 8
- ☒ **B. 10, 24, 26**
- ☐ C. 3, 4, 7
- ☐ D. 8, 9, 12
- ☒ **E. 16, 30, 34**

***B and E***

*$100 + 576 = 676 = 26^2$ and $256 + 900 = 1{,}156 = 34^2$: both, and nothing else.*

## PART B. The Pythagorean Theorem

*Lesson 3 · MA.8.GR.1.1 · 6 questions*

**5.** A right triangle has legs of 12 ft and 5 ft. How long is the hypotenuse?

***13 ft***

*144 + 25 = 169*

**6.** A right triangle has legs of 3 inches and 13 inches. How long is the hypotenuse, exactly and to the nearest tenth?

***√178 in ≈ 13.3 in***

*9 + 169 = 178*

**7.** A right triangle has a hypotenuse of 85 cm and one leg of 84 cm. How long is the other leg?

***13 cm***

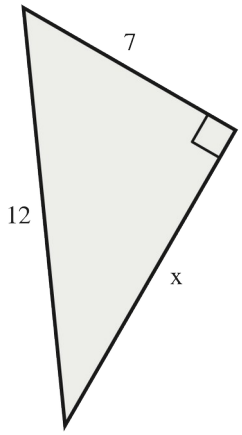
*7,225 − 7,056 = 169*

**8.** A right triangle has a hypotenuse of 11 ft and one leg of 8 ft. How long is the other leg, exactly and to the nearest tenth?

***√57 ft ≈ 7.5 ft***

*121 − 64 = 57*

**9.** Which equation can be used to find x?




- ○ A. $7^2 + 12^2 = x^2$
- ○ B. $7 + x = 12$
- ● **C.** $7^2 + x^2 = 12^2$
- ○ D. $7^2 + x^2 = 2 \cdot 12$

***C***

*The 12 is across from the right angle, so it is c and stands alone: $x^2 = 144 - 49 = 95$.*

**10.** Ana finds the missing leg of a right triangle whose hypotenuse is 5 cm and whose other leg is 3 cm. She writes $3^2 + 5^2 = 34$, so the leg is about 5.8 cm. What did she do wrong? Find the leg.

***She added the squares, and a leg cannot be longer than the 5-cm hypotenuse. Subtract: 25 − 9 = 16, so the leg is 4 cm.***

*the error named and the leg found*

## PART C. The Pythagorean Theorem in Real-World Problems

*Lesson 4 · MA.8.GR.1.1 · 5 questions*

**11.** A 10-ft ladder leans against a wall with its foot 6 ft from the wall. How high up the wall does it reach?

***8 ft***

*the ladder is c: 100 − 36 = 64*

**12.** A rectangular garden is 10 m long and 11 m wide. How long is a path straight across it from corner to corner, to the nearest tenth of a meter?

***About 14.9 m (√221 m)***

*the diagonal is c: 100 + 121 = 221*

**13.** A ramp rises 1 ft over a run of 9 ft along the ground. How long is the ramp, to the nearest tenth of a foot?

***About 9.1 ft (√82 ft)***

*the ramp is c: 1 + 81 = 82*

**14.** A park is a rectangle 9 m by 11 m. How much shorter is it to walk straight across the park from corner to corner than to walk along two of its sides? Give the answer to the nearest tenth.

***About 5.8 m***

*along two sides 20 m; across √202 ≈ 14.2 m*

**15.** A 9-ft ladder stands with its foot 2 ft from a wall. Will it reach a window ledge 8.5 ft up the wall?

***Yes — it reaches √77 ≈ 8.8 ft up the wall, above 8.5 ft.***

*the height is a leg: 81 − 4 = 77*

## PART D. Distance on the Coordinate Plane

*Lesson 5 · MA.8.GR.1.2 · 5 questions*

**16.** Find the distance between the two points.

**a.** (−6, 2) and (5, 2)

***11 units***

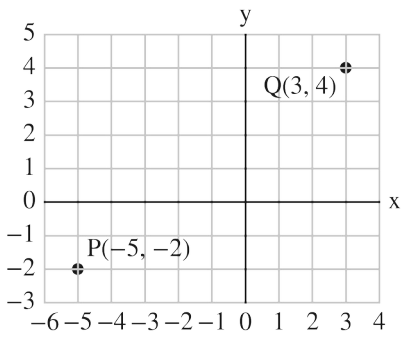
*across zero: 6 + 5*

**b.** (3, −7) and (3, 2)

***9 units***

*across zero: 7 + 2*

**17.** Find the distance between points P and Q.




***10 units***

*across 8, up 6: 64 + 36 = 100*

**18.** Find the distance between (−6, −1) and (1, 5), to the nearest tenth.

***About 9.2 units (√85)***

*across 7, up 6: 49 + 36 = 85*

**19.** How far is the point (−1, 6) from the origin? Round your answer to the nearest tenth.

***About 6.1 units (√37)***

*legs 1 and 6: 1 + 36 = 37*

**20.** What is the distance between (−2, −6) and (−2, 3)?

- ○ A. 3 units
- ● **B. 9 units**
- ○ C. 10 units
- ○ D. −9 units

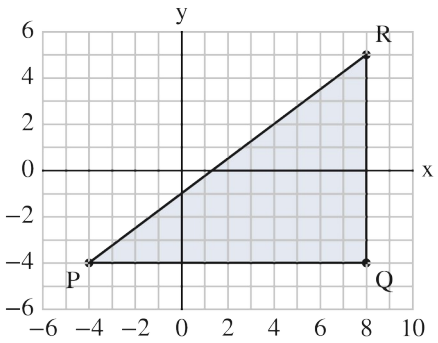
***B***

*3 − (−6) = 9: 6 to the axis and 3 past it.*

## PART E. Coordinate-Plane Problems and Perimeter

*Lesson 6 · MA.8.GR.1.2 · 4 questions*

**21. SSDD.** Every part below is about one triangle on the grid: P(−4, −4), Q(8, −4), R(8, 5). Same surface, four different questions: a distance on a line (across zero), whether the corner at Q is square, the slanted side as a hypotenuse, and a similar copy (which needs QR, the shortest side, first) — the unit's moves on one triangle, so that a student learns to read the question and not the picture.




**a.** Find the length of PQ.

***12 units***

*across zero: 4 + 8*

**b.** Is triangle PQR a right triangle? How do you know?

***Yes — PQ runs straight across and QR straight up, so they meet at a right angle at Q.***

*a horizontal side and a vertical side meet square (and $12^2 + 9^2 = 15^2$)*

**c.** Find the length of PR.

***15 units***

*the hypotenuse of legs PQ = 12 and QR = 5 − (−4) = 9: 144 + 81 = 225*

**d.** A triangle similar to PQR has a shortest side of 3. How long is its longest side?

***5***

*QR = 9 is the shortest side: k = 3 ÷ 9 = 1/3, and 15 × 1/3 — a copy smaller than the original*

**22.** Find the perimeter of the triangle with corners (−4, 2), (6, 2) and (6, −1), to the nearest tenth.

***About 23.4 units***

*10 + 3 + √109; √109 ≈ 10.440 — round once, at the end*

**23.** On a town map in blocks, the school is at (−2, −4) and the park is at (2, 1).

**a.** How far is it from the school to the park along the streets?

***9 blocks***

*4 across + 5 up*

**b.** How far is it in a straight line? Round your answer to the nearest tenth.

***About 6.4 blocks (√41)***

*16 + 25 = 41*

**24.** A rectangle has corners (−3, −2), (5, −2), (5, 4) and (−3, 4).

**a.** Find its perimeter.

***28 units***

*8 + 6 + 8 + 6*

**b.** Find the length of one of its diagonals.

***10 units***

*64 + 36 = 100*

## PART F. Similar Triangles and Indirect Measurement

*Lessons 7–8 · MA.8.GR.2.4 · 6 questions*

**25.** Are the two triangles similar? Show the three ratios.

**a.** Sides 5, 7, 8 and sides 15, 21, 24

***Yes — 15/5 = 21/7 = 24/8 = 3.***

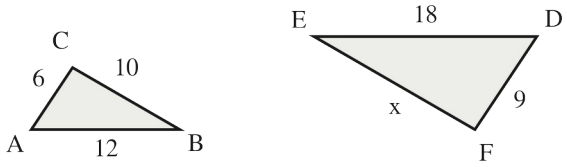
*one k for every side*

**b.** Sides 5, 8, 10 and sides 10, 16, 18

***No — 10/5 = 16/8 = 2, but 18/10 = 1.8.***

*two equal ratios are not enough*

**26.** Triangle ABC is similar to triangle DEF. Find x.




***x = 15***

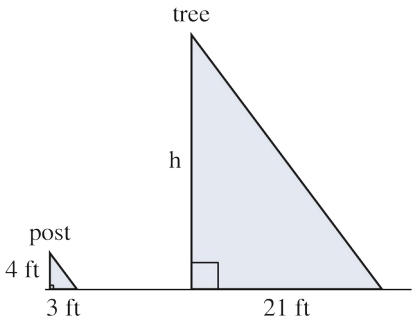
*The names match AC with DF (6 with 9) and BC with EF, where x is: k = 1.5 and x = 10 × 1.5.*

**27.** A triangle has sides 15, 20 and 30. A similar triangle's middle side is 8. Find its other two sides.

***6 and 12***

*k = 8 ÷ 20 = 0.4 — the copy is smaller*

**28.** A 4-ft fence post casts a 3-ft shadow. At the same moment, a tree casts a 21-ft shadow. How tall is the tree?




***28 ft***

*k = 21 ÷ 3 = 7 from the shadows; 4 × 7*

**29.** Kai's eyes are 5 ft above the ground. He stands 3 ft from a mirror on the ground and sees in it the top of a pole that stands 12 ft beyond the mirror. The two right triangles are similar. How tall is the pole?

***20 ft***

*k = 12 ÷ 3 = 4; 5 × 4*

**30.** Which triangle is similar to one with sides 6, 8 and 9?

- ○ A. 8, 10, 11
- ○ B. 12, 16, 20
- ● **C. 9, 12, 13.5**
- ○ D. 6, 8, 12

***C***

*Every side × 1.5.*

GRADE 7 ACCELERATED · UNIT 6 · PRACTICE TEST

# Relationships in Triangles

*18 points*

Name_____
Date_____
Period_____

*Show your work. One point for each lettered part.*

## Part 1. Triangle Inequality and the Converse

*3 points*

**1.** Which set of side lengths can form a triangle? *1 pt*

- ○ A. 4 cm, 7 cm, 11 cm
- ○ B. 12 cm, 5 cm, 6 cm
- ○ C. 6 cm, 8 cm, 13 cm
- ○ D. 3 cm, 10 cm, 15 cm

**2.** Select ALL of the sets of side lengths that make a right triangle. *1 pt*

- ☐ A. 17, 8, 15
- ☐ B. 5, 6, 7
- ☐ C. 20, 21, 29
- ☐ D. 7, 24, 26
- ☐ E. 4, 6, 8

**3.** Two sides of a triangle are 6 inches and 15 inches long. Which inequality shows every possible length x, in inches, of the third side? *1 pt*

- ○ A. $6 < x < 15$
- ○ B. $9 < x < 21$
- ○ C. $9 \leq x \leq 21$
- ○ D. $0 < x < 21$

## Part 2. The Pythagorean Theorem *6 points*

*If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.*

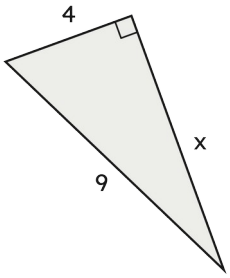
**4.** Find the missing side of each right triangle.

**a.** Its legs are 9 inches and 12 inches. How long is the hypotenuse? *1 pt*

**b.** Its hypotenuse is 61 meters and one leg is 60 meters. How long is the other leg? *1 pt*

**5.** A right triangle has legs of 7 centimeters and 13 centimeters. How long is the hypotenuse, in centimeters? Round your answer to the nearest tenth. *1 pt*

**6.** Which equation can be used to find x? *1 pt*




- ○ A. $4 + x = 9$
- ○ B. $4^2 + 9^2 = x^2$
- ○ C. $4^2 + x^2 = 2 \cdot 9$
- ○ D. $4^2 + x^2 = 9^2$

**7.** A rectangular garden measures 9 meters by 10 meters. A path runs straight across it from one corner to the opposite corner.

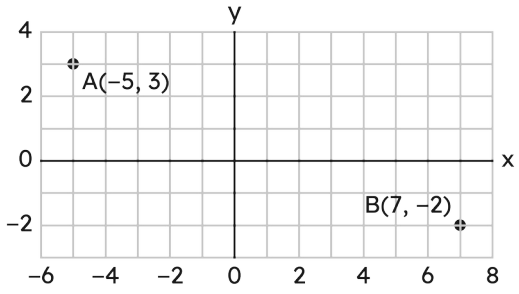
**a.** How long is the path, in meters? Round your answer to the nearest tenth. *1 pt*

**b.** How much shorter is the path than walking along two sides of the garden? Round your answer to the nearest tenth. *1 pt*

## Part 3. Distance on the Coordinate Plane *6 points*

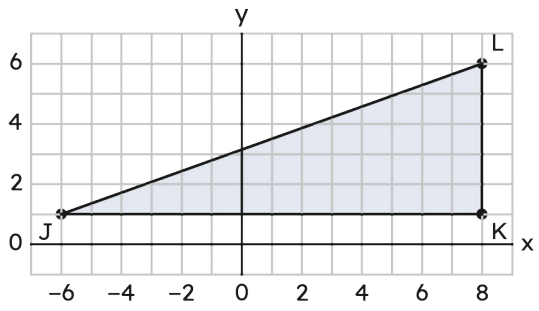
*If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.*

**8.** Find the distance between points A and B. *1 pt*




**9.** Find the distance between (−1, 3) and (4, 1). Round your answer to the nearest tenth. *1 pt*

**10.** Triangle JKL has corners J(−6, 1), K(8, 1) and L(8, 6).




**a.** Find the length of side JK. *1 pt*

**b.** Find the perimeter of triangle JKL. Round your answer to the nearest tenth. *1 pt*

**11.** On a town map, each unit is one block. The library is at (−1, −5) and the pool is at (8, 7).

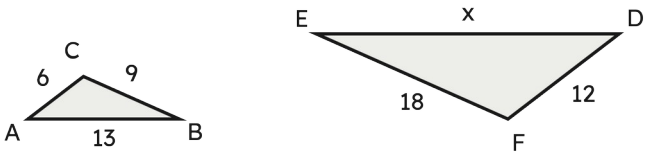
**a.** How far is it from the library to the pool along the streets? *1 pt*

**b.** How far is it in a straight line? *1 pt*

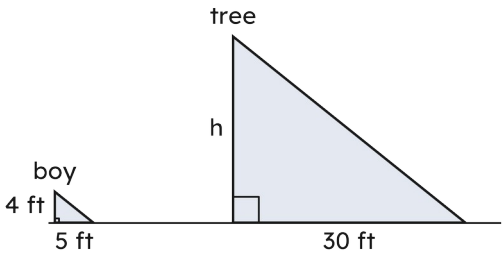
## Part 4. Similar Triangles

*3 points*

**12.** Triangle ABC is similar to triangle DEF. Find x. *1 pt*




**13.** A boy 4 ft tall casts a shadow 5 ft long. At the same moment, a tree casts a shadow 30 ft long. How tall is the tree? *1 pt*




**14.** Which triangle is similar to a triangle with sides 6, 7 and 10? *1 pt*

- ○ A. 12, 14, 20
- ○ B. 10, 11, 14
- ○ C. 12, 14, 18
- ○ D. 6, 7, 12

GRADE 7 ACCELERATED · UNIT 6 · PRACTICE TEST

# Relationships in Triangles

*18 points* **ANSWER KEY**

Name
Date
Period

*Show your work. One point for each lettered part.*

## Part 1. Triangle Inequality and the Converse

*3 points MA.8.GR.1.3*

**1.** Which set of side lengths can form a triangle? *1 pt*

- ○ A. 4 cm, 7 cm, 11 cm
- ○ B. 12 cm, 5 cm, 6 cm
- ● **C. 6 cm, 8 cm, 13 cm**
- ○ D. 3 cm, 10 cm, 15 cm

***C***

*The two shorter sides against the longest: 6 + 8 = 14, more than 13. A lies flat; B and D pass only when the wrong pair is added.*

*Scoring: 1 point.*

**2.** Select ALL of the sets of side lengths that make a right triangle. *1 pt*

- ☒ **A. 17, 8, 15**
- ☐ B. 5, 6, 7
- ☒ **C. 20, 21, 29**
- ☐ D. 7, 24, 26
- ☐ E. 4, 6, 8

***A and C***

*Square all three and test the longest as c: 64 + 225 = 289 = $17^2$ (the longest listed first) and 400 + 441 = 841 = $29^2$. Both and nothing else for the point.*

*Scoring: 1 point.*

**3.** Two sides of a triangle are 6 inches and 15 inches long. Which inequality shows every possible length x, in inches, of the third side? *1 pt*

- ○ A. $6 < x < 15$
- ● **B.** $9 < x < 21$
- ○ C. $9 \le x \le 21$
- ○ D. $0 < x < 21$

***B***

*15 − 6 = 9 and 15 + 6 = 21, never equal to either.*

*Scoring: 1 point.*

## Part 2. The Pythagorean Theorem

*6 points MA.8.GR.1.1*

*If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.*

**4.** Find the missing side of each right triangle.

**a.** Its legs are 9 inches and 12 inches. How long is the hypotenuse? *1 pt*

***15 in***

*$c^2 = 81 + 144 = 225$*

**b.** Its hypotenuse is 61 meters and one leg is 60 meters. How long is the other leg? *1 pt*

***11 m***

*a leg: subtract — 3,721 − 3,600 = 121. Adding the squares gives a 'leg' longer than the hypotenuse.*

*Scoring: a 1 · b 1 = 2 points.*

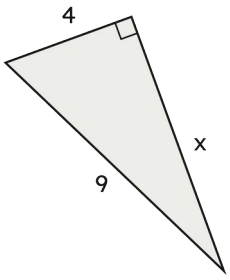
**5.** A right triangle has legs of 7 centimeters and 13 centimeters. How long is the hypotenuse, in centimeters? Round your answer to the nearest tenth. *1 pt*

***14.8 cm (exactly √218 cm)***

*$c^2 = 49 + 169 = 218$; √218 = 14.76…*

*Scoring: 1 point.*

**6.** Which equation can be used to find x? *1 pt*




- ○ A. $4 + x = 9$
- ○ B. $4^2 + 9^2 = x^2$
- ○ C. $4^2 + x^2 = 2 \cdot 9$
- ● **D.** $4^2 + x^2 = 9^2$

***D***

*The 9 is across from the right angle, so it is c and stands alone: $x^2 = 81 − 16 = 65$.*

*Scoring: 1 point.*

**7.** A rectangular garden measures 9 meters by 10 meters. A path runs straight across it from one corner to the opposite corner.

**a.** How long is the path, in meters? Round your answer to the nearest tenth. *1 pt*

***13.5 m  (exactly √181 m)***

*the path is the hypotenuse: 81 + 100 = 181*

**b.** How much shorter is the path than walking along two sides of the garden? Round your answer to the nearest tenth. *1 pt*

***5.5 m***

*9 + 10 = 19 m along two sides; 19 − 13.5. Follow-through on part (a).*

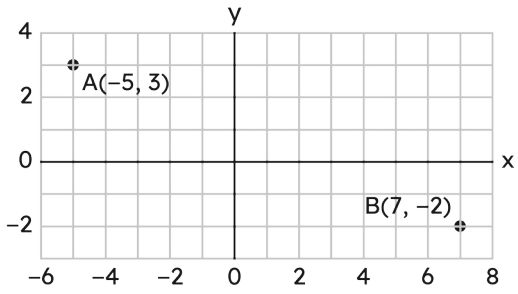
*Scoring: a 1 · b 1 = 2 points.*

## Part 3. Distance on the Coordinate Plane

*6 points  MA.8.GR.1.2*

*If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.*

**8.** Find the distance between points A and B. *1 pt*




***13 units***

*Across 7 − (−5) = 12, down 3 − (−2) = 5: $c^2$ = 144 + 25 = 169.*

*Scoring: 1 point.*

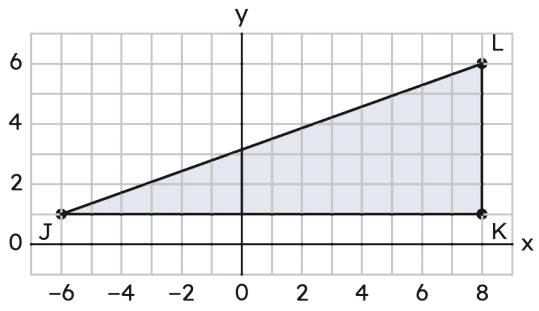
**9.** Find the distance between (−1, 3) and (4, 1). Round your answer to the nearest tenth. *1 pt*

***5.4 units  (exactly √29)***

*Across 5, down 2: $c^2$ = 25 + 4 = 29.*

*Scoring: 1 point.*

**10.** Triangle JKL has corners J(−6, 1), K(8, 1) and L(8, 6).




**a.** Find the length of side JK. *1 pt*

***14 units***

*Across zero: 6 + 8 — never 8 − 6.*

**b.** Find the perimeter of triangle JKL. Round your answer to the nearest tenth. *1 pt*

***33.9 units***

*KL = 5; JL = √(196 + 25) = √221 ≈ 14.866; 14 + 5 + 14.866 = 33.866 — round once, at the end. Follow-through on part (a).*

*Scoring: a 1 · b 1 = 2 points.*

**11.** On a town map, each unit is one block. The library is at (−1, −5) and the pool is at (8, 7).

**a.** How far is it from the library to the pool along the streets? *1 pt*

***21 blocks***

*The streets follow the grid: 9 across and 12 up.*

**b.** How far is it in a straight line? *1 pt*

***15 blocks***

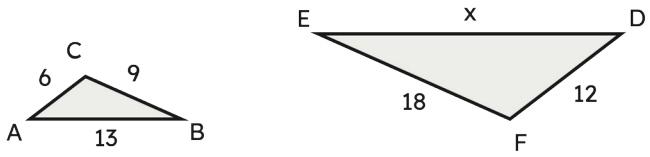
*The hypotenuse of legs 9 and 12: 81 + 144 = 225.*

*Scoring: a 1 · b 1 = 2 points.*

## Part 4. Similar Triangles

*3 points MA.8.GR.2.4*

**12.** Triangle ABC is similar to triangle DEF. Find x. *1 pt*

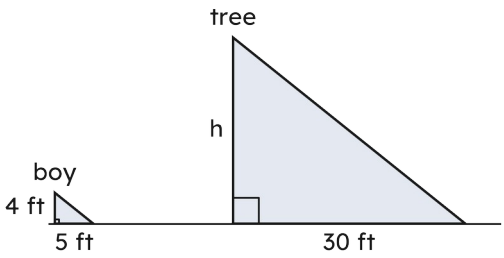



***x = 26***

*The names match AC with DF and BC with EF — 12 with 6 and 18 with 9, as by size — so k = 2, and x is on DE, which matches AB: 13 × 2. 39 matched by place on the page (18 with the 6); 13 took similar as equal; 19 added the difference 12 − 6.*

*Scoring: 1 point.*

**13.** A boy 4 ft tall casts a shadow 5 ft long. At the same moment, a tree casts a shadow 30 ft long. How tall is the tree? *1 pt*




***24 ft***

*k = 30 ÷ 5 = 6 from the shadows; 4 × 6. 37.5 matched height and shadow crosswise; 120 multiplied by the boy's height, not by k.*

*Scoring: 1 point.*

**14.** Which triangle is similar to a triangle with sides 6, 7 and 10? *1 pt*

- ● **A. 12, 14, 20**
- ○ B. 10, 11, 14
- ○ C. 12, 14, 18
- ○ D. 6, 7, 12

***A***

*Every side × 2: 12/6 = 14/7 = 20/10.*

*Scoring: 1 point.*

## Score Tracker

| Benchmark | Questions | Possible | Earned |
|---|---|---|---|
| MA.8.GR.1.3 | 1, 2, 3 | 3 | |

| Benchmark | Questions | Possible | Earned |
|---|---|---|---|
| MA.8.GR.1.1 | 4, 5, 6, 7 | 6 | |
| MA.8.GR.1.2 | 8, 9, 10, 11 | 6 | |
| MA.8.GR.2.4 | 12, 13, 14 | 3 | |
| **Total** | | **18** | |

*If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page. Nothing is weighed and nothing is arguable: apply the part's operation to the student's own earlier value and tick if that is what is written.*

GRADE 7 ACCELERATED · UNIT 6 · LESSONS 1–2

# Triangle Inequality Theorem; Converse of the Pythagorean Theorem

*Independent Set 6.01* **ANSWER KEY**

Name
Date
Period

*Six questions, on your own, in silence. Show the step that does the work.*

**1.** Can sides of 4 cm, 8 cm and 13 cm make a triangle?

***No — 4 + 8 = 12 is less than 13.***

**2.** Can sides of 9 inches, 9 inches and 18 inches make a triangle?

***No — 9 + 9 = 18 lies flat.***

*The equal case is not a triangle.*

**3.** Two sides of a triangle are 5 m and 11 m. Write an inequality for the length x, in meters, of the third side.

***6 < x < 16***

*11 − 5 and 11 + 5; never equal.*

**4.** Do sides of 28 cm, 45 cm and 53 cm make a right triangle?

***Yes — 784 + 2,025 = 2,809 = $53^2$.***

**5.** Do sides of 17 ft, 10 ft and 14 ft make a right triangle?

***No — $10^2 + 14^2 = 296$, but $17^2 = 289$.***

*The longest side is listed first.*

**6.** A carpenter joins boards of 30 inches, 40 inches and 50 inches into a triangle. Is one corner a right angle?

***Yes — 900 + 1,600 = 2,500 = $50^2$.***

# Triangle Inequality Theorem; Converse of the Pythagorean Theorem

*Question Bank 6.01* **ANSWER KEY**

Name
Date
Period

*Show your work. Circle your final answer.*

**For questions 1 through 3, can the three lengths make a triangle?**

**1.** 3 m, 9 m, 11 m

***Yes — 3 + 9 = 12 > 11***

**2.** 5 inches, 13 inches, 7 inches

***No — 5 + 7 = 12 < 13***

*The longest is listed second.*

**3.** 8 ft, 4 ft, 4 ft

***No — 4 + 4 = 8 lies flat***

*The equal case.*

**For questions 4 and 5, two sides of a triangle are given. Write an inequality for the third side, x.**

**4.** 3 cm and 8 cm

***5 < x < 11 (centimeters)***

**5.** 1 cm and 5 cm; the third side is a whole number of centimeters — which one?

***4 < x < 6, so x = 5 cm — the only whole number between***

**For questions 6 through 8, do the lengths make a right triangle?**

**6.** 4.5 cm, 6 cm, 7.5 cm

***Yes — $20.25 + 36 = 56.25 = 7.5^2$***

**7.** 26 inches, 24 inches, 10 inches

***Yes — $10^2 + 24^2 = 676 = 26^2$***

*The longest is listed first.*

**8.** 7 m, 10 m, 12 m

***No — 49 + 100 = 149, not 144***

**Choose one.**

**9.** Which set of lengths makes a right triangle?

- ○ A. 4, 5, 6
- ○ B. 2, 3, 5
- ○ C. 6, 8, 12
- ● **D. 11, 60, 61**

***D***

*121 + 3,600 = 3,721 = $61^2$.*

# Triangle Inequality Theorem; Converse of the Pythagorean Theorem

*Question Bank - Additional 6.01* **ANSWER KEY**

Name
Date
Period

*Show your work. Circle your final answer.*

**For questions 1 through 3, can the three lengths make a triangle?**

**1.** 4 m, 10 m, 12 m

***Yes — 4 + 10 = 14 > 12***

**2.** 6 inches, 17 inches, 9 inches

***No — 6 + 9 = 15 < 17***

*The longest is listed second.*

**3.** 14 ft, 7 ft, 7 ft

***No — 7 + 7 = 14 lies flat***

*The equal case.*

**For questions 4 and 5, two sides of a triangle are given. Write an inequality for the third side, x.**

**4.** 6 cm and 9 cm

***3 < x < 15 (centimeters)***

**5.** 1 inch and 8 inches; the third side is a whole number of inches — which one?

***7 < x < 9, so x = 8 in — the only whole number between***

**For questions 6 through 8, do the lengths make a right triangle?**

**6.** 2.4 m, 3.2 m, 4 m

***Yes — 5.76 + 10.24 = 16 = $4^2$***

**7.** 39 inches, 36 inches, 15 inches

***Yes — $15^2 + 36^2 = 1{,}521 = 39^2$***

*The longest is listed first.*

**8.** 8 m, 9 m, 12 m

***No — 64 + 81 = 145, not 144***

**Choose one.**

**9.** Which set of lengths makes a right triangle?

- ○ A. 5, 6, 7
- ● **B. 14, 48, 50**
- ○ C. 9, 12, 16
- ○ D. 3, 4, 7

***B***

*196 + 2,304 = 2,500 = $50^2$.*

GRADE 7 ACCELERATED · UNIT 6 · LESSON 3

# Pythagorean Theorem

*Independent Set 6.03* **ANSWER KEY**

Name
Date
Period

*Six questions, on your own, in silence. Show the step that does the work.*

**1.** A right triangle has legs of 12 m and 5 m. How long is the hypotenuse?

***13 m***

*144 + 25 = 169 = $13^2$.*

**2.** A right triangle has legs of 3 ft and 8 ft. How long is the hypotenuse, exactly and to the nearest tenth?

***√73 ft ≈ 8.5 ft***

*9 + 64 = 73.*

**3.** A right triangle has a hypotenuse of 39 cm and one leg of 36 cm. How long is the other leg?

***15 cm***

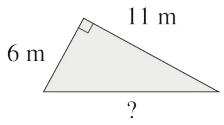
*1,521 − 1,296 = 225 = $15^2$.*

**4.** A right triangle has a hypotenuse of 11 inches and one leg of 4 inches. How long is the other leg, exactly and to the nearest tenth?

***√105 in ≈ 10.2 in***

*121 − 16 = 105.*

**5.** A right triangle has legs of 6 m and 11 m (drawn turned on its side on the paper). How long is its hypotenuse, to the nearest tenth?




***√157 m ≈ 12.5 m***

*36 + 121 = 157; c is across from the square corner.*

**6.** A right triangle has a hypotenuse of 25 yd and one leg of 24 yd. The other leg is x. Write the equation that finds x, then find x.

***$24^2 + x^2 = 25^2$; x = 7 yd***

*625 − 576 = 49 = $7^2$.*

GRADE 7 ACCELERATED · UNIT 6 · LESSON 3

# Pythagorean Theorem

*Question Bank 6.03* **ANSWER KEY**

Name
Date
Period

*Show your work. Circle your final answer.*

**For questions 1 through 3, find the hypotenuse. Give it exactly, and to the nearest tenth when it is not a whole number.**

**1.** Legs of 8 cm and 6 cm

***10 cm***

**2.** Legs of 4 inches and 11 inches

***√137 in ≈ 11.7 in***

*16 + 121 = 137.*

**3.** Legs of 7 m and 8 m

***√113 m ≈ 10.6 m***

*49 + 64 = 113.*

**For questions 4 through 6, find the missing leg. Give it exactly, and to the nearest tenth when it is not a whole number.**

**4.** Hypotenuse 13 ft, one leg 12 ft

***5 ft***

*169 − 144 = 25.*

**5.** Hypotenuse 8 cm, one leg 5 cm

***√39 cm ≈ 6.2 cm***

*64 − 25 = 39.*

**6.** Hypotenuse 61 inches, one leg 60 inches

***11 in***

*3,721 − 3,600 = 121 = $11^2$.*

**7.** A rectangle is 5 inches wide and 12 inches long. How long is its diagonal?

***13 in***

*The diagonal is the hypotenuse of a 5-by-12 right triangle.*

**Choose one.**

**8.** A right triangle has a hypotenuse of 25 m and one leg of 24 m. How long is the other leg?

- ○ A. 1 m
- ○ B. 34.7 m
- ● **C. 7 m**
- ○ D. 49 m

***C***

*625 − 576 = 49 = $7^2$.*

GRADE 7 ACCELERATED · UNIT 6 · LESSON 3

# Pythagorean Theorem

*Question Bank - Additional 6.03* **ANSWER KEY**

Name
Date
Period

*Show your work. Circle your final answer.*

**For questions 1 through 3, find the hypotenuse. Give it exactly, and to the nearest tenth when it is not a whole number.**

**1.** Legs of 9 ft and 12 ft

***15 ft***

**2.** Legs of 3 inches and 10 inches

***√109 in ≈ 10.4 in***

*9 + 100 = 109.*

**3.** Legs of 6 m and 13 m

***√205 m ≈ 14.3 m***

*36 + 169 = 205.*

**For questions 4 through 6, find the missing leg. Give it exactly, and to the nearest tenth when it is not a whole number.**

**4.** Hypotenuse 10 cm, one leg 8 cm

***6 cm***

*100 − 64 = 36.*

**5.** Hypotenuse 7 inches, one leg 4 inches

***√33 in ≈ 5.7 in***

*49 − 16 = 33.*

**6.** Hypotenuse 37 m, one leg 35 m

***12 m***

*1,369 − 1,225 = 144 = $12^2$.*

**7.** A rectangle is 6 cm wide and 8 cm long. How long is its diagonal?

***10 cm***

*The diagonal is the hypotenuse of a 6-by-8 right triangle.*

**Choose one.**

**8.** A right triangle has a hypotenuse of 26 ft and one leg of 24 ft. How long is the other leg?

- ○ A. 100 ft
- ● **B. 10 ft**
- ○ C. 2 ft
- ○ D. 35.4 ft

***B***

*676 − 576 = 100 = $10^2$.*

GRADE 7 ACCELERATED · UNIT 6 · LESSON 4

# Pythagorean Theorem in Real-World Problems

*Independent Set 6.04* **ANSWER KEY**

Name
Date
Period

*Six questions, on your own, in silence. Show the step that does the work.*

**1.** A tablet screen is 5 inches wide and 8 inches tall. How long is its diagonal, to the nearest tenth of an inch?

***About 9.4 in***

*25 + 64 = 89; √89 ≈ 9.43.*

**2.** A fire truck's 50-ft ladder reaches 48 ft up a building. How far from the building is its foot?

***14 ft***

*2,500 − 2,304 = 196; the ladder is c.*

**3.** A ramp rises 3 ft over a run of 14 ft. How long is the ramp, to the nearest tenth of a foot?

***About 14.3 ft***

*9 + 196 = 205; √205 ≈ 14.32.*

**4.** A park is 7 m by 11 m. How much shorter is the walk straight across, corner to corner, than the walk along two sides? Give it to the nearest tenth of a meter.

***About 5.0 m shorter***

*Along two sides: 18 m. Across: √170 ≈ 13.04 m. 18 − 13.04 = 4.96, so 5.0.*

**5.** Will a 10-in pencil lie flat in a box 7 inches by 8 inches, corner to corner?

***Yes — the diagonal is about 10.6 in.***

*49 + 64 = 113; √113 ≈ 10.63 > 10.*

**6.** A kite flies at the end of 25 m of string, straight above a spot 20 m from where the string is held. How high is the kite?

***15 m***

*625 − 400 = 225; the string is c.*

GRADE 7 ACCELERATED · UNIT 6 · LESSON 4

# Pythagorean Theorem in Real-World Problems

*Question Bank 6.04* **ANSWER KEY**

Name
Date
Period

*Show your work. Circle your final answer.*

**For questions 1 through 7, sketch the right triangle first. Round to the nearest tenth when the answer is not a whole number.**

**1.** A TV screen is 9 inches tall and 12 inches wide. How long is its diagonal?

***15 in***

*81 + 144 = 225.*

**2.** A 14-ft ladder leans on a wall with its foot 3 ft from the wall. How high does it reach?

***About 13.7 ft***

*196 − 9 = 187; √187 ≈ 13.67.*

**3.** A ramp rises 2 m over a run of 9 m. How long is the ramp?

***About 9.2 m***

*4 + 81 = 85; √85 ≈ 9.22.*

**4.** A field is 6 m by 8 m. How much shorter is the walk straight across, corner to corner, than the walk along two sides?

***4 m***

*Along two sides 14 m, across 10 m.*

**5.** Will a 12-in ruler lie flat in a box 6 inches by 11 inches, corner to corner?

***Yes — the diagonal is about 12.5 in.***

*36 + 121 = 157; √157 ≈ 12.53, so 12.5.*

**6.** A kite's 13-m string is held at a spot 5 m from the point straight below the kite. How high is the kite?

***12 m***

*169 − 25 = 144; the string is c.*

**7.** A wire runs from the top of an 8-m pole to a stake in the ground 6 m from the pole's base. How long is the wire?

***10 m***

*64 + 36 = 100.*

**Choose one.**

**8.** A 13-ft ladder reaches 12 ft up a wall. How far from the wall is its foot?

- ○ A. 1 ft
- ● **B. 5 ft**
- ○ C. 17.7 ft
- ○ D. 25 ft

***B***

*$169 - 144 = 25 = 5^2$.*

**9.** A ramp rises 3 ft over a run of 20 ft. Which length is the hypotenuse?

- ○ A. the run
- ○ B. the rise
- ● **C. the ramp**
- ○ D. none of them

***C***

*The square corner is under the high end; across from it is the ramp.*

GRADE 7 ACCELERATED · UNIT 6 · LESSON 4

# Pythagorean Theorem in Real-World Problems

*Question Bank - Additional 6.04* **ANSWER KEY**

Name
Date
Period

*Show your work. Circle your final answer.*

**For questions 1 through 7, sketch the right triangle first. Round to the nearest tenth when the answer is not a whole number.**

**1.** A window is 8 ft wide and 6 ft tall. How long is its diagonal?

***10 ft***

*64 + 36 = 100.*

**2.** An 11-ft ladder leans on a wall with its foot 4 ft from the wall. How high does it reach?

***About 10.2 ft***

*121 − 16 = 105; √105 = 10.246…, so 10.2.*

**3.** A ramp rises 3 ft over a run of 10 ft. How long is the ramp?

***About 10.4 ft***

*9 + 100 = 109; √109 ≈ 10.44.*

**4.** A garden bed is 3 m by 4 m. How much shorter is the walk straight across, corner to corner, than the walk along two sides?

***2 m***

*Along two sides 7 m, across 5 m.*

**5.** Will a 15-in rod lie flat in a box 9 inches by 11 inches, corner to corner?

***No — the diagonal is only about 14.2 in.***

*81 + 121 = 202; √202 ≈ 14.21.*

**6.** A kite's 17-m string is held at a spot 8 m from the point straight below the kite. How high is the kite?

***15 m***

*289 − 64 = 225; the string is c.*

**7.** A wire runs from the top of a 9-m pole to a stake in the ground 12 m from the pole's base. How long is the wire?

***15 m***

*81 + 144 = 225.*

**Choose one.**

**8.** A 10-ft ladder reaches 8 ft up a wall. How far from the wall is its foot?

- ● **A. 6 ft**
- ○ B. 2 ft
- ○ C. 36 ft
- ○ D. 12.8 ft

***A***

*$100 - 64 = 36 = 6^2$.*

**9.** A wire runs from the top of a pole to a stake in the ground. Which length is the hypotenuse?

- ○ A. the pole
- ● **B. the wire**
- ○ C. the ground from the pole to the stake
- ○ D. none of them

***B***

*The pole stands square on the ground; across from that corner is the wire.*

GRADE 7 ACCELERATED · UNIT 6 · LESSON 5

# Distance on the Coordinate Plane

*Independent Set 6.05* **ANSWER KEY**

Name
Date
Period

*Six questions, on your own, in silence. Show the step that does the work.*

**1.** Find the distance between $(-6, 3)$ and $(7, 3)$.

***13 units***

*Same y: 7 − (−6) = 13.*

**2.** Find the distance between $(5, -7)$ and $(5, 4)$.

***11 units***

*Same x: 4 − (−7) = 11.*

**3.** Find the distance between $(-7, 1)$ and $(5, -4)$.

***13 units***

*Across 12, down 5; 144 + 25 = 169.*

**4.** Find the distance between $(-2, 5)$ and $(1, -2)$, to the nearest tenth.

***$\sqrt{58} \approx 7.6$ units***

*Across 3, down 7; 9 + 49 = 58.*

**5.** How far is $(-4, 9)$ from the origin, to the nearest tenth?

***$\sqrt{97} \approx 9.8$ units***

*Legs 4 and 9; 16 + 81 = 97.*

**6.** On a map in kilometers, a pool is at $(-2, -3)$ and a park at $(7, 9)$. How far apart are they in a straight line?

***15 km***

*Across 9, up 12; 81 + 144 = 225.*

GRADE 7 ACCELERATED · UNIT 6 · LESSON 5

# Distance on the Coordinate Plane

*Question Bank 6.05* **ANSWER KEY**

Name
Date
Period

*Show your work. Circle your final answer.*

**For questions 1 through 7, find the distance between the two points. Round to the nearest tenth when it is not a whole number.**

**1.** $(-8, 4)$ and $(2, 4)$

***10 units***

*Same y.*

**2.** $(3, -6)$ and $(3, 5)$

***11 units***

*Same x.*

**3.** $(-1, -1)$ and $(2, 3)$

***5 units***

*Across 3, up 4.*

**4.** $(-6, 2)$ and $(6, -3)$

***13 units***

*Across 12, down 5.*

**5.** $(-3, -2)$ and $(4, 1)$

***$\sqrt{58} \approx 7.6$ units***

*Across 7, up 3.*

**6.** $(0, 0)$ and $(-5, 7)$

***$\sqrt{74} \approx 8.6$ units***

*Legs 5 and 7.*

**7.** $(2, -4)$ and $(-1, 6)$

***$\sqrt{109} \approx 10.4$ units***

*Across 3, up 10.*

**Choose one.**

**8.** What is the distance between $(-6, -1)$ and $(2, -1)$ ?

- ○ A. 4
- ○ B. −8
- ○ C. 9
- ● **D. 8**

***D***

*Same y: 2 − (−6) = 8.*

**9.** Which gives the distance between $(5, 2)$ and $(-2, -3)$ ?

- ● **A.** $\sqrt{7^2 + 5^2}$
- ○ B. $\sqrt{3^2 + 5^2}$
- ○ C. $\sqrt{5^2 + 2^2}$
- ○ D. $7 + 5$

***A***

*Across 7, down 5: √74.*

GRADE 7 ACCELERATED · UNIT 6 · LESSON 5

# Distance on the Coordinate Plane

*Question Bank - Additional 6.05* **ANSWER KEY**

Name
Date
Period

*Show your work. Circle your final answer.*

**For questions 1 through 7, find the distance between the two points. Round to the nearest tenth when it is not a whole number.**

**1.** $(-5, -2)$ and $(4, -2)$

***9 units***

*Same y.*

**2.** $(-1, 7)$ and $(-1, -6)$

***13 units***

*Same x.*

**3.** $(-4, 1)$ and $(2, 9)$

***10 units***

*Across 6, up 8.*

**4.** $(-3, -5)$ and $(6, 7)$

***15 units***

*Across 9, up 12.*

**5.** $(-2, 3)$ and $(4, -2)$

***$\sqrt{61} \approx 7.8$ units***

*Across 6, down 5.*

**6.** $(0, 0)$ and $(8, -3)$

***$\sqrt{73} \approx 8.5$ units***

*Legs 8 and 3.*

**7.** $(-5, 4)$ and $(2, -5)$

***$\sqrt{130} \approx 11.4$ units***

*Across 7, down 9.*

**Choose one.**

**8.** What is the distance between $(-7, 3)$ and $(-7, -4)$ ?

- ○ A. 8
- ○ B. −7
- ○ C. 1
- ● **D. 7**

***D***

*Same x: 3 − (−4) = 7.*

**9.** Which gives the distance between $(-4, 5)$ and $(0, -4)$ ?

- ○ A. $\sqrt{4^2 + 5^2}$
- ● **B.** $\sqrt{4^2 + 9^2}$
- ○ C. $\sqrt{4^2 + 1^2}$
- ○ D. $4 + 9$

***B***

*Across 4, down 9: √97.*

GRADE 7 ACCELERATED · UNIT 6 · LESSON 6

# Real-World Coordinate-Plane Problems; Perimeter

*Independent Set 6.06* **ANSWER KEY**

Name
Date
Period

*Six questions, on your own, in silence. Show the step that does the work.*

**1.** Find the perimeter of the triangle $(-2, -7)$, $(-2, 5)$, $(3, 5)$.

***30 units***

*12 and 5 on grid lines; the slanted side 13.*

**2.** Find the perimeter of the triangle $(-4, 1)$, $(2, 1)$, $(2, -4)$, to the nearest tenth.

***18.8 units***

*6 + 5 + √61 ≈ 18.81.*

**3.** Find the perimeter of the parallelogram $(-4, -5)$, $(2, -5)$, $(7, 7)$, $(1, 7)$.

***38 units***

*Two sides of 6; two slanted sides, across 5 and up 12, of 13.*

**4.** On a map in blocks, Kai's home is at $(-6, -5)$ and his school at $(3, 7)$. How much shorter is the straight line than the streets?

***6 blocks***

*Streets 9 + 12 = 21; straight 15.*

**5.** On a map in miles, a dock is at $(-4, 3)$ and an island at $(3, -5)$. How far apart are they, to the nearest tenth?

***√113 ≈ 10.6 mi***

*Across 7, down 8; 49 + 64 = 113.*

**6.** A rectangle has corners $(-3, -2)$, $(2, -2)$, $(2, 10)$ and $(-3, 10)$. How long is a diagonal?

***13 units***

*Sides 2 − (−3) = 5 and 10 − (−2) = 12; 25 + 144 = 169.*

GRADE 7 ACCELERATED · UNIT 6 · LESSON 6

# Real-World Coordinate-Plane Problems; Perimeter

*Question Bank 6.06* **ANSWER KEY**

Name
Date
Period

*Show your work. Circle your final answer.*

**For questions 1 through 4, find the perimeter of the triangle with the given corners. Round to the nearest tenth when it is not a whole number.**

**1.** $(-1, -2)$ , $(8, -2)$ , $(8, 10)$

***36 units***

*9, 12 and 15.*

**2.** $(-3, 6)$ , $(2, 6)$ , $(-3, -6)$

***30 units***

*5, 12 and 13.*

**3.** $(-2, -3)$ , $(4, -3)$ , $(4, 4)$

***22.2 units***

*6 + 7 + √85 ≈ 22.22.*

**4.** $(-5, -2)$ , $(5, -2)$ , $(0, 10)$ — a triangle with no right angle

***36 units***

*Base 10; the height 12 splits it 5 and 5, so each slanted side is 13.*

**For questions 5 through 7, each grid unit is one block or one meter.**

**5.** Home is at $(-6, 2)$ and school at $(3, -10)$ . How much shorter is the straight line than the streets?

***6 blocks***

*Streets 21; straight 15.*

**6.** On a park map in meters, a bench is at $(-2, -4)$ and a slide at $(6, -1)$ . How far apart are they, to the nearest tenth?

***√73 ≈ 8.5 m***

*Across 8, up 3.*

**7.** How long is a diagonal of the rectangle with corners $(-3, -1)$, $(9, -1)$, $(9, 4)$ and $(-3, 4)$?

***13 units***

*Sides 12 and 5.*

**Choose one.**

**8.** What is the perimeter of the triangle with corners $(-3, -1)$, $(0, -1)$ and $(0, 6)$, to the nearest tenth?

- ○ A. 10
- ● **B. 17.6**
- ○ C. 20
- ○ D. 68

***B***

*3 + 7 + √58 ≈ 17.62.*

**9.** How much shorter is the straight path from $(-4, -2)$ to $(5, 10)$ than the path along the grid lines?

- ○ A. 15
- ○ B. 21
- ● **C. 6**
- ○ D. 36

***C***

*21 along the grid lines, 15 straight.*

GRADE 7 ACCELERATED · UNIT 6 · LESSON 6

# Real-World Coordinate-Plane Problems; Perimeter

*Question Bank - Additional 6.06* **ANSWER KEY**

Name
Date
Period

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*Show your work. Circle your final answer.*

**For questions 1 through 4, find the perimeter of the triangle with the given corners. Round to the nearest tenth when it is not a whole number.**

**1.** $(-4, -1)$ , $(1, -1)$ , $(1, 11)$

***30 units***

*5, 12 and 13.*

**2.** $(6, -3)$ , $(-3, -3)$ , $(6, 9)$

***36 units***

*9, 12 and 15.*

**3.** $(-5, 2)$ , $(3, 2)$ , $(3, -1)$

***19.5 units***

*8 + 3 + √73 ≈ 19.54.*

**4.** $(-9, -4)$ , $(9, -4)$ , $(0, 8)$ — a triangle with no right angle

***48 units***

*Base 18; the height 12 splits it 9 and 9, so each slanted side is 15.*

**For questions 5 through 7, each grid unit is one block or one meter.**

**5.** Home is at $(-5, 6)$ and the store at $(7, 1)$ . How much shorter is the straight line than the streets?

***4 blocks***

*Streets 17; straight 13.*

**6.** On a park map in meters, a bench is at $(-6, 1)$ and a fountain at $(2, -4)$ . How far apart are they, to the nearest tenth?

***√89 ≈ 9.4 m***

*Across 8, down 5.*

**7.** How long is a diagonal of the rectangle with corners $(-4, -3)$ , $(5, -3)$ , $(5, 9)$ and $(-4, 9)$ ?

***15 units***

*Sides 9 and 12.*

**Choose one.**

**8.** What is the perimeter of the triangle with corners $(-1, 2)$ , $(4, 2)$ and $(4, -5)$ , to the nearest tenth?

- ○ A. 12
- ○ B. 24
- ● **C. 20.6**
- ○ D. 86

***C***

*5 + 7 + √74 ≈ 20.60.*

**9.** How much shorter is the straight path from $(-3, -5)$ to $(9, 0)$ than the path along the grid lines?

- ● **A. 4**
- ○ B. 17
- ○ C. 13
- ○ D. 30

***A***

*17 along the grid lines, 13 straight.*

GRADE 7 ACCELERATED · UNIT 6 · LESSONS 7–8

# Similar Triangles; Indirect Measurement

*Independent Set 6.07* **ANSWER KEY**

Name
Date
Period

*Six questions, on your own, in silence. Show the step that does the work.*

**1.** Are triangles with sides 4, 10, 12 and 6, 15, 20 similar? Show the three ratios.

***No — 6/4 = 15/10 = 1.5, but 20/12 ≈ 1.67***

*Two ratios agree and the third does not.*

**2.** Triangle PQR is similar to triangle XYZ. PQ is 8, QR is 10 and PR is 13, and XY is 20. Find YZ and XZ.

***YZ = 25 and XZ = 32.5***

*The names match PQ with XY: k = 20 ÷ 8 = 2.5; YZ matches QR and XZ matches PR.*

**3.** A triangle has sides 15, 25 and 30. A similar triangle's longest side is 6. Find its other two sides.

***3 and 5***

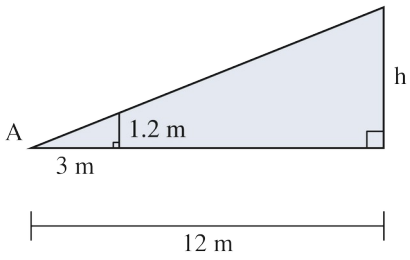
*k = 6 ÷ 30 = 0.2.*

**4.** A 1.5-m child casts a 2-m shadow. At the same moment a tree casts a 14-m shadow. How tall is the tree?

***10.5 m***

*k = 14 ÷ 2 = 7; 1.5 × 7.*

**5.** The two right triangles are similar and share corner A. The small one runs 3 m along the ground from A and stands 1.2 m high. The big one runs 12 m along the ground from A. How high is it?




***4.8 m***

*Both from A: k = 12 ÷ 3 = 4, and 1.2 × 4 = 4.8.*

**6.** Which is similar to a triangle with sides 8, 10 and 14: one with sides 10, 12, 16, or one with sides 12, 15, 21? Show why.

***12, 15, 21 — every side × 1.5***

*10, 12, 16 adds 2 to every side.*

GRADE 7 ACCELERATED · UNIT 6 · LESSONS 7–8

# Similar Triangles; Indirect Measurement

*Question Bank 6.07* **ANSWER KEY**

Name
Date
Period

*Show your work. Circle your final answer.*

**For questions 1 and 2, decide whether the triangles are similar. Show the three ratios.**

**1.** Sides 5, 7, 8 and 15, 21, 24

***Similar — every ratio is 3***

*15/5 = 21/7 = 24/8 = 3.*

**2.** Sides 6, 9, 10 and 8, 12, 15

***Not similar — 8/6 = 12/9, but 15/10 = 1.5***

*Two ratios are 4/3; the third is 3/2.*

**For questions 3 through 5, the two triangles are similar. Find the missing sides.**

**3.** Sides 4, 9 and 11; the copy's 4-side is 10.

***22.5 and 27.5***

*k = 2.5.*

**4.** Sides 18, 24 and 30; the copy's 30-side is 5.

***3 and 4***

*k = 5/30 = 1/6.*

**5.** Sides 3, 7 and 8; the copy's 7-side is 28.

***12 and 32***

*k = 4.*

**For questions 6 and 7, the triangles are similar.**

**6.** A 6-ft man's shadow is 8 ft long. At the same moment a pole's shadow is 20 ft. How tall is the pole?

***15 ft***

*k = 20/8 = 2.5; 6 × 2.5.*

**7.** Ty's eyes are 4 ft up, 2 ft from a mirror on the ground. In it he sees the top of a tree 15 ft beyond the mirror. How tall is the tree?

***30 ft***

*k = 15/2 = 7.5; 4 × 7.5.*

**Choose one.**

**8.** Which triangle is similar to one with sides 4, 6 and 9?

- ○ A. 6, 8, 11
- ● **B. 8, 12, 18**
- ○ C. 8, 12, 16
- ○ D. 4, 6, 12

***B***

*Every side × 2.*

**9.** A 5-ft girl casts a 4-ft shadow. At the same moment a tree's shadow is 24 ft. How tall is the tree?

- ○ A. 19.2 ft
- ● **B. 30 ft**
- ○ C. 25 ft
- ○ D. 120 ft

***B***

*$k = 24 \div 4 = 6$; $5 \times 6$.*

GRADE 7 ACCELERATED · UNIT 6 · LESSONS 7–8

# Similar Triangles; Indirect Measurement

*Question Bank - Additional 6.07* **ANSWER KEY**

Name
Date
Period

*Show your work. Circle your final answer.*

**For questions 1 and 2, decide whether the triangles are similar. Show the three ratios.**

**1.** Sides 6, 8, 9 and 15, 20, 22.5

***Similar — every ratio is 2.5***

*15/6 = 20/8 = 22.5/9 = 2.5.*

**2.** Sides 4, 5, 7 and 6, 8, 10.5

***Not similar — 6/4 = 1.5, but 8/5 = 1.6***

*The middle ratio differs.*

**For questions 3 through 5, the two triangles are similar. Find the missing sides.**

**3.** Sides 5, 6 and 8; the copy's 5-side is 15.

***18 and 24***

*k = 3.*

**4.** Sides 20, 30 and 35; the copy's 20-side is 4.

***6 and 7***

*k = 0.2.*

**5.** Sides 2, 5 and 6; the copy's 6-side is 21.

***7 and 17.5***

*k = 3.5.*

**For questions 6 and 7, the triangles are similar.**

**6.** A 1.8-m woman's shadow is 1.2 m long. At the same moment a building's shadow is 16 m. How tall is the building?

***24 m***

*Each height is 1.8 ÷ 1.2 = 1.5 times its shadow: 16 × 1.5.*

**7.** Mia's eyes are 5 ft up, 2 ft from a mirror on the ground. In it she sees the top of a building 30 ft beyond the mirror. How tall is the building?

***75 ft***

*k = 15; 5 × 15.*

**Choose one.**

**8.** Which triangle is similar to one with sides 3, 5 and 6?

- ● **A. 9, 15, 18**
- ○ B. 5, 7, 8
- ○ C. 6, 10, 14
- ○ D. 3, 5, 9

***A***

*Every side × 3.*

**9.** A 6-ft man casts a 5-ft shadow. At the same moment a tree's shadow is 35 ft. How tall is the tree?

- ○ A. 29.2 ft
- ○ B. 36 ft
- ○ C. 210 ft
- ● **D. 42 ft**

***D***

*$k = 35 \div 5 = 7$; $6 \times 7$.*

GRADE 7 ACCELERATED · UNIT 6 · THREAD B · DAY 1

# Angle Pairs: Supplementary, Complementary, Vertical, Adjacent

*Independent Set 6.T1* **ANSWER KEY**

Name
Date
Period

*Six questions, on your own, in silence. Show the step that does the work.*

**1.** What is the complement of 61°?

***29°***

*90 − 61.*

**2.** What is the supplement of 76°?

***104°***

*180 − 76.*

**3.** Two lines cross, and one of the four angles is 58°. Find the other three.

***122°, 58° and 122°***

*The vertical angle is equal; each neighbour makes 180 with it.*

**4.** A linear pair measures $(5x + 10)^\circ$ and $70^\circ$. Find x and the first angle.

***x = 20; 110°***

*5x + 80 = 180.*

**5.** Vertical angles measure $(7x - 9)^\circ$ and $(3x + 31)^\circ$. Find the measure of each angle.

***61°***

*7x − 9 = 3x + 31, x = 10.*

**6.** $\angle PQR$ and $\angle RQS$ share the side QR and together make a straight line, and $m\angle PQR = 65^\circ$. Name the pair and find $m\angle RQS$.

***A linear pair (adjacent and supplementary); 115°***

*180 − 65.*

GRADE 7 ACCELERATED · UNIT 6 · THREAD B · DAY 1

# Angle Pairs: Supplementary, Complementary, Vertical, Adjacent

*Question Bank 6.T1* **ANSWER KEY**

Name
Date
Period

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*Show your work. Circle your final answer.*

**For questions 1 through 4, find the angle.**

**1.** The complement of 17°

***73°***

*90 − 17.*

**2.** The supplement of 95°

***85°***

*180 − 95.*

**3.** The angle vertical to a 128° angle

***128°***

*Vertical angles are equal.*

**4.** The angle that makes a linear pair with a 41° angle

***139°***

*180 − 41.*

**For questions 5 through 7, find x and the angles.**

**5.** $(x+12)^\circ$ and $(2x)^\circ$ are complementary.

***x = 26; 38° and 52°***

*3x + 12 = 90.*

**6.** $(4x-6)^\circ$ and $(2x+18)^\circ$ are a linear pair.

***x = 28; 106° and 74°***

*6x + 12 = 180.*

**7.** Vertical angles measure $(8x-7)^\circ$ and $(5x+20)^\circ$.

***x = 9; each 65°***

*3x = 27.*

**Choose one.**

**8.** Angles M and N are supplementary, and angle M is 64°. What is angle N?

- ○ A. 26°
- ● **B. 116°**
- ○ C. 64°
- ○ D. 244°

***B***

*180 − 64.*

**9.** Vertical angles measure $(3x+8)^\circ$ and $(5x-14)^\circ$. A student answers 11. What did the student find?

- ○ A. the angle
- ● **B. the value of x**
- ○ C. the supplement
- ○ D. the complement

***B***

*x = 11; each angle is 3(11) + 8 = 41°.*

# Angle Pairs: Supplementary, Complementary, Vertical, Adjacent

*Question Bank - Additional 6.T1* **ANSWER KEY**

Name
Date
Period

*Show your work. Circle your final answer.*

**For questions 1 through 4, find the angle.**

**1.** The complement of 43°

***47°***

*90 − 43.*

**2.** The supplement of 151°

***29°***

*180 − 151.*

**3.** The angle vertical to a 66° angle

***66°***

*Vertical angles are equal.*

**4.** The angle that makes a linear pair with a 117° angle

***63°***

*180 − 117.*

**For questions 5 through 7, find x and the angles.**

**5.** $(3x)^\circ$ and $(x+18)^\circ$ are complementary.

***x = 18; 54° and 36°***

*4x + 18 = 90.*

**6.** $(5x+4)^\circ$ and $(3x-8)^\circ$ are a linear pair.

***x = 23; 119° and 61°***

*8x − 4 = 180.*

**7.** Vertical angles measure $(9x-15)^\circ$ and $(4x+30)^\circ$.

***x = 9; each 66°***

*5x = 45.*

**Choose one.**

**8.** Angles E and F are complementary, and angle E is 37°. What is angle F?

- ○ A. 143°
- ○ B. 37°
- ○ C. 127°
- ● **D. 53°**

***D***

*90 − 37.*

**9.** Vertical angles measure $(6x-5)^\circ$ and $(2x+27)^\circ$. A student answers 8. What did the student find?

- ● **A. the value of x**
- ○ B. the angle
- ○ C. the complement
- ○ D. the supplement

***A***

*x = 8; each angle is 6(8) − 5 = 43°.*

GRADE 7 ACCELERATED · UNIT 6 · THREAD B · DAY 2

# Interior and Exterior Angles of a Triangle

*Independent Set 6.T2* **ANSWER KEY**

Name
Date
Period

*Six questions, on your own, in silence. Show the step that does the work.*

**1.** Two angles of a triangle are 63° and 79°. Find the third angle.

***38°***

*180 − 142.*

**2.** An isosceles triangle's two base angles are equal. The angle between its two equal sides is 40°. Find each base angle.

***70°***

*(180 − 40) ÷ 2.*

**3.** A triangle's remote interior angles are 46° and 58°. Find the exterior angle.

***104°***

*46 + 58.*

**4.** An exterior angle of a triangle is 140°. One remote interior angle is 62°. Find the other.

***78°***

*140 − 62.*

**5.** A triangle's angles are $(2x)^\circ$ , $(3x)^\circ$ and $(4x)^\circ$ . Find each angle.

***40°, 60° and 80°***

*9x = 180, x = 20.*

**6.** An exterior angle of a triangle measures $(4x+10)^\circ$ ; its remote interior angles measure $50^\circ$ and $(2x)^\circ$ . Write an equation and find x.

***4x + 10 = 50 + 2x; x = 20***

*2x = 40.*

GRADE 7 ACCELERATED · UNIT 6 · THREAD B · DAY 2

# Interior and Exterior Angles of a Triangle

*Question Bank 6.T2* **ANSWER KEY**

Name
Date
Period

*Show your work. Circle your final answer.*

**For questions 1 through 4, find the angle.**

**1.** The third angle of a triangle with angles of 34° and 101°

***45°***

*180 − 135.*

**2.** The third angle of a right triangle with an angle of 27°

***63°***

*180 − 90 − 27.*

**3.** The exterior angle whose remote interior angles are 52° and 61°

***113°***

*52 + 61.*

**4.** The other remote interior angle, when an exterior angle is 118° and one remote angle is 47°

***71°***

*118 − 47.*

**For questions 5 through 7, find x and the angles.**

**5.** A triangle's angles are $x^\circ$, $(x+30)^\circ$ and $(2x-10)^\circ$.

***x = 40; 40°, 70° and 70°***

*4x + 20 = 180.*

**6.** An isosceles triangle's base angles are each $(3x+5)^\circ$ and the third angle is $50^\circ$.

***x = 20; 65° and 65°***

*6x + 10 + 50 = 180.*

**7.** An exterior angle is $(6x-4)^\circ$; its remote interior angles are $(2x+8)^\circ$ and $40^\circ$.

***x = 13; the exterior angle is 74°***

*6x − 4 = 2x + 48.*

**Choose one.**

**8.** A triangle's angles are 30°, 70° and 80°. What is the exterior angle beside the 80°?

- ○ A. 80°
- ○ B. 280°
- ● **C. 100°**
- ○ D. 70°

***C***

*30 + 70.*

**9.** Which equation finds x when an exterior angle is $(4x)^\circ$ and its remote interior angles are $45^\circ$ and $(x+15)^\circ$ ?

- ○ A. $4x+x+15=180$
- ● **B.** $4x=45+x+15$
- ○ C. $4x=x+15$
- ○ D. $4x+45+x+15=180$

***B***

*Outside equals the two far ones: x = 20.*

# Interior and Exterior Angles of a Triangle

*Question Bank - Additional 6.T2* **ANSWER KEY**

Name
Date
Period

*Show your work. Circle your final answer.*

**For questions 1 through 4, find the angle.**

**1.** The third angle of a triangle with angles of 29° and 88°

***63°***

*180 − 117.*

**2.** The third angle of a right triangle with an angle of 44°

***46°***

*180 − 90 − 44.*

**3.** The exterior angle whose remote interior angles are 33° and 79°

***112°***

*33 + 79.*

**4.** The other remote interior angle, when an exterior angle is 125° and one remote angle is 68°

***57°***

*125 − 68.*

**For questions 5 through 7, find x and the angles.**

**5.** A triangle's angles are $x^\circ$, $(2x+15)^\circ$ and $(x+25)^\circ$.

***x = 35; 35°, 85° and 60°***

*4x + 40 = 180.*

**6.** An isosceles triangle's base angles are each $(4x-2)^\circ$ and the third angle is $64^\circ$.

***x = 15; 58° and 58°***

*8x − 4 + 64 = 180.*

**7.** An exterior angle is $(5x+2)^\circ$; its remote interior angles are $(3x+10)^\circ$ and $36^\circ$.

***x = 22; the exterior angle is 112°***

*2x = 44.*

**Choose one.**

**8.** A triangle's angles are 25°, 65° and 90°. What is the exterior angle beside the 65°?

- ● **A. 115°**
- ○ B. 65°
- ○ C. 295°
- ○ D. 25°

***A***

*25 + 90.*

**9.** Which equation finds x when an exterior angle is $(3x+6)^\circ$ and its remote interior angles are $54^\circ$ and $(x)^\circ$ ?

- ○ A. $3x+6=x$
- ○ B. $3x+6+x=180$
- ○ C. $3x+6+54+x=180$
- ● **D.** $3x+6=54+x$

***D***

*Outside equals the two far ones: x = 24.*